

SKN COLLEGE OF AGRICULTURE :JOBNER
(SKN AGRICULTURE UNIVERSITY:JOBNER)

PRODUCTION TECHNOLOGY FOR VEGETABLES AND SPICES

(HORTI- 211) 2(1+1)

Name :

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LECT. 1 Importance of vegetables & spices in human nutrition and national economy

Vegetable:

Any part of crop consumed as fresh or after cooking is called as "vegetable".

Vegetable: The term vegetable is applied to the edible herbaceous plant or plant parts thereof, which are consumed generally in the unripe stage after cooking.

Difference between Fruits and Vegetables:

Following are important differences between fruits and vegetables:

1. In case of fruits only flower or fruit is consumed While in vegetable all parts from roots to fruits are consumed (roots / leaves / stem / flower / fruit / seed / buds).
2. Fruits are consumed a fresh without cooking eg. Banana, apple, guava while vegetables are consumed a fresh as salad and also after cooking few are eaten both way fresh and after cooking eg. Tomato, onion, cucumber. etc.
3. . Biennial Fruits are perennial and woody in nature with exception that some are annual and biennial and e.g. papaya, banana, while majority of vegetables are annual while few are biennial (onion, Cole crops, root crops) and few perennial (asparagus, parwal, tondali, hadga, drumstick).
4. Orchard management: Tranning, pruning and certain specific operations are performed in fruit crops and not in gegetable crops.
5. Fruit crops are mostly propagated by vegetative method except papaya, phalsa, Coconut and kagzi lime which are propagated by seed while majority of vegetable crops are propagated by seed with exception of tondali aspaaragus, sweet potato, parwal, potato.

Olericulture: is one of the branches of Horticulture that deals with the vegetables. The word olericulture is derived from the Latin word **Oleris** which means pot herb and the English word **culture** which means cultivation.

Thus olericulture means cultivation of pot herbs. However, in the present days, it is bradly used to indicate the cultivation of vegetables. The term vegetable gardening is more popular to signify olericulture in the present context.

Importance of vegetables in human nutrition:

- The balanced diet contain adequate energy source, nutrients and vitamins, mineras, carbohydrates, fats, protein etc.
- Vegetable are the reliable source for many dietary factors.
- As vegetable contain many of the dietary factors like vitamins, minerals and amino acids they are considered as **protective supplementary food**.
- They produce taste, increase appetite and produce fair amount of fibres.
- They maintain good health and protect against degenerative diseases.

- They can neutralise the acids produced during digestion of proteins and fats.
- Nutrients which are present in vegetables vary from crop to crop.
- Peas and beans are enriched with **proteins**.
- Root crops like Tapioca, Sweet potato and potato are well known for **carbohydrates**, calcium K, Fe are the important minerals which are lacking in cereals and these are available in abundant quantities in the vegetables like peas, beans, spinach and bendi.
- Amaranth, cabbage, beans contain large quantity of cellulose which aid in digestion. All the leaf and fruit vegetables possess the required quantities of vitamins.

S.No	Dietary factors	source vegetables
1	Calories	sweet potato, tapioca, yam, colacasia corms, potato, Brussels Sprouts, onion and garlic, immature seeds of broad bean and peas <i>Phaseolus lunatus</i> (Lima bean), <i>Pussia fada</i> (Broad bean).
2	Proteins	peas, double bean, winged bean (<i>Psochocarpus tetragonolobus</i>), Garlic, Brussels sprouts, cowpea, lema bean seeds, amaranthus Leaves, drumstick leaves and menthe.
3	Vitamin A (Beta carotene)	Carrot, spinach, turnip green, palak, mustard green, amaranth, coriander, colacasia leaves, sweet potato, pumpkin, tomato
4	Vitamin B complex	Peas, broad bean, lema bean, garlic, asparagus, colacasia and Tomato.
5	Vitamin C	Turnip green, green chillies, Brussels sprouts, mustard green, Amaranth, coriander, drumstick leaves, cauliflower, knoll khol Spinach, cabbage, bitter gourd and reddish leaves.
6	Calcium	curry leaves, amaranth leaves, drumstick leaves, menthi, turnip, Mustard green, coriander and palak.
7	Iron	drum stick leaves and fruits, amaranth, menthi, mint, coriander, Spinach, palak and mustard green. Spinach, lettuce, cabbage
8	Roughages	Amaranth and root vegetables.
9	Vegetable milk	Pea pods and cabbage leaves

Importance of vegetables in National economy:

1. Annually we produce about **129 million tonnes** of vegetables from an area of **7.98** million ha, contributing **13.4** percentage to the world's production. (2009 NHB Data).
2. Vegetables crops have high export potential and vegetables worth of Rs 4431 crores are being exported annually both in fresh form or processed form. APEDA is the nodal organization involved in regulation of export and import of various food products.
3. Traditional vegetables like onion, potato, bhendi, bitter gourd and chillies and non traditional vegetables like asparagus, celery, paprika, sweet corn, baby corn, beans, peas and cherry tomato have been short listed by APEDA for export. Among the different vegetables, **77%** of the onion produced is being exported, thus earning valuable foreign exchange to the country. onion and traditional vegetables are being imported by gulf countries, Singapore, Malaysia, Srilanka, Bangladesh and Nepal. Non traditional vegetables are being imported by Australia, European countries, South East Asian and Gulf countries.
4. Out of the total out put of vegetables, **only 1%** is being utilized by the processing industry. Inadequate preservation techniques, and storage facilities contribute to 25% loss of vegetables produced. By improving post harvest techniques, we can ensure year round availability of vegetables.

Spices:

Spices are those plants, the products of which are made use of as food adjuncts o add aroma and flavour (**ex.** Pepper, Cardamom, Clove, Nutmeg etc), while **condiments** are those plants, the products of which are used as food adjuncts to add **taste** only.

Both spices and condiments contain essential oils, which provide the flavour and taste. They are of little nutritive value. They are used as whole, ground, paste or liquid form, mainly for flavouring and seasoning food. Most spices increase the shelf-life of food.

Uses of spices:

1. The principal use of spices is to season the insipid foods to impart flavour, aroma and taste.
2. They are also used as preservatives and fumigants.
3. Spices also find use in pharmacy and indigenous medicines.
4. Spices are also used in perfumery, soaps, cosmetics, tooth paste, confectionery, insenses, dyes, etc.

Properties of spices:

1. Spices are well known as appetizers.
2. They add a tang (taste) and flavour to otherwise insipid foods.
3. Some of them also possess anti-oxidant properties.
4. Some of them have preservative qualities (clove and mustard)
5. Some have strong anti-microbial and anitbiotic activities.
6. Several of them possess medicinal properties.

Importance of spice industry in India:

India is often referred as “**Home of spices**”. This is because many of the 70 spices grown in the world, are **native to India**. Further, since antiquity, India pioneered in growing spices and exported. India has enjoyed virtual monopoly in the International Spice trade since ancient times.

Out of the **70 spices**, several of them can be grown in India, whereas in other countries a few spices are only grown. This is because, India has a great extent of diversity in the climate and soils, which enables to grow a variety of spices. Spices were always export – oriented crops not only in India, but also in other spice producing countries.

India produces annually about 5.1 million tones of different spices valued at about Rs. 10200 crores, contributing to 25 to 30 per cent of the world production. India is the biggest exporter of spices and annually exporting about 2 million tonnes of different spices and spice products earning a foreign of about Rs. 5240 crores.

Classification of spices:

There are about 70 spices grown all over the world, which are classified in different ways.

I. Botanical classification (family wise)

Piperaceae - pepper.

Zingiberaceae – Ginger, Cardamom,
Turmeric

Apiaceae – Coriander, Fennel, Cumin.

Myrtaceae – Clove, Nutmeg

Fabaceae – Fenugreek

Lauraceae – Cinnamom

II. Plant part useful as spice:

Root spice - Angelica, Horse radish.

Rhizome spice – Turmeric, Ginger.

Bulbous spices – Onion, Garlic.

Bark spice – Cinnamom, Cassia

Leafy spices – Mints, Coriander, Methi

Aril spices – Mace.

Seedy spices – Coriander, Celery, Methi

Fruit spice – i) Capsules – Cardamom,
Chillies.

ii) Berries – Pepper, Allspice.

III. Depending upon the longevity of spice plants.

Annual spice – Coriander, Mints, Methi

Biennial spices – Onion, Garlic

Perennial spices – Clove, Nutmeg, Pepper etc

IV. Depending upon the type of the plant:

Tree spices – Clove, Nutmeg, Cinnamom, Cassia

Bush spices – Cardamom

Herbaceous spices – Coriander, Fenugreek, fennel, Cumin.

Climber spices – Pepper, Vanilla.

V. Economic importance: Depending upon the magnitude of trade, earnings and use.

Major spice: Pepper, Cardamom, Turmeric, Ginger, Clove, Nutmeg.

Minor spice: Coriander, Fenugreek, Cumin, Fennel.

LECT.2 Classification of Vegetables

I. Botanical classification:

All vegetable belongs to Division Angiospermae. The Angiospermae has 2 classes mono and dicotyledone. They are further divided into families, genus, species, subspecies and botanical varieties. The grouping of vegetables therefore is as follows:\

A. Monocotyledonae:

1. Amaryllidaceae – Onion, Garlic, Leak, Shallot, Chive.
2. Liliaceae – Asparagus
3. Araceae – Colocasia esculenta
4. Dioscoreaceae – Yam

B. Dicotyledonous plants (vegetables):

1. Chenopodiaceae – Palak, Beetroot, Spinach
2. Compositae – Lettuce, chikori
3. Convolvulaceae – Sweet potato.
4. Brassicaceae – Cabbage, Cauliflower
5. Cruciferae – Brussels, Cole rabi or knol-khol, radish, mustard.
6. Cucurbitaceae – All cucurbitaceous vegetables
7. Euphorbiaceae: Tapioca (Manihot esculenta)
8. Leguminosae :
 - Pea (*Pisum sativum*)
 - French bean (*Phaseolus vulgaris*)
 - Lima bean
 - Asparagus bean (*Vigna unguiculata* var *sesquipedalis*)
 - Cluster bean (*Cyamopsis tetragonoloba*)
 - Gem (*Dolichos lablab*)
 - Winged bean (*Psochocarpus tetragonolobus*)
 - Soya bean (*Glycine max*)
 - Methi (*Trigonella fonumgreacum*)
9. Malvaceae: Okra (Bendi)
10. Polygonaceae: Sorrel, Rhubarb (*Rheum rhapontium*)
11. Solanaceae
 - Potato: *Solanum tuberosum*
 - Brinjal: *Solanum melongena*
 - Tomato: *Lycopersicon esculentum*
 - Chillies: *Capsicum frutescence*
 - Pepper: *Capsicum annum*
12. Umbelliferae (Apiaceae)
 - Carrot, parsley,
13. Rutaceae: Curry leaf *Murraya koenigii* (Munaga)

II. Vegetables classification based on plant parts used:

According to parts used for consumption

1. **Leafy vegetables:** Amaranthus, Cabbage and Palak
2. **Fruts :** Tomato, Brinjal, Bhendi, Peas, Beans, Cucurbits
3. **Flower parts:** Cauliflower and Broccoli
4. **Under ground parts:**
 - a) **Stem tubers:**Potato and colocasia
 - b) **Root tubers:** Carrot, Sweet potato and Tapioca
 - c) **Bulbs:** Onion and garlic

The cultural requirements of crops in each group are not same i.e tomato, bhendi and cucurbhits. Hence this method is also not of much value.

III. Classification based on season of growing:

1. **Kharif season** –June to September – Cucurbits, Brinjal, Okra
2. **Rabi season** - October to January – Cabbage, cauliflower, Beet, Peas, tomato
3. **Summerseason** – February to May – Brinjal, Chilli, cluster beans, Bottle gourd

IV. Classification based on method of cultivation:

This is a very convenient method. In this method all the crops that have similar cultural requirements are grouped together. Therefore it is possible to recommend general cultural practices for all vegetables, which are grouped together. Some groups like cucurbits, cold crops, bulb crops are not only have similar cultural requirements for the group but the crops in each group belong to same family. The groups like greens, salad crops, bulb crops, the parts edible in each group are also same.

Therefore this system of classification has been found to be more satisfactory in understanding the principles of vegetable growing than any other methods

According to this method the vegetables are grouped in to 13 groups (Table)

Group No.	Group name	Examples
Group 1	Perennial vegetables	Asparagus, Coccinea
Group 2	Greens	Spinch, Palak
Group 3	Sald crops	Celery, Lettuce
Group 4	Cole crops	Cabage, Cauliflower
Group 5	Root crops	Beet root, Carrot, Radish
Group 6	Bulb crops	Onion, Garlic, Leek
Group 7	Potato	
Group 8	Sweet Potato	
Group 9	Peas and beans	Pea, cowpea, French bean
Group 10	Solanacious vegetables	Tomato, Brinjal, Chilli
Group 11	Sweet corn, Okra	
Group 12	Cucurbits	Bottle gourd, Pumpkin
Group 13	Yam, Tapioca	

Other Basis of Classification of Vegetables

A) Thermo Classification of Vegetables:

Vegetables are classified as cool and warm season crops. Depending upon temperature requirement, cool season crops require cool weather,

i) *Cool Season*: They are characterized by :

- a. In all cases vegetative parts are edible ex. Leaves (Palak), roots (carrot). stem (kale) except peas. Fibrous root system.
- b. Need constant supply of moisture.
- c. Careful application of nutrients. Ex. Onion, garlic, Cole crops, root crops, Palak, potato, peas.

ii) *Warm Season Crops*: They are summer vegetables and characterized by:

- i) In this group reproductive parts of vegetable are used for consumption example. Tomato, Brinjal, Cucurbits and beans. (Except - Sweet Potato).
- ii) Tap root system.
- iii) Water requirement is less.
- iv) Tolerate more Temperature. Ex. Solanaceou8s except potato, sweet potato, okra, beans.

B) Miscellaneous Classification:

- a. **Direct sown:** Okra, Beans
- b. **Transplanted:** Brinjal, Tomato Onion, Cole crops
- c. **By cuttings only:** Tondali, Sweet Potato
- d. **By bits of tubers :** Potato, Yam

C) According to Lime Requirement:

- a. Require no Lime : Potato, Tomato, and Radishb.
- b. Require lime medium : Pumpkin, Pea, and Cauliflower
- c. Require more lime : Brinjal, Cabbage, and Cauliflower
- d. Need much lime : Onion, Chilies

D) According to Forcing:

- a. **Cool forcing** : Onion, Pea, cauliflower
- b. **Warm forcing** : Cucumber, brinjal, Tomato

E) Rate of Respiration:

- a. **Very high rate of respiration** : Pea, spinach
- b. **High rate of Respiration** : Beans
- c. **Moderate rate of respiration** : Carrot, Tomato, Chilli
- d. **Low and very low** : Cabbage, Onion, and Potato.

F) Tolerance to Soil Reaction:

- a. **Slightly Tolerant (pH 6.8.6) :** Cauliflower, Palak, Musk melon,
- b. **Moderately Tolerant (pH 6.8.6):** Bean, Carrot, Garlic, Tomato,
- c. **Highly Tolerant (pH 6.8 - 5-5):** Potato, Sweet Potato, Watermelon.

G) Classification Based on Salt Tolerance:

- a) Semi tolerant 0.25 : Tomato,
- b) Medium Tolerant 0.50 - 0.75 : Chilies, Okra, Cabbage, Sweet potato, Onion,
- c) Highly tolerant 0.1 - 1.20 : Ridge gourd, Bitter gourd However, no one method satisfies all groups of workers.

Lect.3 Types of vegetable gardening with special reference to kitchen gardening

Types of vegetable gardens: Vegetable gardens can be classified into 7 different types according to the **purpose for which** they have been developed.

These are home-gardens or kitchen-gardens, market-gardens, truck-gardens, gardens for processing, gardens for vegetable-forcing, gardens for seed production and Floating vegetable garden.

1. **Home or Kitchen garden:** is a vegetable garden where vegetable crops are grown in the backyard of a house or any available space in the home compound to meet the daily requirement of the family. The layout of a home garden will differ from individual to individual. However, broadly, a city home gardener will follow a very intensive method of vegetable-growing compared with that followed by a home-gardener in a village.

Advantages of kitchen garden:

1. It is best means of recreation and exercise
2. An excellent hobby and healthy occupation for young and old during their leisure time
3. Cut down the expenditure on purchase of vegetables
4. An ideal medium for training children in duty and order
5. Vegetables grown in kitchen garden are fresh and are free from market infection

2. **Market garden:** A market garden produces vegetables for the local market. Most of such types of gardens are located within 15 to 20 km from a city. The cropping pattern depends on the demands of the local market. The land being costly, intensive methods of cultivation are followed. A market gardener will like to grow early varieties to catch the early market. The high cost of land and labour is compensated by the availability of city compost, sludge, and water near cities and high return on the produce.

3. **Truck garden:** A truck-garden produces selected crops in a relatively large quantity for distant markets. It generally follows a more extensive method of cultivation than the market-garden. The commodities raised are usually sold through middlemen.

The location is determined by soil and climatic factors suitable for raising particular crops. Truck gardener should be a specialized person and expert in the large scale production and handling of some special crops.

The cost of labour and land may be cheap and he follows mechanized method of cultivation, hence his cost of cultivation is less. The net income is also less as this includes cost of transport and charges of middlemen.

4. Vegetable garden for processing: A vegetable garden for processing develops around the processing factories and is mainly responsible for supplying vegetables to the factories regularly.

This type of garden grows particular varieties suitable for canning, dehydration or freezing. These gardens specialized in growing only a few vegetables in bulk.

They choose heavier soil since their main consideration is a high and continuous yield rather than an early yield. The prices are paid on fixed contract basis on weight and quality of the product. The return may be low but the cost of marketing and the transport charges are negligible.

5. Vegetable-forcing: Vegetable-forcing is concerned with the production of vegetables out of the normal season. The commonest forcing structures are glass and plastic houses. The vegetables commonly grown under these are tomato and cucumber. These are mostly used during winter in the temperate region.

Crops like cucumber and tomato can not be grown outside during colder months and as they are required throughout the year, they are grown under protection. Special varieties have been developed which do better under these structures.

The growing of summer vegetables on river beds during winter months with the help of organic manure and wind breaks of dry grass is also a type of vegetable forcing. Sometimes for an early produce seedlings of crops like tomato or brinjal are forced to germinate in small protected structures. This may also be regarded as a type of vegetable forcing.

6. Vegetable garden for seed production: Vegetable seed production is rather a specialized type of vegetable-growing. A thorough knowledge of a vegetable crop in respect of its growth habits, mode of pollination, proper isolation distance, etc. are of prime importance in the production of quality seed.

The handling of the seed-crop, its curing, threshing, cleaning, grading, packing and storage need specialized knowledge. A vegetable garden for seed production is, therefore, considered a special type of garden. The third and fourth stages of seed multiplication i.e. registered and certified seeds are usually multiplied by growers. This is an expanding industry in India and has good future.

7. Floating vegetable garden: Floating vegetable garden is seen on the Dal Lake of the Kashmir valley. Most of the summer vegetables are supplied to Srinagar from these gardens.

A floating base is first made from the root of Typha grass which grows wild in some parts of the lake. Once this floating base is ready, seedlings are transplanted on leaf compost made of the vegetation growing wild in the lake. All the intercultural operations and occasional sprinkling of water are done from boats.

Cultivation of Tomato

Botanical name: *Lycopersicon esculentum*

Family: Solanaceae

Chromosome no $2n = 24$

Origin: Peru, Ecuador – Bolivia

The word tomato is said to have derived from **Zitomate** or **Xitotamate** used to denote “plants for food”.

Origin and distribution:

Cultivated tomato originated from Peru, Ecuador, Bolivia. Domesticated place of tomato lies in Mexico. The ancestor of cultivated tomato is cherry type (*Lycopersicon esculenta var cerasiformae* - cherry tomato). From Mexico is distributed to Spain, Europe, Britain, France .

Species:

Tomato belongs to the family **solanaceae**. The genus **Lycopersicon** differs from Solanum by the absence of spines and anther tips are narrow and exhibit longitudinal divisions. The genus *Lycopersicon* was divided into two sub genera.

1. **Eu lycopersicon.**
2. **Erio persicon.**

Eu lycopersicon is characterized by red fruited edible species with carotinoid pigmentation and they are annuals. Cultivated tomato is included in this. The species included are

1. Cultivated tomato - *Lycopersicon esculentum* and.
2. Small fruited tomato- *Lycopersicon pimpinellifolium*

Eriopersicon is characterized by green fruited species, anthocyanin pigmentation. The species included are peruvianum, hirsutum, persicae and glandulosum, cheesmanii.

Growth habit:

Tomato can also be classified depending on the growth habit.

1. **Indeterminate:** terminal buds ends with a leafy bud and continue it vegetative growth. **Ex:** Pusa ruby.
2. **Determinate fruits:** terminal buds ends with floral bud and further its vegetative growth is checked and are called as Self topping or self pruning types. **Ex.** Pusa Early Dwarf
3. **Semi-determinate:** have semi dwarf growth and it is between determinate and indeterminate. Number of nodes between two consecutive inflorescences will be around one. **Ex:** S-12, Roma.

Importance and Nutrition value:

The tomato is one of the most important "**protective foods**" both because of its special nutritive value and also because of its widespread production. It is the world's largest vegetable crop after potato and sweet potato, but it tops the list of canned vegetables.

Tomatos are used for soup, salad, pickles, ketchup, puree, sauces and in many other ways.

Tomato is a major source of vitamins and minerals. It is widely used a salad vegetable. In England, it referred as “**love of apple**’ or ‘**love apple**’.

In India it is commonly referred as ‘**poor mans orange**’ (Ascorbic acid 15 mg to 20 mg/100g edible portion).

It is also rich in Citric acid and Mallic acid. Glutamic acid is an amino acid mostly present in tomato. Tomato contains many important minerals like Na, K, Ca, Mg, P, K, Fe, Zn, Boron.

The alkaloid present in tomato is called **tamatin** and the coloured pigment is called **Lycopene**. Lycopene content is high at 700 F or 210 C.

Varieties:

In certain varieties of tomato, the vegetative growth automatically stops, giving rise to bushy growth. They are called self pruning varieties.

Certain varieties like Pusa **rubi**, **Pusa early dwarf**, **Marutham**, **Arka vital**, **Pusa 120**, **HS 10**, **sweet 72**, **S-12**, **Co I** are suitable for cultivation in **plains**.

Varieties like **Sioux**, **Best of all**, **Pusa early dwarf** are suitable for **hilly areas**.

Varieties like **Roma** and **Punjab chuharra** are suitable for **processing**.

Varieties released by IIHR **Arka Abha**, **Arka Abhizit**, **Arka Ahuthi**, **Arka Aloukik**, **Arka Meghali**, **Arka Sourab**, **Arka Srasika**, **Arka vartnan**, **Arka vikas**, **Arka visal**.

Pusa Sheetal : cold resistant variety

Best of all : Mid season variety

IVRI -2 : variety developed from IVRI, Varanasi

Floradade : it is a variety brought from Florida

SL – 120 : resistant to nematode and released by IARI, New Delhi.

S-12 : Evolved by PAU, Luthiana, fruit round to flatish with persistent pedicel suitable for summer crop all over the India.

Sioux : American variety, resistant to growth cracks

Hybrids:

S.No	Hybrid	Parentage	Characters
1	Pusa rubi	Sioux X Improved meeruti	Indeterminate
2	Pusa Early Dwarf	Improved meeruti X Red cloud	Determinate and slightly furrowed
3	Pusa Red Plum	L. esculentum X L. Pimpinellifolium	

Co 1, Co -2 – released from TNAU, Coimbatore. Co3 it is a mutant of Co1 Gulmohar (MTH 6) – released from Maharashtra hybrid Seed Company Punjab chuharra, Ox heart, Punjab kesari, Pusa early dwarf, Pusa rubi, Pusa red plum, Pusa sada bahar, Sweet – 72, Roma, Yasvanth 2 are other hybrids..

Climate:

Tomato is a warm season vegetable require a long season optimum temperature is 21 to 24°C. Temperature and light intensity effect the fruit set, pigmentation and nutritive value.

Optimum temperature for seed germination is 30° C. Maximum fruit set occurs at a night temperature of 15 to 20° C. High temperature (38° C) accompanied by low humidity and dry winds adversely affect the fruit set.

Tomato has a yellow pigment *Carotene* and red pigment (at ripened stage) called *Lycopene* and at very high temperature formation of Lycopene is inhibited.

Soil:

Tomato can be grown in a wide range of soils from sandy to heavy soils.

A well drained, fairly fertile loam with fair moisture holding capacity is ideal for growing a good crop of tomato. Tomato is highly susceptible to **water logging**.

Well drained soils are highly necessary. The preferable p^H range is 6 to 7. If the soil is acidic i.e pH 5.0 or lower liming is advocated.

Seed sowing:

It is grown almost the year round. In north India seed sowing is done in November and transplanting during the month of January. In case of South India sowing is done in the month of August. In case of Summer sowing is done in the month of March; it is transplanted in the month of April- May.

Seed rate:

Varieties : 300-350 g / ha

Hybrids : 100-150 g / ha.

Nursery practices:

Tomato is a transplanted crop. Seeds are sown in the area of 225 m² . A raised bed prepared by well decomposed FYM is mixed @ 3 kg FYM per m² of nursery bed. A fertilizer dose of 0.5 kg N, P, K per bed is also mixed in the soil.

Seeds are treated with fungicides and 40% formalin solution at 500 ml / m² area of nursery bed sterilisation. 10% formaldehyde is also used for fumigation. After fumigation the beds are covered with polythene for 24 hours. Seeds are sown 4 to 5 days after removal of polythene sheets.

In line sowing 7.5 cm distance is kept between the rows. The beds are covered with straw or polythene till the seeds germinate. Seedlings are protected against wind, exposure to sun and excess rainfall. Fungicides are sprayed weekly to avoid of damping off. Nursery can also be grown in poly house.

The seedlings should be hardened before setting them into field.

Transplanting:

Seedlings are transplanted at 25 to 30 days and 10-15cm height, on the evening of sunny day. Whole day transplanting is done in a cloudy day. In some of the areas tomato is directly sown.

Direct sowing is reduce the infestation of root knot the nematode, bacterial wilt and damping off. The seedlings are transplanted at the side of ridge later earthing up is done to keep the plant in the middle of the ridge.

For indeterminate varieties and hybrids, row to row spacing of 60 to 120 cm and plant to plant distance from 45 to 75 cm is adopted. In case of determinate types spacing is 45 to 60 cm x 30 to 40 cm is adopted.

Nutritional management:

Nitrogen is the most important nutrient. Deficiency of **nitrogen** reduces endogenous auxins and blossom end rot disease increase with increased levels of nitrogen. Phosphorus is essential for rapid root development.

Deficiency of **phosphorus** leads to the development of purple colour on the underside of leaflets. Potassium is involved in the synthesis of proteins and organic acids. Deficiency leads to yellowing at the margins of leaf. A high level of potassium improves the shape of fruit.

In general NPK @ 75 to 150 kg: 60 kg: 60 kg per ha has been recommended for various tomato varieties. The quantity of Nitrogen to be applied varies greatly depending on the variety as well as soil conditions.

Nitrogen is the best applied in the form of NH_4SO_4 , Urea, CAN and Ammonium chloride. Besides the above inorganic fertilizers combination of well decomposed FYM and groundnut cake is recommended.

FYM is to be added @ 20-25 tons per ha at the time of last ploughing and incorporated into soil. Regarding the inorganic chemical fertilizers half N, entire P and K should be applied as basal dose, half N is given in 2 to 3 splits. 30, 45, 60 days after sowing.

Micro nutrients like B, Zn also need to be applied and lime is essential under acidic soil. Availability of Boron is considered to be essential for production of large size fruit with high vitamin content and prevent fruit cracking, while Zinc for higher ascorbic acid content and tolerances to diseases.

Irrigation:

Tomato is a deep rooted crop. Roots will grow to a depth of 120 to 150 cm and it has some drought tolerance. They require adequate moisture for their fair growth. Excess as well as insufficient moisture is harmful.

First irrigation is given immediately after transplanting afterwards care should be taken not to apply too much water as it makes the plant to run and drops the blossom.

However light irrigation should be given at 3 to 4 days interval in summer 10 to 15 day interval in winter. Furrow irrigation is the most widely used. Drip irrigation is fairly recommended as it can save more water compared to furrow irrigations.

Intercultural operations:

Tomato is subjected to pinching, the lateral shoots are pinched to improve more bushy growth but little foliage is to be kept. Frequently shallow hoeings are necessary to improve the yield it also reduce the weed growth.

Mulching should be done 15 to 20 days after planting. 2 to 3 weeding before flowering encourages good crop growth. Application of a weedicide, basalin or pendimethalin @ 1 kg a.i. / ha plus one hand weeding at 45 days after transplanting was recommended. Staking is very essential for indeterminate group of varieties because it improves yield and quality protection of fruits. In pest and diseases, easy harvesting and easy spraying of chemicals.

Harvesting:

Stage maturity at which tomato should be harvested depends upon the purpose for which they are used and the distance of transportation.

1. **Immature:** means before the seeds have fully developed and before the jelly like substance around the seeds are fully formed
2. **Mature green:** fully grown fruit shows a brown ring at stem scar. It has light green colour at blossom end and seeds are surrounded by jelly like substance.
3. **Turning or breaker stage:** 1/4th of the surface at blossom end shows pink
4. **Pink stage:** 3/4th of the surface shows pink.
5. **Hard ripe:** all the surface turn to pink or red but flesh is firm.
6. **Over ripe:** fully coloured and flesh is also soft.

For distant market mature green stage fruit can be harvested and for a local market, they can be harvested at hard ripe stage. Fruits at fully ripe or over ripe stages are utilized within 24 hours for processing.

Grading:

Fruits are graded based on size as Super A, Super, Fancy and Commercial according to IIHR.

Yield:

Depends on various factors on an average an open pollinated variety will give 300 to 500 Q per ha. Hybrids can give up to 1000 Q per ha.

Storage: tomato can be stored either in mature green or breaker stage of maturity. Fruit remain firm up to 21 days when kept at 200 C for the cultivars like Florida MH and Floradade. Temperature of 100 C cause moderate chilling injury and Alternaria root rot.

Physiological disorders in Tomato:

1. **Blossom end rot:** it is more serious, ground discoloration starts. In blossom end of the fruit. Black spot develops to encompass 1/2 to 2/3 rd portion of the fruit. Later the tissues shrink and skin becomes dark grey to black. It may lead to secondary infection by fungus and unfit for consumption.

Causes: use of Ammonium sulphate, imbalance of Mg & K; deficiency of calcium

Remedies: cultural practices that concern soil moisture and maintain uniform moisture supply. Transplanting in early April instead of early June. Foliar spray of 0.5% CaCl₂. Apply Nitrogen in the form of Urea.

2. **Fruit cracking:** occurs for middle of the May. Reduced transpiration has increased cell turgidity and contributed to tomato fruit cracking. Reduced transpiration occurs even in summer when fruit are grown in green house. Cracking also occurs in rainy season when rains fall in long dry spell.

Presence of water on the surface of fruit is more conducive in cracking than high soil moisture. **Boron** deficiency in the soil also causes fruit cracking.

They are two types of cracking.

i. *radial cracking*: occurs mostly at ripe stage.

ii. *Concentric cracking*: it is common in mature green stage.

Remedies: use of resistant cultivars like Sioux, Punjab chuhara. Picking of the fruit before the full ripe stage. Soil application of Borax @ 10 – 15 kg per ha. Regulation of soil moisture. Misting (spray of cool water).

2. **Puffiness:** commonly known as hallowness. Tomato puffs, puffy tomatoes, puffs and pockets. As the fruit reaches about 2/3rd normal size outer wall continues to develop normally but remaining internal tissue growth is retarded. Fruit become lighter in weight and partially filled. Very high or vary low temperature and low soil moisture conditions will lead to puffiness.

3. **Cat facing:** a large scar is formed at the blossom end portion of the fruit. Such fruits have ridges and furrows and blotches at blossom end.

Reasons: low temperature, faulty pollination, application of nitrogen during transition from vegetative to reproductive phase.

Remedies: balanced fertilizer application; regulation of temperature.

4. **Sunscald:** when fruits and leaves are exposed to the sun, there is appearance of yellow, white patches on green and ripen fruits. These patches may have secondary infection of fungus and start rotting varieties with sparse foliage will suffer more sunscald especially in the month of May and June.

Remedies: prefer the varieties having more foliage and follow appropriate cultural practices.

5. **Flower and fruit drop:** higher incidence of flower and fruit drop is resulted because of fluctuations in temperature poor water management and soil moisture.

Remedies: good package of practices should be followed. Moisture stress should be avoided. Spraying of planofix or NAA @ 1 ml in 4.5 lt of water. Control of fluctuations in temperature.

6. **Blotchy ripening:** greenish yellow; maintain balance between Nitrogen and Potassic fertilizers.

7. **Bronzing or internal browning:** also known as grey wall. Characterized by death of tissues with in the fruits associated by vascular browning variety EL 235673 is found resistant to this disorder.

8. **Green back:** stem and portion of the fruit turns green. At high temperature ripening is inhibited and green band is expected. Reduce temperature by artificial means.

Lect.5 Brinjal, Chilli, Capsicum,

Botanical name: *Solanum melongena*
Family: Solanaceae
Chromosome no $2n = 24$
Origin: Indo-Burma region.
Common name: egg plant

It is a non-tuberiferous species of solanum. In India it might have spread to African and European countries. It is main vegetable in plains areas of India and almost available throughout the year. 8% of total area under vegetables in the country is occupied by brinjal. *Solanum auriculatum* is immune to little leaf.

Importance and Nutritive value :

Brinjal is a stable vegetable high in nutritive value. It is rich in minerals is Ca, Mg, P, K and Fe. It is also a good source of Vitamin A and C.

Bitterness in Brinjal is due to presence of glycoalkaloids. Glycoalkaloids content vary from 0.4 to 0.5 mg per 100 g of fresh weight.

Purple variety has higher copper content and polyphenol oxidase activity where as iron and catalase activity is the highest in the green cultivars. Amino acid content is higher in purple variety.

Uses:

Used as cooked vegetables. Used in pickle making, dehydration in dustury. Frut is employed as a cure for toothache. Excellent remedy those suffering from liver complaints.

Green leaves of brinjal plant are good appetizers, aphrodisiac and cardiotonic. It also beneficial in **vaata and kapha**. In unani system roots are used to alleviate pain.

Growth habit:

1. Based on growth habit brinjal can be classified as below:

- i) *Solanum melongena var esculentum*; fruit is long, round and oval.
- ii) *Solanum melongena var serpentine* known as snake brinjal. Fruit is extra ordinarily long. Leaves are prickly.
- iii) *Solanum melongena var depressum*: plant is extensively short and dwarf.

2. Classification on the basis of fruit shape:

- i) Pusa purple long (PPL): long brinjal
- ii) round brinjal – pusa purple round
- iii) oval brinjal – pusa kranti.

3. Classification on the basis of fruit colour:

Purple brinjal: they have no anthocyanins. Eg: PPL.

Green brinjal: more of chlorophyll. Eg: Arka kusumakar.

Botany:

Leaves are alternate, solitary, flowers are stalked, leaf oppressed. Calyx is five lobed, corolla is gamopetalous, fruit is berry and berries are variable in shape and colour.

Flowers in brinjal are of different styles.

a. long style **b.** medium style **c.** short style **d.** pseudo short style

Brinjal is self pollinated.

Only **long style and medium style flowers** will set fruits. Entire plant surface is covered by 'hairy layer called 'tomentum'

Long styled flowers: they have a big ovary, stigma is swollen, long anthers.

Medium styled flowers: it has medium, long style, anthers are of same length, ovary is also medium sized.

Short styled flowers: they have rudimentary ovary.

Pseudo short styled flowers: ovary is rudimentary. Style should be shorter than the anther. No swelling of anther is observed.

Varieties:

Brinjal varieties are grouped on the basis of colour and shape of fruit.

1. **Long fruit varieties:** ex. **Pusa purple long:** evolved as a selection from mixed batia.
2. **Pusa purple cluster long:** is an early maturing variety.
3. **Long green varieties:** arka kusumakar, arka shirish, Krishna nagar green long.
4. **Round purple:** ex: **Pusa purple round:** is resistant to fruit borer and little leaf of brinjal. Selection-6, suphala, arka navaneet, krihsnanagar purple round, pant ritu raj, vijaya hybrid, shyamala.
5. **Round green:** banarasi gaint, round striped.
6. **Roundish white:** some varieties under this group have purplish tinge with white stripes. Ex: Manjeri, Visali.
7. **Oval or Oblong fruited varieties:** Junagad oblong, Bhagyamati, H4., Pusa anmol (**Pusa anmol** is a hybrid variety between **pusa purple long and hyderpur**).
8. **Cluster fruited varieties:** fruits born in cluster. Ex: pusa purple cluster, arka kusumakar, **Bhagyamathi (APAU variety)**.
9. Spiny varieties: H-4, Manjeri

Hybrids:

S.No	Hybrid	Parentage
1	Pusa Anmol	Pusa purple long X Hyderpur
2	Arka Navaneet	IIHR22-1 X Supreme

Climate:

Brinjal is the warm season crop. It is susceptible to severe frost. It requires a long warm season, before fruit maturity. Optimum temperature is 22 to 33 °C. Late round varieties are more tolerant to frost than early long varieties. Under very cool seasonal conditions, the ovaries are split leading to the development of abnormal fruits.

Soil:

Brinjal can be grown on a wide range of soils. The ideal soils should be a deep, fertile and well drained. The pH should be not more than 5.5 to 6 for better growth and development. Light soils are good for a healthy crop but heavy soils are suited for higher yields.

Time of sowing:

In plains crop is grown in three seasons, summer crop is sown during February - March and rabi crop is October to November. In hills seed is sown in April-May and the seedlings transplanted in May-June

Seed rate

Varieties : 400 g / ha

Hybrids : 200 g / ha

Seed treatment

Treat the seeds with *Trichoderma viride* @ 4 g / kg or *Pseudomonas fluorescens* @ 10 g / kg of seed. Treat the seeds with *Azospirillum* @ 40 g / 400 g of seeds using rice gruel as adhesive. Irrigate with rose can. In raised nursery beds, sow the seeds in lines at 10 cm apart and cover with sand. Transplant the seedlings 30 – 35 days after sowing at 60 cm apart in the ridges.

Nursery practices: nursery practices are almost similar to **Tomato**.***

Transplanting:

Seedlings are of 8 to 10 cm height with 2 to 3 true leaves are ready for transplanting. Seedlings should be hardened before lifting for transplanting.

The summer crop may be transplanted on ridges and furrow system for effective use of water. Hardening of seedlings is achieved by withholding water for 4 to 6 days before transplanting.

Light irrigation should be given on due day of nursery pulling. Seedlings are pulled without any injury to the root.

At the time of transplanting soil around the seedlings is pressed firmly. Distance of transplanting depends on soil fertility, climatic conditions and varieties. Long fruited varieties are transplanted at a spacing of 60 x 60 cm. Round fruited varieties at 75 x 75 cm.

Manuring:

Brinjal occupies the land nearly 6 to 8 months, about 25 to 30 tons of well decomposed FYM is incorporated in the soil before transplanting. NPK @ 100, 80, 60 kg per ha is generally applied, ½ of nitrogen full quantities of P and K is applied at the time of transplanting while the remaining quantity of nitrogen may be applied either twice or thrice depending upon soil conditions at 30 days, 45 days after transplanting.

Irrigation:

Brinjal is a shallow rooted crop it needs frequent irrigation. The crop is irrigated at 3 to 4 day interval during summer season 12 to 15 days during winter season. However during rainy spells irrigation is not needed. Brinjal is generally irrigated by furrow system of irrigation. Drip irrigation is recommended to improve water use efficiency and also to reduce weed growth.

Inter culture:

Generally manual weeding is done to remove weeds. Shallow cultivation is followed to put down the weed growth. Weeds can be controlled by applying herbicides like fluchloralin @ 1.5 kg a.i. / ha.

Harvesting:

Fruits are harvested when they attain good size and when the surface is bright and glossy appearance. If the fruit is too immature we press the fruit the pressed portion springs back.

Yield:

In case of open pollinated variety 200 to 500 Q per ha. Hybrids 300 to 700 Q per ha.

Storage:

under ordinary conditions the fruits can be stored for 1 to 2 days in summer, 3 to 4 days in winter at temperature of 7.2° C at 85 – 90% RH. The brinjal fruits can be stored for around 10 days.

Grading: three categories 1. super 2. fancy 3. commercial.

Packing: fruits are packed loosely in gunny bags, net bags, wooden trays and bamboo baskets.

Diseases of Brinjal:**1) Damping Off:**

It is a serious disease of brinjal seedlings and mainly occurs in nursery bed. The disease infected seedlings rot at ground level and then the plants fall over ground. The seedlings die in patches.

Control Measures:

1. The seed bed should be treated with Formalin before sowing of seeds.
2. The seeds should be treated without water (30 minutes at 52° C) or Cerasan or Agrosan G.N. before sowing of seed.
3. The seedlings in the nursery should be sprayed with any fungicides at an regular interval.

2) Phomopsis Blight and Fruit Rot:

It is a serious disease of brinjal. The fungus attacks all parts of the plants above ground. Dark brown lesions appear on the stem and round to oval spots are formed on the leaves. Disease fruits show short and watery lesions which later on become black and mummified.

Control Measures:

- i. Use of disease free seeds, seeds treatment with some fungicide and long crop rotation are the most common remedial measures of this disease.
- ii. Disease resistant variety (Such as Pusa Bhairab, Pusa Cluster etc.) should be Cultivated,
- iii. The disease can effectively be controlled by weekly spraying of nursery and field with Zineb (Dithane-Z-78) or Mancozeb (Dithane M-45) @ 2.5 gm per liter of water.

3) Little Leaf of Brinjal:

CHILLI

Botanical name: *Capsicum fruitiscens* – Bell pepper
Capsicum annum - Chilli
Family : Solanaceae
Chromosome no $2n = 24$
Origin: Bell pepper from South America
Chilli from Peru

Chilli is also called as pepper.

Importance and nutritive value:

Green chillies are rich in proteins 2.9 g per 100 g. Ca, Mg, P, K, Cu and S. vitamins like Thiamine, Riboflavin and Vitamin C. Chillies are the major ingredients in curry powder.

In powdered form it is mixed in red or cayenne pepper. Chilli pulp is pickled in strong vinegar or brine. Extracts of chillies are used in the production of Ginger beer and other beverages. Cayenne pepper is incorporated in poultry feeds, green chillies are rich in Rutin which has pharmaceutical use. Pungency of chillies is due to **capsaicin**. The pigment (colour) in chillies is due to **capsanthin** also contains many other oleoresins.

Botany and floral biology:

Genus capsicum 20 wild species have been reported at only few are cultivated.

1. **Capsicum annum (sweet pepper and chilli)**: it has blue anthers, milky white corolla.
2. **C. baccatum**: it has yellow or brown spots on corolla. Its cultivation is restricted to South America.
3. **C. fruitescens**: tobacco pepper. It has blue anthers. Milky yellowish white corolla.

Chilli Varieties and hybrids:

Andhra Jyothi or G5 (G2 x Bihar variety): released from Lam Guntur. Fruits are short and called as Gundu types.

Bhagya laxmi (G4): selection from thohian chillies grown largely for green chillies.

Sindhuri: Tall growing and less pungent variety suit for green chillies

Baskar/ CA -235: released from Lam, Guntur. It is a cross between G4 x yellow anther mutant.

Prakash (LCA 206): developed from RARS, Lam Guntur.

Hissar sakthi: multiple resistant variety developed at hissar.

N.P46 A: Medium, early prolific and pungent variety of IARI (N.P means New Pusa)

Arka lohit: highly pungent variety released from IIHR, Bangalore.

Hybrids:

S.No	Hybrid	Parentage
1	Pusa Jwala	Pusa Red X N.P 46 A
2	Pusa sadabahar	Pusa jwala X IC 31339
3	Punjab lai	Perennial X long red
4	Kiran (x235)	G4 X anther mutant

Climate:

Chilli is grown in both tropical and sub-tropical areas. It can grow up to 1000 MSL altitude. For vegetative growth it requires warm humid climate. For fruit maturity it requires warm dry weather.

It requires a well distributed annual rainfall of about 800 – 1200 mm. Heavy rainfall leads to poor fruit set and high humidity leads to fruit rot. The crops continue to develop at high temperature but root development is retarded at a temperature of 300 C.

Fruit development is adversely retarded at 380 C. Average night temperature favours high capsaicin content. Day length of 9 to 10 hours light stimulate plant growth. In general capsicum is grown at low temperature conditions than chillies.

Soil:

Chilli can be grown on a wide variety of soils provided. They are well drained, well aerated and rich in organic manure. In ill drained soils plants shed their leaves and turn sick. Cannot tolerate water logging conditions.

Sandy loam soil with adequate irrigation and manuring can support better crop of chilli.

Black soils also preferable to grow chillies as rainfed crop. Strongly acid soils and alkaline soils are not suitable. Chilli can be grown in saline soils. Seed germination and plant vigour affected by salinity. Ideal pH 6 to 7.

Time of sowing:

Chilli seeds are sown in nursery beds during May-july. Sowing is little early in the north east India. In south states where rainfed cultivation is in vogue chillies can be in may-june and September to October. In hills it is sown during march to april.

Seed rate:

Varieties : 1.0 kg / ha.

Hybrids : 200 - 250 g / ha.

Nursery area : 100 sq.m / ha.***

Seed treatment

Treat the seeds with *Trichoderma viride* @ 4 g / kg or *Pseudomonas fluorescens* @ 10 g/ kg and sow in lines spaced at 10 cm in raised nursery beds and cover with sand. Watering with rose can has to be done daily. Drench the nursery with Copper oxychloride @ 2.5 g/l of water at 15 days interval against damping off disease. Apply Carbofuran 3 G at 10 g/sq.m. at sowing.\

Transplanting:

Seedlings ready for transplanting 35 to 45 days. Short thick stem seedlings are preferred for better establishment. In older seedling topping has to be done one week prior to transplant. Early seedlings are transplanted singly different spacing 30 x 30 cm 45 x 30 cm, 45 x 45 and even 30 x 20 were tried in chillies. Spacing varies with variety, rainfall, seasonal conditions etc. Immediately after transplanting field should be irrigated.

Manuring:

It needs good fertile soils supplied humus. Excess nitrogen lead to increase the vegetative growth and delays maturity. 10 to 15 tones of well decomposed FYM need to be applied in the last ploughing.

Besides that 75 kg N, 75 kg P, 55 kg K per ha is to be applied. Entire quantity of FYM, Phosphorus, potassium and half of nitrogen is to be applied at the time of field preparation. Remaining half nitrogen is to be given as top dressing in two equal splits at one month interval of transplanting.

Irrigation:

first irrigation is given just after transplanting for better establishment in the soil. Second irrigation is given 10 days after transplanting. During this time gap filling can be taken up. After wards irrigation is given as per the requirement. Generally 8 to 9 irrigations have given depending on rainfall, soil type, humidity and temperature. Method of irrigation adopted is ridges and furrows.

Interculture:

Chilli is a slow growing crop cannot compete with aggressive weeds hand weeding or hoeing or application of herbicides need to be done in order to ensure weed free conditions. Frequent shallow conditions are under taken to facilitate soil aeration and proper root development. However deep cultivation should be avoided because, it damages roots. Herbicides like Alachlor 2.5 kg per ha can be used on chillies.

Layout & Spacing:

Ridges and furrow type of layout is used. Seedlings are raised on raised bed. Spacing for rained crops is 60 x 45 cm & for irrigated crops is 60 x 60 cm.

Weed control

Spray Fluchloralin @ 1 lit a.i./ha or Pendimethalin @ 1 kg a.i/ ha. or Oxyflourfen @ 0.15 kg a.i./ha as pre-emergence herbicide and may be combined with hand weeding once and earthing up 45 days after planting. Raise intercrop of onion in paired row system to get additional income.

Diseases

Damping off: Treat the seeds with *Trichoderma viride* 4 g/kg or *Pseudomonas fluorescens* 10 g /kg of seed 24 hours before sowing. Apply *Pseudomonas fluorescens* as soil application @ 2.5 kg/ha mixed with 50 kg of FYM. Stagnation of water should be avoided. Drench with Copper oxychloride at 2.5 g /lit at 4 lit /sq.m.

Leaf spot: Spray Mancozeb @ 2 g/lit or Copper oxychloride @ 2.5 g/lit.

Powdery mildew: Spray Wettable sulphur @ 3 g/lit or Carbendazim @ 1 g/lit, 3 sprays at 15 days interval from the first appearance of symptom.

Die-back and fruit rot: Spray Mancozeb 2 g/lit or Copper oxychloride @ 2.5 g/lit thrice at 15 days interval starting from noticing the die-back symptoms.

Chilli mosaic: Raise 2 rows of maize or sorghum for every 5 rows of chilli crop against wind direction. Spray recommendations given for controlling the vector.

Yield

Varieties : 2 - 3 t/ha of dry pods or 10 - 15 t/ha of green chillies.

Hybrids : 25 t / ha of green chillies

CAPSICUM,

Capsicum (Sweet pepper / Bell pepper): *Capsicum annuum* ; Solanaceae

Varieties : Arka Basant, Arka Gaurav, Arka Mohini, Green Gold, Bharath.

Soil : Well drained loamy soil rich in organic matter.

Season of sowing : September - February

Seed rate

Varieties : 1.25 kg/ha

Hybrids : 200 g / ha

Nursery : 3 cents /ha'

Treat the seeds with of Carbendazim @ 2 g /kg of seed and sow in lines across the bed at a spacing of 2.5 cm and then cover with top soil and then paddy straw. Watering with rose can has to be done daily. On 20th day of sowing, 300g of carbofuran 3G granules have to be applied in between the seedling lines across the bed, the soil has to be stirred and then the beds are irrigated.

Preparation of field

Plough the field to a fine tilth. Form ridges and furrows 45 or 60 cm apart. Transplant 40-45 days old seedlings at 30 cm spacing.

Irrigation

Irrigation at weekly or 10 days interval.

Weed control

On 30th day, hoeing and weeding has to be done once and the plants are earthed up.

Growth regulator

Spray 1.25 ppm Triaccontanol (12.5 mg /10 l of water) on 20th, 40th, 60th and 80th day after transplanting. Spray NAA 10 ppm (10 mg/l of water) on 60 and 90 days after planting.

Diseases

Powdery mildew: Spray wettable sulphur @ 2g /l

Dieback and fruit rot: Spray Mancozeb @ 2g /l

Harvest and yield

Harvest fully matured green fruits before ripening. Yield: 15 tonnes/ha in 150-160 days.

Lect.6. Cucumber, Melons, Gourds, Pumpkin,

Cucurbits-introduction-flowering-sex expression-and modification

Cucurbits are the largest group of summer vegetables crops belonging to the family cucurbitaceae. Most of the cucurbits are monocious in nature except coccinia and pointed gourd (Parval) which are dioecious. A number of hermaphrodite and andromonocious cultivars are also available in some crops (melons). Fruit is **Pepo** botanically. All are susceptible to frost. The cultural requirements of all crops in this group are more or less similar.

Flowering:

Most of the cucurbits are monocious in nature except coccinia and pointed gourd (Parval) which are dioecious. A number of hermaphrodite and andromonocious cultivars are also available in some crops (melons).

Sex expression:

The cucurbitaceous plants are generally monocious i.e they bear male and female flowers separately on the same plant. Male flowers open first and later the female flowers. Generally there are more male flowers.

An female to male flower ratios goes on increasing with the age of the plant. The lower the nodes number on which the female flowers appear on main vine the earlier is the variety.

Though the sex expression and sex ratio are the varietal characters, they are modified by environment. Lower fertility, high temperature, longer light period all induces maleness. Both plant regulators (auxins and antiauxins) and nutrient at proper concentrations modify sex. GA at higher concentrations 10ppm increases the number of female flowers.

Two sprays, one at 2 leaf stage again at 4 leaf stage with 50 to 100 ppm of MH or 250 ppm Etherel or 100 ppm NAA or 25 to 50 ppm of TIBA or 3ppm of Boron or Molybdenum or 20 ppm Calcium can suppress the number of male flowerds and increase the number of female flowers, fruit set and ultimate yield.

S.No	Crop	Botanical Name
1	Cucumber	<i>Cucumis sativus</i>
2	Pumpkin	<i>Cucurbita moschata</i>
3	Summer squash	<i>Cucurbita pepo</i>
4	Bottle gourd	<i>Lagenaria siceraria</i>
5	Bitter gourd	<i>Momordica charantia</i>
6	Ridge gourd	<i>Luffa acutangula</i>
7	Sponge gourd	<i>Luffa cylindrica</i>
8	Ash (wax) gourd	<i>Benincasa hispida</i>
9	Snake gourd	<i>Trichosanthes anguina</i>
10	Pointed gourd	<i>Trichosanthes dioca</i>
11	Round (squash melon)	<i>Citrullus lanatus</i>
12	Musk melon	<i>Cucumis melo</i>
13	Snap melon	<i>Cucumis melo var. momordica</i>
14	Long melon	<i>Cucumis melo var. utilissimus</i>
15	Coccinia	<i>Coccinia indica</i>
16	Chow - chow	<i>Sechium edule</i>

CUCUMBER

Botanical name: <i>Cucumis sativus</i> Family : Cucurbitaceae Chromosome no $2n = 24$ Origin: India
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Cucumber is a native of India *C. sativus* is originated in North India. *C. hardwicki* occurs as wild species.

Importance:

It is rich in vitamin B and C. Cucumber is known by presence of bitter principle called as cucurbitacins, which are chemically tetra cyclic triterpenes. When a bitter pollen fertilizer non-bitter ovules the resulting fruit is found to bitter. This phenomenon is called metaxenia.

Varieties and hybrids:

Japanese long green: released from IARI. It is an extra early maturing variety. It matures within 45 days after sowing.

KTCH – 8: poinsettia x LC – 3. it is an early maturing hybrid – released by IARI regional station, katra. It is resistant to angular leaf spot, downy mildew.

KTCH – 11: it is a gynodioecious line, developed at IARI, matures in about 55 days.

Pusa samyog: it is a hybrid between Japanese gynodioecious line x green long naples released by IARI.

Straight – 8: released by IARI regional station, katra. Fruit is light green in colour.

Poinsettia: originally developed in US multiplied by national research station, New Delhi. Resistant to angular leaf spot, downy mildew.

Gemini: IARI released variety, prolific bearer, fruits are crispy and excellent for salads

Climate:

It is a warm season crop. It cannot tolerate high cold and frost. The condition of high humidity and short day length will promote female flower production. Cucumber requires minimum temperature of 18°C for seed germination 20°C growth and development.

Soil:

For an early and good crop, it requires sandy loam soils, heavy soils will give high yield. The crop can be grown successfully in slightly acidic soils pH 5.5 to 6.7. The soil should be with proper drainage facility.

Time of sowing:

Region: north Indian plains: February to March

Hills: April – May;

South India: October to November.

Seeds and sowing:

Seed rate of 3 to 5 kg per ha. Sowing is done on raised beds or in furrows or in pits. Two seeds per hill are generally sown in both sides of bed. In case of pit systems pits of 45 x 45 x 60 cm are dug up filled with FYM, cattle manure and soil in equal proportions 3 to 4 seeds per pit are sown. Spacing adopted is 1.5 to 3 m from row to row 60 to 90 cm from plant to plant.

Manuring:

25 to 50 q per ha of FYM is to be mixed with soil or applied to the pit besides that about 100 kg N, 50 kg each of P and K per ha need to be applied. Half of Nitrogen, entire P and K is applied in the pits or at the time of sowing. Remaining quantity of nitrogen is applied at 30 days after sowing.

Irrigation:

Cucumber cannot withstand water logging over irrigation should be avoided in dry weather crop is irrigated in every 5th day. Flower initiation and full bloom stage are critical. For irrigation fruit enlargement requires sufficient water. Moisture stress, during flowering results in non-viable pollen grains leading to reduced yield. Foliage becomes chlorotic or yellow. Growth is retarded. If irrigation water over flows and inundate plants for any length of time. If irrigation water over flows and inundate plants for any length of time.

Inter culture:**Training and pruning:**

Vines are trained on bamboo with the rope or wire for high production and quality. Cucumber can be trained in **bower system**. The height of Bower should be about 1.5 m and spacing is kept as 2 m x 1 m between row to row and plant to plant respectively. Japanese long green and hybrids like Pusa samyog are suitable for bower system.

Pruning of secondary shoots up to 10 nodes increase the fruit production. Field should be kept weed free throughout cropping season. First weeding may be given 15 to 20 days after sowing. Two more weedings are given at 25 to 30 day interval.

Harvesting:

Crop becomes ready for harvesting in about 60 to 70 days after sowing. Fruit takes 7 to 10 days from setting to reach marketable size. Cucumber is harvested when it is till tender and green. Over mature fruit will fetches less price. Picking can be done at an interval of 2 days yield varies from 80 to 120 Q/ha.

WATER MELON

Botanical name: *Citrullus lanatus*– Water melon

Family : Cucurbitaceae

Chromosome no $2n = 22$

Origin: South Africa

Importance:

Water melon contains 95% of water and is a **richest iron** containing cucurbitaceous crop. Most of cultivars have deep pink or pale pink coloured flesh which largely out **lycopene and anthocyanin** pigments. It is relished by poor and rich, young and old as a table fruit.

Varieties:

Arka jyothi: F1 hybrid by crossing IIHR – 20 with crimson sweet. Fruit is dark green in colour with blue angular stripes.

Arka manik: F1 hybrid IIHR – 21 and crimson sweet. Fruits are green with dark green stripes.

Asahi yamato: Japanese introduction released by IARI, New Delhi. Rind colour is light green and non-striped variety. It gives 22 tonnes yield in 95 days.

Pusa bedana: released by IARI, New Delhi. It is a cross between **tetra – 2 (4x) x Pusa rasard (2x)**. fruits are dark green with faint stripes. It takes 120 days for first harvest.

Sugar baby: introduction from US. Fruits are bluish black in colour. Fruits will ripe in 85 days and gives 150 Q per ha.

Climate:

Water melon is a warm season crop. It cannot tolerate high cold and frost. It requires a minimum of 18°C temperature for seed germination and 24 to 27°C for growth and development. There should be high temperature at the time of ripening for producing good quality fruits.

Soil:

Watermelon require sandy to sandy loam soil for an early and good crop. Soil should be well drained. Alluvial river beds are good for production of water melon. Soil pH: 6.7 to 7.

Varieties like Jobner – 21, Jobner 18-1 are capable of growing on high pH soil.

Time of sowing:

In north Indian plains, it is sown in February to March. In south and central India, it is sown in December to January. In Rajasthan, rainy season of crop is sown in August to September.

Seeds and sowing:

Seed rate is 3 to 5 kg per ha. Sowing may be done on raised beds. Furrows or pits. Direct sowing is also practiced. 2 seeds per hill on both sides of raised beds and 3 to 4 seed per hill in case of river beds are sown. In case of pit method 60 cm³ pits are dug. 4 seeds per pit are sown. Recommended spacing is 2.4 to 4 m between row to row and 60 to 120 cm between plant to plant.

Manuring:

Recommendation of fertilizers should depend up on soil test. FYM 15 to 20 to/ha is applied at 15 to 20 days before sowing. It is mixed thoroughly with soil. Besides that 60:40:40 kg per ha. NPK in Punjab conditions; 100 : 80: 80 kg per ha NPK in Karnataka conditions are recommended. Half nitrogen, entire P and K are applied to the pits or furrows at the time of sowing. Rest of nitrogen is applied 30 DAS.

Irrigation:

Watermelon planted on river beds does not need any irrigation but under light soils conditions irrigation is required once in 7 to 10 days is required. Crop is irrigated in furrows or basin irrigation in India, pitcher irrigation is also used in water melon.

Interculture:

Apical shoots should be removed. Side shoots should be allowed in getting good yield and quality, only 3 to 5 shoots are allowed. Field should be kept weed free by performing 2 to 3 weeding or by applying herbicides like Trifluralin.

Harvesting:

Crop will be ready for harvesting about 100 days after sowing depending upon cultivar and season. Fruits will be ready 40 days after anthesis. Maturity is judged by

1. withering of tendril
2. change in belly colour or ground spot into yellow and
3. by thumping test, on thumping the fruit it gives a dull sound
4. on pressing the fruit ripe fruit will give a crisp cracking noise. Fruit is separated for vine with the help of knife.

Yield varies between 40 to 60 tonnes per ha. Water melon requires careful handling as they are easily damage. Water melon cannot be stored for more than 2 to 3 weeks.

- The seed rate is 3 to 4 kg/ha.
- The average yield is 200 to 300 quintal/ha.

MUSKMELON:

Botanical name: *Cucumis melo* – Musk melon

Family : Cucurbitaceae

Chromosome no $2n = 22$

Origin: China

Importance:

Fruits are sweet, musky in flavour, fruits are rich in protein, minerals like. Calcium, Iron, Phosphorus and Vitamin A. Immature fruits are cooked as vegetable. Ripe fruits are consumed whole some. They have cooling effect defend constipation. Quality of musk melon is determined by colour, thickness, texture and absence of fibrous material along with aroma. Softening of fruits is accompanied by the change of pectin from insoluble to soluble form. In temperate regions it is cultivated in the name of cantaloupe.

Varieties:

Arka jeet: it was released from IIHR, Bangalore. Selection from Bati strain of UP. Fruit skin is orange coloured yields about 160 Q per ha in 90 days.

Arka rajahans: selection from local collection of Rajasthan. It has fruit weighing up to 1 kg fine net. Fruits are white and sweet variety resistant powdery mildew. Yield is 280 Q per ha.

Har madhu: selection from UP. Released by PAU, Ludhiana. Leaves are sparsely lobed. Fruits pale yellow at maturity.

Pusa madhuras: it is released by IARI, Delhi.

Pusa ras raj: released by IARI, Delhi.

Pusa sharbati: released by IARI, Delhi.

Climate:

It requires hot and dry climate. Optimum temperature 27 to 30 °C. short days promote female flowers. High temperature, low humidity, plenty of sun shine are essential for proper ripening, high sugar content. Plants are sensitive to low temperature and frost.

Soil:

it requires sandy soils. Loamy soils are good. Preferable pH 6 to 6.8. soil should be rich in organic manure, proper drainage facility.

Sowing time:

In plains of North India best time is February to March. In south India best time is October to November.

Seed rate and method of sowing:

4 to 6 kg per ha; broad cast sowing. In case of transplanted crop 1 kg per ha is sufficient.

Seeds can be directly sown in fields or first raised in polythene bags @ two seeds per bag and water daily with enough care.

Seedlings will be ready for transplanting in about 30 days. In case of river beds trenches are dug at 1 m deep and specially fertilized with FYM and fertilizers other wise pits of 60 cm³ are dug and filled with mixture of FYM and then sowing can be done. Recommended spacing 2.5 m x 1 m

Manuring:

Manures and fertilizers should be given to musk melon as per the availability in the soil FYM @ 25 to 45 t per ha is applied. It is mixed with the soil at the time of last ploughing or applied 10 to 15 days before sowing.

In case of Karnataka and himachal Pradesh 100: 75: 50 kg per ha of NPK should be applied half N, full quantity each of P and K are applied in pits or furrows at the time of sowing.

Remaining quantity may apply 30 DAS or visit before flowering.

Irrigation:

Pre sowing irrigation is given, if there is insufficient moisture in soil. Musk melon plants require abundant moisture when the vines are rapidly developing and up to the time the melons are grown. Crop require eleven irrigations at the interval varying 4 to 9 days. Furrow and basin irrigation is commonly adopted for watering in musk melon. Pitcher method of irrigation is also used in water scarcity areas

Interculture:

Secondary growth up to 7th node in 3rd node in Punjab sunheri, 4th node in Punjab hybrid and Pusa madhuras, 6th node in Pusa sarbati it is to be removed in order to enhance fruit yield. First weeding. Is done 15 to 20 DAS or DAT. 2 to 4 weedings are required before the vine covers the whole area. Weedicides like fluchloralin, Nitrofen can also be used for weed control.

Harvesting, yield and storage:

Crop is already for harvesting about 90 days after sowing depending upon variety. Fruits of musk melon takes 30 days from fruit set to maturity.

Maturity is determined by

1. a change in outer colour to yellowish green or brown.
2. Fruits will also slip from vine. This is due to development of abscission layer resulting in automatic detachment of the fruit from the vine when fully mature. This is called **slip stage**.

Varieties like karanapur there is no development of slip in such case only indication is change in colour. For local market fruits are harvested at full stage where as for **distant markets** they should be harvested at **half slip stage** or even earlier yield varies for 10 to 15 t/ ha. Fruits can be stored at room temperature i.e. 32 + 80 C up to 8 days and + 1 0 C up to 25 days.

The seed rate is 2 to 3 kg/ha.

The average yield is 100 to 150 quintal / ha.

Gourds

Botanical name:	<i>Luffa acutangula</i> - ridge gourd <i>Luffa cylindrical</i> - Sponge gourd
Family	: Cucurbitaceae
Chromosome no	2n = 26
Origin	: Spain

The word **Luff/ Loofah** is of Arabic origin.

Importance and nutritive value:

Compared to ridge gourd, sponge gourd rich in carotene and fibre. Ridge gourd contain high amount of minerals. Both of them have delightful compound called as Luffein.

Botany: Fruits are club shaped and 10 ribbed. Fruits of ridge gourd are ribbed where as the fruits of sponge gourd are smooth.

Varieties:

Sponge gourd:

1. *Kalyanpur chikni*: developed from VRC, Kalyan pur. Average yield is 35 to 40 t/ha.
2. *Pusa chikni*: selection from Bihar collection. Released by IARI, New Delhi.
3. *Pusa sneha*: released from IARI, New Delhi. Fruits are suitable for long distance transportation. Crop is ready for 40 to 65 days.
4. *Pusa supriya*: fruits become ready for picking in about 45 to 60 days.

Ridge gourd:

1. *Arka swathi*: cross between medium fruits at IIHR – 54 and long fruitvat IIHR 18. Average yield is 52 tonnes per ha.
2. *Arka sumeeth*: it is a cross between early fruiting IIHR – 54 and long fruited IIHR. Suitable for cultivation round the year on Karnataka.
3. *Pusa nasadar*: selection from Neemuch variety. Released from IARI, Delhi 15 to 16 t/ha.
4. *Punjab sadabahar*: released from PAU, Ludhiana
5. *Satputia*: it is a cultivar from Bihar, bears hermaphroditic flowers, fruits produced in clusters.

Climate:

it requires long and warm climate. Humid warm climate is preferable. It grows best in rainy season. Optimum temperature is 24 to 37°C.

Soil:

Loamy soil is the best. Soil is rich in organic manure is preferable. Optimum pH is 5.5 to 6.7.

Time of sowing:

Summer crop is sown during February to March. In south India rabi crop is sown during the months of December. Rainy season crop is produced in June to July. However, in hilly regions optimum time of sowing is April to May.

Seeds and sowing:

Seed rate is 5 to 7 kg per ha. Sowing can be done either on raised beds or in pits. Spacing is 1.5 to 3 m rows 60 to 120 cm between plants.

Manuring:

Nutrients have to be applied based on soil tests. 15 to 20 t/ha well decomposed FYM is to be applied at 10 to 15 days before sowing. NPK are applied in the ratio of 100, 60 and 60. Half nitrogen, entire P and K are applied at the time of sowing in raised beds or pits. Remaining nitrogen is applied 30 to 45 days after sowing.

Irrigation:

First irrigation may be given soon after planting later frequent irrigation. Once in 4 to 5 days are given. Flowering and fruit development are susceptible to soil moisture stress. Rainy season crop hardly requires any irrigation if there is good rain fall. The crop is normally raised in basin or furrow system of irrigation.

Interculture:

First weeding is 20 days after sowing another weeding is given to 30 to 45 days after first weeding.

Harvesting:

The crop is ready for first harvest about 60 days after planting. Fruits attain marketable maturity in 5 to 7 days. Pickings are taken up at an interval of 3 to 4 days. Fruits should not be left on the plant till they get over mature.

Mature or just to mature are slowly picked. Average yield is 15 to 20 tonnes per ha. Fruits can be stored at 3 to 4 days at room temperature.

Botanical name: *Lagenaria sicerarea* – Bottle gourd
 Family : Cucurbitaceae
 Chromosome no 2n = 22
 Origin: Malabar region of India

Importance:

Fruit is rich in protein, thiamin, riboflavin and vitamin C. it is easily digestible fruit. Pulp is good for overcoming constipation, cough and light blindness. Used as antidotes against certain poisons. A decoction from leaf is used to cure Jaundice and also in digestion. Bottle gourd is strictly cross pollinated.

Varieties:

Pusa manjari: hybrid between **pusa summer prolific round X selection 11.**

Pusa meghdoot: hybrid between **pusa summer prolific long X selection – 2**

Pusa summer prolific long: it is suited for growing as a **summer** crop.

Climate:

Bottle gourd is warm season crop cannot tolerate cold and frost. Highly sensitive to photo period, short days and humid climate favour femaleness, requires 18 degrees Centigrade, minimum temperature for seed germination and about 20 to 30 degrees Centigrade for growth and development.

Soil:

It can be grown on all types of soil. Sandy to sandy loam soils with high amount of organic matter are considered best with good drainage facility. It can be raised on river beds. Optimum soil pH is 5.7 to 6.7.

Time of sowing:

In case of plain areas Kharif crop : June to July

Rabi crop : October to November

Summer crop : February to March.

For hilly areas - March to June

For river beds - November to December.

Seeds and sowing: *Seed rate: 3 to 6 kg per ha.*

Crop is mainly sown directly in the field on raised beds or furrows or pits. Generally 2 seeds per hill are sown otherwise. Pits of 60 cm³ are dug and filled with well decomposed FYM along with top soil. Spacing 1.8 to 3 m row to row; 0.6 to 1.5 m plant to plant. For taking early crop of bottle gourd seeds are sown in plastic bags. Bags are filled with FYM and soil in 1: 3 ratio. Seeds treated with Capton are sown in bags. After germination plants are transplanted in February to March.

Manuring:

Well decomposed FYM @ 15 to 20 tonnes per ha is given in 20 days before sowing. Recommendations of NPK varies from place to place, 100: 50: 50 kg per ha in Punjab conditions, 32: 24: 24 kg NPK per ha under TN conditions are recommended. Entire P and K half of Nitrogen are applied to pits or furrows at the time of sowing. Remaining quantity of nitrogen is applied to 30 to 40 DAS.

Irrigation:

Summer crop requires frequent irrigations for every 3rd or 4th day. By the crop sown in rainy season, it is irrigated as per requirement. Winter crop requires irrigated once in 10 days. Moisture stress during fruit development reduces fruit size and yield. Over watering should be avoided as it will result in fruit drop. Basin and furrow irrigation is commonly used to raise bottle gourd.

Drip system of irrigation was found to increase yield of about 47% compared to sprinkler and furrow system of irrigation.

Interculture:

For high quality as well as quantity of fruits, vine should be **trained** on bamboo with rope or wire in bower system. Bower system is laid as per the procedure described in bottle gourd. In bower system of training plant height is kept at 2.5 m planting distance 2.5 x 0.6 m. Varieties like PSPL, Pusa naveen, Arka bahar, Punjab long, Punjab round, Punjab komal suitable for Bower system. Bottle gourd trained on bower yields about 60 tonnes per ha. Field should be kept free of weeds through cropping season. Weedicides like fluchloralin, butachlor can be used.

Harvesting:

Crop will be ready for about 60 to 100 days after seed sowing. Bottle gourd takes about 15 days after fruit set to reach marketable stage. Fruit is harvested when it is still tender and medium in size. Pickings are taken at an interval of 3 to 4 days. Tenderness and edible maturity are judged by pressing the skin little pubescence persisting on the skin. Seeds should be soft smaller fruits generally fetch in better price on market.

Yield:

20 to 23 tonnes per ha for open pollinated variety and 60 to 65 tonnes per ha for hybrids. Fruits can be stored under 100 C and 60 to 70 % RH for few weeks.

Botanical name: *Trihcosanthes anguina* – Snake gourd
 Family : Cucurbitaceae
 Chromosome no $2n = 22$
 Origin: China

Importance:

Young tender fruits used as vegetable. Fruit is rich in vitamin A. Crop is highly cross pollinated due to its monoecious nature. Flowers are unisexual. Female flowers are solitary. Ratio of male and female ranges from 31 : 1 to 22 : 1

Varieties: The cultivate dvarieties are two types

1. Light green – with white stripes
2. Dark green – with palegreen stripes

Climate:

Snake gourd can be grown in tropical and sub tropical climates. High humidity is favourable for growth. Snake gourd may not be successfully grown above 1500 m altitude.

Soil:

It can be grown on wide range of soils. However soil which is well drained. Rich in organic manure is suitable for snake gourd.

Time of sowing:

It can be sown in April to July and also October to November. Seeds are sown n hills near edge of raised beds or flat beds at 1.5 to 2 m between row to row 60 to 90 cm between plant to plant. 2 to 3 seeds per hill should be planted. Seed rate is 3 to 6 kg per ha.

Manuring:

10 to 15 tonnes of well decomposed FM per ha is incorporated. 60 kg N, 50 kg P and 40 kg per ha is the blanket recommendation. Half nitrogen is top dressed at a stage when the plants start bearing.

Irrigation:

It is given immediately after sowing. Afterwards light irrigation is gien every 4th or 5th day. Frequent irrigations are needed during flowering and fruit development stage. Any stress during this period will reduce fruit size and yield. Over watering is very harmful. Snake gourd is irrigated usually by basin or furrow method.

Inter cultivation:

Shallow cultivation is done to make the soil loose and to keep down. Weeds the vine is trained on sakes, patches, which will facilitates hanging of fruit. Fruit will be tried with some weight to encourage linear growth. Earthing up of individual vines is an essential operation to avoid exposure of roots.

Harvesting:

Young tender half grown fruits are ready for harvest in about 8 weeks after planting. Fully ripe fruits are collected for seed purpose.

Yield: average yield is 13 to 20 tonnes per ha.

Storage: snake gourd cannot store well for more than 2 days.

Botanical name: *Momordica charantia* – Bitter gourd
 Family : Cucurbitaceae
 Chromosome no $2n = 22$
 Origin: Indo – Burma region

Importance:

It occupy **first rank** among the cucurbits in respect of iron, vitamin C. Alkaloid **momordicasoides** gives the bitter taste to the fruit. Fruit is rich in P, Ca and Vitamin A. Fruit is wormicidal and cure stomach disorders and has beneficial effect on the persons suffering from diabetes, arthritis, rheumatism and asthmatic complaints. Juice from leaves are used against Leishmaniasis and eruptions and voice.

Powder prepared for plants is useful against ulcers. Protein of bitter gourd inhibit the growth of HIV 1, viruses in human cell cultures. Bitter gourds can be used as fresh vegetables can be canned and pickled and used as dried vegetables.

Varieties:

Arka harith: selection from Rajasthan. Collection released by IIHR, Bangalore. Yield is 130 Q/ ha in 120 days and suitable for **both summer and rainy** season.

Coimbatore long: it is a selection from National seeds corporation, New Delhi. Yield is 150 Q per ha.

Pusa do mausami: it is a selection from IARI, New Delhi. Suitable for cultivation in **both season (summer and rainy)**.

Pusa vishesh: it is dwarf vine variety suitable for **picking and dehydration**. It is released by IARI, New Delhi.

Climate:

Bitter gourd can be grown in both tropical and sub-tropical climate. Warm climate is considered best. Bitter gourd cannot tolerate, frost and cold. Require minimum temperature 18°C for seed germination and 30°C for growth and development. Short day conditions will help in improving female flower production.

Soil:

For an early and good crop bitter gourd needs sandy loam soil. pH is 5.5 to 6.7 an ideal soil for bitter gourd cultivation should be rich in organic manure and should be well drained soil.

Time of sowing:

Summer crop is sown in February to March. In South India bitter gourd is generally sown in October to November. Rainy season crop is sown during June to July. Hills of North India. Bitter gourd can be sown in the month of April to May.

Seed rate: 4 to 6 kg per ha.

Spacing:

1.5 to 2 m between rows and 60 to 120 cm between plants. Sowing can be done on raised beds or furrows. Another method of sowing is by digging pits 60 x 60 x 45 cm and refilling them with top soil and FYM.

Manuring:

Well decomposed FYM of 20 to 25 t/ha was applied 15 to 20 days before sowing. Besides that 100, 50, 50 kg per ha of NPK. Under Punjab conditions, 20 to 30 kg per ha of NPK under

Tamil Nadu conditions is generally recommended. Half nitrogen, entire P and K is applied in pits or furrows at the time of sowing. Remaining Nitrogen is applied at 30 days after sowing. Seed treatment with Boron at the rate of 3 to 4 mg per litre is also useful for high fruit yield.

Irrigation:

Bitter gourd both in rainy and summer season. Rainy season crop do not require much watering. Summer crop is irrigated immediately after planting. Light irrigation is given on every 4th or 5th day until flowering. Maintenance of optimum moisture during flowering and fruit development stage is highly essential. Crop is irrigated through basin and furrow system.

Inter culture:

Vines should be trained on bamboo with rope or wire on **bower system**.

Training as the following **advantages** over ground method.

1. expansion of fruit area results in increased number of fruits.
2. fruits become bigger in size.
3. intercultural operations can be done easily
4. better pollination and fruit set.
5. fruit becomes attractive due to proper exposure to sun light.
6. low incidence of fruit fly or other insects and diseases.
7. spraying and resting operations can be done easily.
8. 100 per cent clean harvesting can be easily done
9. no scab or rotting of fruits.

For preparation of bowers bamboo, cement or iron poles, poles connecting wire of 16 gauge are quite useful. Poles are kept at a spacing of 3 x 3 m. poles are buried up to 50 cm. apart from this a heavy stone may be kept in 1 cu m pit to tie with the corner by connecting wire.

10 gauge GI wire is used to stretch and form criss cross squares at a distance of 60 cm. to prepare bower system for one ha. Following material are required 1500 x 1500. cement pole 200 kg supporting wire of 16 gauge, 400 kg criss cross netting wire of 10 gauge, 120 kg jute or coconut fibre, 25 kg plastic rope. Height maintained is 2.5 to 5 m

Spacing adopted is 1.5 m between row to row; 0.6 m between plant to plant. Variety is **pusa do mousami** is suited for bower system of training. Field should be kept free throughout cropping season.

Harvesting:

Crop takes 55 to 110 days from sowing to harvesting. Picking is mainly done when the fruits are tender and green. Pickings can be done at an interval of 2 to 3 days.

Bitter gourd fruits mature very fast and turn red. Irregular harvest may delay the formation of successive fruits effect their growth and development. Irregular harvest also decrease number of fruits per pit. Average yield of bitter gourd vary from 11Q (open pollinated) to 300 Q (hybrid) per ha. After harvesting all the undesirable fruits like pest effected, disease effected, deformed, over matured and dull coloured. Fruits should be sorted out. Fruits are washed with water and shade dried. They are arranged properly. In bamboo baskets on news papers. Fruits in baskets can be kept for 2 to 3 days.

Pumpkin - *Lagenaria steraia*.

Local Name: Dudhya Sweet- Marathi, Lauki- Hindi.

Family: Cucurbitaceae. Bottle gourd is a very important crop in India. The fruit in green stage and leaves with stem are used as vegetable. The hard shell of the fruit is used for different purpose.

Climate: Hot and moist climate is favorable for its cultivation. It cannot withstand frost.

Soil and its Preparation:

Bottle gourd can be grown in any types of soil. But sandy loam soils are best suited for its cultivation. The land should be prepared thoroughly by five to six ploughings.

Time of Sowing and Layout:

The seed is sown from January to end of February for summer crops. June – July for rainy season crop in the plains and April in the hills. Layout is ring and basin used.

Seed Rate: The seed rate is 3 to 6 kg/ ha.

Methods of Sowing and Spacing:

The seed is sown by dibbling method at spacing of 2 to 3 X 1.0 to 1.5 m. Generally three to four seeds are sown in a pit at 2.5 to 3.0 cm depth. Manures and fertilizer same as for pumpkin.

Intercultural Operation:

Two to three hoeing is given to keep down the weeds during the early stage of growth. The rainy season crop is usually stalked, often trained on a bower made of bamboos and sticks.

Irrigation:

The summer crop requires frequent irrigation at an interval of 4 to 5 days. The winter crop is irrigated as and when needed.

Harvesting:

The fruits should be harvested when they are still green. Delay in harvesting causes the fruit to become unit for marketing.

Yield:

The average yield is 90 to 120 quintal / ha.

Varieties:

There is some varieties of bottle gourd, viz summer prolific Round, summer prolific long, Pusa Navven, Arka Bahar, Pusa Manjiri, Pusa Meghdoot etc.

Lect.7 French bean, Peas and Okra

Beans and peas belongs to the family legumineseae. Most of the cultivated varieties of beans and few of peas are **vine types** except the French bean where the cultivated varieties are **bush types**.

Botanical name: *Phaseolus vulgaris* - **FRENCH BEAN**
Family : Leguminaseae
Origin : Central America
Chromosome number: $2n = 22$
Also called as : Kidney bean or snap bean

Botany:

Generally three types of beans are recognized by their growth habit and requirement of day length.

1. **Dwarf bush types:** Day neutral plants. They have short erect stem with the main axis consisting of 4- 8 shortened internodes.
2. **Semi pole or runner types:** These are short day plants which have to 4-8 inter nodes , longer than those in bush types.
3. **Pole type:** are long day plants which have longer internodes and their number is depend upon the length of the growing season.

Varities:

1. **arka komal:** plant is bushy. 70 days for flowering.
2. **blue pod medium:** it is a white seed variety. It has bluish black spots on pods.
3. **bountiful:** introduction from US comes up in September to February. Brown sweedish. Seeds are. It is resistant to common disease.
4. **contender:** tolerant to powdery mildew and mosaic.
5. **Kentucky wonder:** whole variety
6. **Pant anupama:** developed from pantnagar. Bushy type
7. **Pusa parvathi:** developed by **X ray radiation** of American variety IARI

Climate:

French bean is a cool season crop. It gave good yield under mild warm season. It is sensitive to frost. French bean has the cultivars of long day, short day and day neutral varieties. Most of them are day neutral. Seeds will not germinate, very high temperature. Continuous rains, resulting in the breakage of pods.

Soil:

French bean can be grown on wild range of soils, sandy loams are best. Heavy soil gives good yield. Ideal pH range is between 6 to 7. French bean mature early on sandy soils compared to other soils.

Time of sowing:

French bean is sown twice in a year. First sowing is done in July to September and second time in January to February. In case of hills sowing is done from March to June.

Seed rate:

Cultivars spacing

Bush type of cultivars - 95 kg/ha

Pole type of cultivars - 25 to 30 kg/ha

Spacing:

Bush type- 60 x 15 cm

Pole type- 1 m x 60 – 75 cm x 30 cm

Seeds are sown 5 cm deep. Two seeds are sown at one place and one of the weak seedling is removed after completion of germination. Seeds germinate slowly, at soil temperature of 15 °C and they germinate 10 days after sowing under favourable conditions.

Manuring:

It has poor nodule formation capability, well decomposed FYM up to a dosage of 20 to 30 t/ha is given, besides that 60 kg N, 40 kg P and K are applied. Nitrogen is applied in two splits. Half nitrogen first split is given at the time of sowing whole quantity of P and K. Second split is given at flowering stage.

Irrigation:

French bean is said to be a drought resistant crop. For better growth and yield proper supply of moisture is essential. Irrigation is required in the early phases of crop growth during blooming and pod development period. Plants are susceptible to water stress. Irrigation at regular intervals are necessary. Lack of adequate soil moisture results in reduced percentage of pod setting, reduced length of pods, reduced number of seeds per pod and high fibre content in pods.

Inter culture:

Shallow cultivation is to be done at early stages of crop growth. Deep cultivation will disturb plant growth because of root pruning. Cultivation in early stages helps to keep down the weeds. Large leaves form dense canopy which act as weed suppressor.

Weedicides like penta chlorethanol, alachlor can be applied. Dwarf varieties do not need support. Climbing varieties need to be supported with bamboo sticks. Single stick is fixed near each plant. The frame work consists of line of pairs of caves not less than 42 cm apart at ground level. Earthing up around the base of the plant gives additional support and encourages root growth.

Harvesting:

Pods are ready for harvest, 2 to 3 weeks after first blossom. Pods are harvested when they are young, tender and delicate. Seeds start forming when the pods are little matured. They are cooked after cutting into pieces.

Mature pods contain developing seeds. These are separately picked and shelled. Green of seeds are also cooked as vegetables. Harvesting of tender parts should be done at frequent intervals otherwise it will suppress formalin and development of new pods.

In order to obtain dry beans harvesting is done when pods are fully ripe and about to shatter. In this case whole plants are harvested either by hands or by machine. Harvested plants are staked for 7 to 10 days and then they are crushed either by bullocks or by machines. All possible care should be taken that seeds are not injured. Pods and grain yields of French bean vary considerably with locality, soil fertility, variety and management practices.

Average yield of bushy varieties is 4 to 5 t/ha

Pole varieties 7 to 10 t / ha

Yield of dry seeds varies from 1.2 to 1.8 t / ha

Storage: it is completely dried seeds are stored in glass container with tight podding varieties or polythene bags.

PEAS

Botanical name:	<i>Pisum sativum</i> – Garden pea
Family	: Leguminosae
Origin	: Ethiopia
Chromosome number:	2n =14

Importance:

Garden pea is rich in digestible protein, vitamin A and C. It is also rich in minerals like calcium, potassium, iron and phosphorus. Fresh green pea is excellent food for human consumption taken as vegetable or in soup. Large proportion of pea are processed either canned, frozen, dehydrated. The straw of pea is a nutritious fodder. The pea protein is moderately **deficient** in sulphur containing **amino acids methionine, cysteine**.

Varieties:

Alderman: suitable for freezing.

Arkel: early wrinkled seeded variety introduced from England. It was tested by IARI.

Asauji: this was a selection from the material. Collected from Amritsar area made by IARI, New Delhi.

Bonneville: wrinkled seeded variety introduced from US

Early badger: dwarf early wrinkled seeded variety evolved at Wisconsin. Resistant to fusarial wilt.

Climate:

Crop grows best in relatively cooler situations. Seed germination takes place up to the minimum temperature of 5° C. However, optimum temperature for seed germination is 22° C. Warm weather speeds up the rate of maturity. Pea is not susceptible to cold but severe frost causes injury to freshly opened flowers and young pods. Pods grow best in the regions where there is slow transition from cool to warm weather in spring.

Soil:

Pea can be grown on many types of soils from light sandy to clay soils. Highly organic soils are unsuitable for pea as their moisture reserve leads to excess vegetative growth and poor pod formation. It is very sensitive to saline and alkaline conditions. Most favourable range of pH is between 6 to 7.5.

Time of sowing:

Pea is grown generally as a rabi crop. It is sown from the beginning of October to middle of November. In the hills of South India the crop is sown from March to May. In Darjeeling it is sown from June to August.

Seed rate:

If sowing is done in ridges and furrows method, seed rate adopted is 50 to 60 kg per ha. Seed rate also varies with growth habit of varieties for early varieties 100 to 120 kg per ha. Mid late varieties 80 to 90 kg per ha is the general recommendation.

Seeds can be sown on flat beds or raised beds either by broadcasting or behind the plough. Seeds are also drilled through poras (tubes attached to desi plough).

Depth of sowing: 5 to 7.5 cm

Spacing: 30 cm x 5 cm

Manuring:

Well decomposed FYM is applied at a dosage of 8 to 10 t/ha at least 15 days before sowing. NPK should be applied in the ratio of 30: 50: 25 kg per ha. Foliar application of 0.1% ammonium molybdate is given to increase the number of root nodules, yield, TSS and number of grains per pod.

Irrigation:

Crop requires 2 or 3 irrigations. First irrigation is given 40 DAS, second irrigation at bloom stage 60 DAS.

Method of irrigation – furrow method

Inter culture:

Manual or mechanical weed control has to be taken care of chemical weed control i.e. gaining dominance because it is cheaper and less time consuming. Various herbicides recommended are lasso (alachlor) at the rate of 0.75 kg a.i./ha, basalin at the rate 2 kg a.i./ha.

Harvesting:

Generally 3 to 4 pickings are to be given in the season. For good quality of produce harvesting should be done either early in morning or late in the afternoon. Quality is ascertained by tenderometer or motovometer. Usually periodical pickings 6 to 8 day interval are taken. Care should be taken not to jerk the plants till they get injured.

Harvesting is generally done by manual method employing female pickers is advisable compared to males. Green pods are packed in gunny bags or baskets.

Yield:

Early varieties 30 to 40 Q/ha

Late and Mid season varieties 60 to 70 Q/ha

Shelling percentage from 35 to 50 depending on variety, agro-climatic conditions, management practices, grain yield about 15 to 20 Q per ha.

Storage:

Fresh unshelled pea can be stored for 2 weeks at a temperature of 32° F. Relative humidity is 85 to 90%. Peas can also be stored in crushed ice for about 2 to 3 weeks. However pods will freeze at – 100° C.

Botanical name: *Vigna unguiculata*– Cow pea
 Family : Leguminosae
 Origin : India
 Chromosome number: 2n =22

Importance:

Cow pea is nutritive vegetable rich in protein, calcium, magnesium and other minerals. It is used as food both green and dry stages. It is also grown for hay, silage, Pasture and for all types of straw and as source of protein especially lysine.

Varieties:

Arka garima: variety was bred at IIHR, Bangalore. Plants are tall, vigorous. Pods are of light weight.

Pusa barsati:

Pusa dofsali: cross between pusa phalguni x Philippines early. It is suitable for sowing both in **summer and rainy season**. It is a **photo-insensitive**.

Pusa phalguni: it is selected from Canadian cultivar. It is a bushy type and broad variety. It is suitable for cultivation in February – March. It is of 60 days duration.

Pusa komal: it was developed from IARI, New Delhi.

Pusa Ritu Raj: it was developed from NBPGR, New Delhi. It is **photoinsensitive**. Immature grown in summer and rainy seasons.

Climate:

Cow pea is warm season crop. More tolerant to heavy rainfall compared to other pulses. It suffers from water stagnation and drought thrives between 21 to 30°C temperature. It is grown in rainy season as a pure crop or mixed with Jowar.

Soil:

Cow pea grows well on all types of soil and yields well on loamy soils. Time of sowing: there is two main cropping seasons. Summer crop generally sown towards the end of February. It can be continued till the middle of April. Sowing time for monsoon crop commences from the middle of June and extends up to the end of July. It is sown in December to January in plains and varieties which are photo-insensitive can be sown both in July and February to March.

Seed rate and sowing:

In summer – 20 to 25 kg per ha. Rainy -15 kg per ha

Spacing – 60 x 15 cm – kharif or rainy season crop. 30 cm between rows and no distinct spacing between plant to plant within a row. Seed treatment with Thiram is done at the rate of 4 g per kg of seed.

Manuring:

Being a leguminous crop, cow pea requires less Nitrogen, 10 to 20 kg Nitrogen is incorporated in the soil before sowing. Phosphorus promotes the multiplication of Rhizobia, P and K 50 to 70 kg per ha are to be drilled in the soil. Cow pea is highly susceptible to zinc deficiency is more 10 to 15 kg Zn SO₄ per ha is beneficial. Cow pea seed should be inoculated with Rhizobia before sowing.

Irrigation:

Cow pea is shallow rooted crop. Light irrigation are advisable. Rainy season crop do not need any irrigation, except in long dry spells. Summer crop require irrigation once in a week during March to April. Early sown rainy season. Crop needs 1 or 2 irrigations in pre monsoon period. water logging should be avoided as cow pea is sensitive to it.

Interculture:

Effective control of leaves. The first 20 to 25 days of the crop is essential. 2 to 3 weeding and hoeings are required to check the weed growth. One kg a.i. Treflin is recommended for control of weeds as a soil incorporation.

Harvesting, yield and storage:

Pods are ready for harvesting after 40 to 50 DAS. Pods are picked up when they are tender and half grown. Pods of cow pea develop very quickly and if they are not picked at right stage they tend to become puffy unlike peas. Cow pea has to be more frequently picked.

OKRA

Botanical name: *Abelmoschus esculentus*

Family : Malvaceae

Chromosome no $2n = 72 \text{ to } 144$ (poly ploidy)

Origin: South Africa

Importance and nutritive value:

Bendi is rich in calcium, magnesium, phosphorus etc, and vitamin A, B, C. Bendi is good for people suffering from renal colic, lecorrhoea and general weakness. Bendi has high iodine content and it has ability to control goiter.

Bendi leaves are used in turkey to reduce inflammation. Bendi can be fried in butter or ghee. It can be put as a sole vegetable. The roots and stem are used for clarifying cane juice and in the manufacture of gur or brown sugar. The plants are soaked in water and resulting solution is used as clarifier in **jaggery** manufacture.

Mature fruits and stems contain fruit fibre and used in paper industry.
Ripe seeds are roasted, ground, used as substitute for coffee.

Botany:

The genus *Abelmoschus* is distinct from *Hibiscus* in having deciduous type of calyx where as are persistent in the genus Hibiscus. Fruit is **capsule**, oblong, longitudinally narrow and dehisces longitudinally. The greatest increase in fruit weight, length and diameter occurs during 4-6th day after flowering. At this stage the quality will be high usually fibre formation in the pod starts from the 5th to 6th day of formation and a sudden increase in fibre content may be seen from **9th day**. A new variety **IHR-2- 31** is however retains its tenderness up to the 10th day.

Varieties:

Abelmoshus manihot, a wild species was found to be almost immune to YVMV

Arka Abhaya: it is released from IIHR, Bangalore. Fruits are 18 cm in length, duration of crop is 120 to 130 days.

Arka Anamika: IIHR, Bangalore. It is of inter specific origin between *Abelmoschus esculentus* x *A. manihot* sp *tetraphyllus*.

DUR – 1(seetla jyothi): released from IIVR Varanasi.

DVR – 2 (seetla uphar): released from IIVR, Varanasi, resistant to YVMV.

EMS – 8: it is a mutant derived from Pusa sawani. Pusa sawani was treated with 1 % Ethyl methane sulphonate. Final selection was made in M8 generation..

Harbajan: it was originally perkins long green (selection – 6). It was named in the memory of Dr. Harbajan Singh.

P-7: released from Punjab AU, Ludhiana. Evolved from a cross between *Abelmoshus esculentus* cultivar pusa sawani and *A. manihot* sub species *manihot*

Parbani kranthi: evolved by MPAU, Parbani (Mahatma pule Agricultural University).

Interspecific cross between *A. esculentus* sp *pusa sawani* x *A. manihot*.

Pusa makhmali: developed from plant introduction section, IARI, New Delhi. It is susceptible to YVMV.

Pusa sawani: evolved by Dr Harbajan singh at IARI, New Delhi. Derived from an inter varietal hybridization between IC 1542, a field resistant cultivar to YVMV and Pusa makhmali (CIC 1542 x Pusa mukhmali – Pusa sawani). It yields about 175 Q per ha.

Perkins long green: Recommended for hilly areas

Climate:

Okra prefers warm climate. It is sensitive to frost and requires frost free growing period. optimum temperature 24 to 27°C. seed does not germinate when the temperature is below 20°C. plant growth is greater in rainy season, compared to summer season. Temperature higher than 42°C may cause flower drop.

Soil:

It can be grown in sandy loam soils. However clay loam soil give good yield. Ideal soils should be rich in organic manure. It should be well drained and have high organic matter content. Roots are sensitive to high water stagnation pH is 6 to 6.8. It can also be grown in saline soils up to EC value of 6 milli mhos per cm.

Time of sowing:

In north India summer crop is sown in February to March. In South India generally sown in November as main crop. However rainy season crop is sown in June-July throughout India.

Seeds and sowing:

20 to 25 kg per ha is required for summer crop and 12 to 15 kg per ha is required for rainy season crop. During summer seed is soaked in water at 12 to 24 hours the field is laid out in to ridges and furrows in flat bed. Sowing can be done by broad cast, line sowing and dibbling method. Sowing of okra is done in pits in west Bengal. Spacing adopted in summer season crop is 30 x 30 cm whereas rainy season crop is sown in rows at a spacing of 60 x 30 cm.

Manuring:

Well decomposed FYM @ 20 to 30 kg per ha s applied thoroughly in soil about 20 days before sowing. Besides this 100 kg N, 60 kg each of P and K are recommended ½ nitrogen, full quantity of P and K are applied at 30 to 45 days of sowing at top dressing.

Irrigation:

First irrigation is given after seeds are germinated subsequently field is irrigated at an interval of 4 to 5 days in summer in case of rainy season there is no much emphasis on extra irrigation. Drip irrigation as given higher yield compared to furrow irrigation.

Interculture:

First weeding is done 15 to 20 days after sowing. A total of 3 to 4 weedings is taken. Herbicide like basalin (fluchloralin) @ 2.5 lit per ha at pre plant incorporation. Lasolin (alachlor) 5 litres per ha as pre emergence application are recommended. Germination commences in 13 to 15 days of sowing and it can be over in another week. The crop is thinned at 30 days after sowing to maintain plant to plant distance of 30 cm.

Harvesting:

Pods are ready for harvesting in about 45 to 65 days after sowing depending on only tender fruits are to be picked. In case of pusa sawani variety the fruits will reach 10 cm about 6 days after anthesis. In case of Punjab padmini variety the fruits will gain 10 cm length in about 5 days of anthesis. Pickings are taken at an interval of 4 days.

Yield:

it varies from 60 to 75 quintols per ha in summer and 120 to 125 quintols per hectare in rainy season. Okra can be stored just for 2 to 3 days under room temperature conditions.

Insect Pests of Bhendi**1) Stem and Fruit Borer:**

The caterpillar bores into the growing points of the of the plant and later on U also flower bud and young fruit.

Control Measures:

Spraying the crop with carbaryl Sevin-50 W, Sevimol 0.1 per cent, or Monocrotophos Nuvacron)0.1 per cent or Endosulfan (Thiodan -35 EC) 0.97 percent on the 40 and 60 days after sowing is found beneficial.

2) Jassids:

The hoppers suck the cell sap from the leaves, causes curling of leaves.

Control Measures:

Spraying the crop with Dicblorovos (Nuvaii) 76 EC or Malathion -50 EC) @ 2 ml per liter of water or Dimethoate (Roger -30 EC) or Oxydemeton Methyl (Metasystox —25 EC) @ 1 ml per liter of water can control this pest.

Lect.8 Cole crops such as Cabbage, Knol-khol, Cauliflower

Cole crops-Cabbage-introduction - importance –varietiesclimate and soil-seeds and sowing-manuring- irrigation-intercultural operations-harvesting – yield

The word **cole** was originated from the word *Caules* means stem. They are originated single wild species wild cliff cabbage known as “**Cole worts**” *Brassica oleracea var silvestris*. They belongs to family Cruciferae and genus Brassica.

Among the cole crops **only cabbage and cauliflower** are grown in India and khol rabi is grown in foreign countries. The vegetable under cole crops grown all over the world is **cauliflower**.

In India cabbage, cauliflower grown states are U. P., Karnataka, Maharashtra, Bihar, West Bengal, Punjab and Haryana. Cole crops occupied in large area in winter with India. They are rich sources of vitamin C. cabbage juice is used as a remedy against poisonous monsoon.

Some extract of these cole crops are used to cover wounds and ulcers. They are eaten raw as well as cooked. They have protective property against bowel cancer.

All brassica species contain glucosinolates, which is crushed leaves is broken down by the enzyme myrosinase giving a bitter taste.

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Six varieties of *Brassica* species are.

1. *Brassica oleracea var capitata* - Cabbage
2. *Brassica oleracea var botrytis* - cauliflower
3. *Brassica oleracea var gongiloides* - knoll khol
4. *Brassica oleracea var gemmifera* - burssel sprout
5. *Brassica oleracea var italica* - sprouting broccoli
6. *Brassica oleracea var acephala* – kale

CABBAGE

Introduction:

Cabbage was described as more or less compact leaf formed head. The main stem axil of the head or true cabbage is sharp and thick. The **head** is formed by the development of densely packed leaves around the growing point. Cabbage is classified as slow growing biennial and has distinct periods of growth cycle.

Importance:

It is used for pickling. Cabbage pickle is called as **Sauerkraut**. Cabbage soup is good and nourished.

Nutritive value:

Protein quantity is high in kale and longest in red cabbage. Carotin was present in kale.

Varieties:

There are eight groups of cabbage as detailed below.

1. **copen hagen market group:** heads are round, large and compact. Leaves lightly hairy bloom, stem is short. Ex: Golden acre.

2. **flat head or drum head group:** heads are flat. Plant is medium to large. Ex. Pusa Drum Head – It is resistant to Black leg and dry rot diseases
3. **Conical head group:** heads are small pointed. This group small pointed heads includes early maturing varieties such Jersey wake field.
4. **Savoyee group:** foliage very much glittered and wrinkled. Leaves are twine with little bloomy. This group include chief drum head savoy.

Red cabbage: it has deep purplish red coloured heads. Yield is less. Ex: red rock, red Danish. Protein content is lowest. Highest in kale.

Early varieties:

Golden acer: IARI variety selection from Copenhagen market. Pride of India

Late varieties:

Pusa drum head: it was released by IARI. It is a selection from an introduction EC 774.

Climate:

It requires mean annual temperature of 15 – 22⁰ C. it can tolerate frost to a great extent better than cauliflower. In Nasik, ootakamand and kerala it can be grown as a kharif crop. Growth exceed arrested more than 25⁰ C. seed germination is normal when soil temperature 12 to 16⁰ C.

Soil:

For an early crop sandy loam considered best. For late crop heavy soil clay loams and silty loams are preferred.

Land preparation:

Soil should be prepared to a fine tilth by giving 4 to 5 ploughing before planting seedlings. It is a hardy cool season crop. It is annual crop vegetable production, biennial crop grown for seed production. It was introduced to India by portugese people. Its edible parts called **head**. Head consists of thick leaves which over lap lightly around the growing bud. The **flavour** of cabbage is due to **glucoside** called **sinigrin**, which is sulphur containing compound.

Nursery:

The nursery soil is preferred to fine tilth. Raised beds are prepared to a height of 10-15 cm and an area of 60 – 80 sq. m nursery is sufficient for one acre. In nursery beds are sown in lines. Beds are covered with organic manures with increase in temperature promotes early germination, water is sprinkled.

Main field preparation:

Land is ploughed many times. 25 tonnes of FYM per ha to be spread at the time of final ploughing. If cabbage is sown directly heads are formed 25 days earlier to the crop sown by transplanting. For direct sowing two seeds per hill are drilled at spacing 60 x 30 cm but thinning is done when plants are at 10 to 12 cm height.

Transplanting:

Maximum yield was obtained on the flat bed at the spacing of 75 x 45 cm. Early varieties are transplanted 75 to 90 cm spacing row to row and 45 cm from plant to plant. The seedlings are transplanted at 6 to 7 week old. Transplanting is done in the evening and then they should be irrigated.

Irrigation:

Irrigation is given immediately after transplanting. Later it is irrigated at 10 to 15 days interval. Starter solution is given to increase the production to an extent of 0.25 tonnes per ha. Starter solution consists urea, potassium sulphate, single super phosphate at 1:1:2. it is given twice first immediately after transplanting and second 15 days after first application.

Foliar application of Molybdenum and cobalt increases cabbage head size. It is done 1 to 2 months after transplanting. At the time of maturity of head watering should be suspended otherwise can be bursting or splitting. Total 6 to 8 irrigations are required.

Manuring:

Azospirillum and phosphobacteria 2 kg each are recommended along with FYM. At high temperature formation of head can be improved by spraying 2 % common salt. Inter culture include hoeing, earthing up, weeding and mulching. Shallow hoeing is to be done for better soil aerator and also put to the weeds.

Intercultural operations:

Earthing up helps to produce healthy heads it is done four to six weeks after transplanting. Removal of weeds is to be done reduce intensity of weed chemical trifluralin, basalin can be used for these purpose. **Mulching** with black polyethylene sheet control weed and conserve moisture is to be done.

Harvesting and yield:

Cabbage heads have a tendency burnt harvest should not be delayed once thus have attained full size still time. Harvesting is done during

December to april in plain and june to july in the hills. Early crop comes to harvesting to first sixty to eight days after transplantation but will fetch higher price the yield less. Late crop come harvested 100 to 120 after transplanting give 30 to 45 tonnes. Entire plant is uprooted the heads are detached only heads are cut leaving stump in field when the crop is meant for seed production.

Early varieties mature relatively a high temperature heads become loose at initial stages bud gradually become harder with cooling temperatures.

Yields in south India are less than 25% winter is mild and duration is short Maturity is faster in south India. Yield are optimum in north India where longer and moderate weather prevails.

CAULIFLOWER

Botanical name:	<i>Brassica oleracea var botrytis</i>
Family	: Cruciferae
Origin	: Mediterranean region
Plant part	: Curds

The haploid number of *Brassica oleracea* is 9. it is **triple tetraploid** ($6 \times 9 = 54$)

Varieties

Early Kunwari : It has bluish green leaves with waxy blooms. The curds are white and not so compact. It is the earliest cultivar released by Punjab Agricultural University.

Pusa Deepali : Released from IARI as a hot weather cultivar. It is suitable for sowing in late May in North India. Its temperature requirement is 20° and 25°C during curd initiation and development. The plant has erect growing habit and a narrower frame. The curd is somewhat round, medium-sized and white to creamy white in colour.

Pusa Himjyoti : It has erect bluish green leaves with a waxy coating. The curd is pure white and retains its colour even after exposure. Recommended for the spring-summer sowings in the hills.

Pusa Shubhra : It is field resistant to black rot and curd blight.

Dania: The cultivar has been successful in the eastern region, particularly in the hills. The plants are very hardy with waxy leaves and medium deep curds.

Pusa snowball K-1: Field resistant to black rot. It has upright waxy light green leaves, self-blanching, snow-white curds.

Climate:

Cauliflower is more exactly climatic requirement than cabbage or broccoli. It is quite responsive to temperature and photoperiod and has specific temperature according to variety. The average optimum monthly temperature is 15-20°C with the average maximum at 25 °C and the minimum at 8 °C.

Early variety require higher temperature in short days. At relatively lower temperature the curds of early varieties become ricy, leafy, loose and yellow resulting in poor quality. At relatively lower temperature the curds of late varieties remains under sized or small, showing some signs of bolting. In hot dry weather, the curd becomes small and hard. Thus it is very essential that the early, main and late varieties are planted at proper time.

Soil:

Fairly deep loamy soils is preferable can be grown in sandy loam, Silt loams and clay loams. Soil should be rich organic matter and well drained. Cauliflower is sensitive to high acidity. It requires an average pH of 5.5 to 6.6

Season

Sep. – Nov. : Early Kunwari

Oct. – Nov. : Pusa Katki, Pusa Deepali.

Nov. – Dec. : Improved Japanese, Pusa Shubhra

Dec. – Jan. : Pusa Synthetic, Pusa Shubhra

Jan. – Feb. : Cvs in this group are of snowball type : Dania, Pusa Snowball -1, Pusa Snowball

Nursery:

100sq.m nursery area /ha. Apply FYM at 300 kg and 10 kg of No. 5 mixture (9:9:9) along with 50 g of Sodium molybdate and 100 g of Borax. Sow the seeds at 10 cm between rows in raised seed beds after drenching it with copper oxychloride (2.5 g/lit). transplant 30 to 40 days old seedlings at a spacing of 45 cm. Avoid land infected with 'club root diseases'.

Seed rate :

375 g /ha. Sow the seeds in raised beds and transplant 25 days (early varieties), 45 days old seedlings (late varieties) at 45 cm apart.

Transplanting:

Normally, a hardened seedling is likely to give better results and since the early crops are raised in the monsoon, 5-6 week old seedlings stand better chance of establishment. Starter solution of ammonium sulphate and superphosphate (1:2) in combination with 0.1 ppm of IBA was best and economical for early and total yield.

Preparation of field :

Bring the soil to fine tilth. Pits should be taken at a spacing of 45 cm either way in hills. Form ridges and furrows at 60 cm in plains.

Manuring:

Hills : Apply 30 t/ha of FYM and 90 kg N, 90 kg P and 90 kg K as basal dose and 45:45:45 kg NPK / ha after 45 days.

Plains: Apply 15 t FYM / ha and 50 kg N, 100 kg P and 50 kg K as basal and 50 kg N after 45 days. Apply 2 kg of Departmental vegetable micronutrient without mixing with the chemical fertilizers.

Irrigation:

Hills : Once in a week during January and February.

Plains: Once in a week.

Inter cultivation:

Gap filling after 20 days of planting to maintain the population and uniform growth. Hoeing and weeding on 30th and 45th day. Avoid deep inter cultivation as it is a shallow rooted crop.

Harvesting:

When harvesting cauliflower, care should be taken to cut the curd immediately in reaches prime condition. If there is any doubt about the maturity, it is better to harvest early than late. If the harvests are late, flower stalk elongates and loose, leafy, reicey or fuzzy curds result.

Such overmature curds wilt rapidly after cutting. It is better to discard any such over mature curds as they will spoil the appearance of the consignment. When harvesting, the curds are cut off the stalk with a large, sharp knife. The large leaves are then trimmed away until only sufficient jacket leaves remain to protect the curd. When transporting loose, more jacket leaves should remain than when packing in crates.

In crate, jacket leaves are trimmed leaving a fringe of leaves projecting 2-3 cm above the curds. When transported loose, the jacket leaves are only lightly trimmed until they reach the market.

Yield:

When the cauliflower curds mature at a higher temperature above 25°C, the curds are small, loose and yellow. In spite of closer spacing which is followed by the growers, the yield hardly exceeds 10 tonnes / ha. Cultivars, slightly later maturing curds between 20 and 25°C, the yield may vary between 12 and 15 tonnes/ha. With cooler season, the midseason cultivars can yield upto 20 tonnes/ha. The highest yield, is however, obtained from Snowball the yield may go upto 25 to 30 tonnes/ha. Snowball group is the only cauliflower which is grown almost under ideal conditions in North India.

Physiological disorders in cauliflower:

1. Buttoning :

This disorder is characterized by development of small curd. Buttoning may be due to overaged seedlings, poor nitrogen supply, planting wrong cultivar, i.e., early cultivar planted late and also due to root injury by insects or by some disease. Generally, if there is any check in the vegetative growth of the seedlings, buttoning may be induced.

2. Riceyness :

Premature initiation of floral buds is characterized by riceyness in cauliflower and is considered to be of poor quality for marketing. Such disorder may result from any temperature higher or lower than the optimum required for a particular cultivar.

3. Browning :

This is caused by **boron deficiency**. It appears as a small water soaked areas in the centre of the curd. Later, the stem becomes hollow with soaked tissue surrounding the walls of the cavity. In more advanced stage, pinkish or rusty brown areas develop on the surface of the curd and hence, it is also known as brown rot or red rot. Affected curd develops a bitter taste.

It can be controlled by application of borax or sodium borate @ 20 kg/ha. In case of acute deficiency, spray of 0.25 to 0.50% solution of borax @ 1 to 2 kg/ha is advocated.

4. Whiptail :

This results due to **deficiency of molybdenum**. The leaf blades do not develop properly and become strap-like. The growing point is severely deformed and no marketable head is formed. It occurs in acidic soil below pH 4.5. Whiptail can be prevented by brining the soil pH to 6.5 by liming with 5-7.5 t/ha of limestone or by applying 2.5-5 kg/ha of sodium or ammonium molybdate. Foliar spray with 0.05% sodium molybdate solution is also recommended.

5. Blindness:

It is supposed to be due to **low temperature** when the plants are small or due to damage to the terminal bud during handling of the plant or due to insect pest like cut worms which eat the terminal bud.

Lect.9 Bulb crops such as Onion and Garlic

Bulb crops:

The group of bulb crops includes onion, garlic, leek, shallot and chive. These belong to the family **Amaryllidaceae** and genus **Allium**. They are grown in India as winter vegetables.

Botanical name: *Allium cepa* - **Onion**

Family : **Amaryllidaceae**

Origin : **North West India**

Plant part : **Bulbs**

Importance:

It is consumed either raw or cooked along with spices and vegetables. Primarily the bulbs used as vegetables. The flowering shoot known as **Scape** is also used as vegetable. It is rich in minerals like phosphorus and calcium and carbohydrates. It also contains Protein and Vitamin C.

Botany:

All the cultivars of common onions are long day plants in respect of bulb formation i.e the bulb more rapidly as day length increase. The typical flavor of onion is due to presence of a volatile oil known as "**Allyl propyl disulphide**" and the red colour is due to the presence of another pigment "**Quercetin**".

Varieties:

Pusa Red: It is selection from local red varieties, yellow colour and mild pungency. It is fairly resistant to onion thrips.

Other local varieties: Bellary Big, Bellary Red, Poona Red, Nasik Red, Patna red

IIHR varieties: Arka Kalyan, Arka Niketan and Arka Pragati

Climate:

Onion is best suited as a tropical crop. It does best where the season is mild without the extremes of heat or cold or excessive rainfall. A relatively high temperature as well as long photo period (i.e a temperature of 15.6 to 21.1°C for about 10 hours a day and about 80% RH) is essential for bulb formation, where as temperature alone is more important than the length of days in seed stalk development. So most of the varieties transplanted early will bold i.e from seed stalk.

Soils:

It grows best in light soils which may be sandy loam or silt loam, but they should be avoided as they not permit it proper bulb formation. It is sensitive to higher acidity and the optimum pH of the soil should range from 5.8 to 6.5 and the land should be prepared to fit tilt for growing onion.

Sowing season:

North India - November

Tamilnadu - June to October

Maharashtra - October to January

Near Bengalore – June to October

Seed rate:

8 -10 Kg of seed is sufficient to transplant seedlings for 1Ha. 6 to 8 weeks old seedlings which are about 15 cm tall are transplanted. For direct sowing the seed rate has to be doubled.

Bulbs:

When bulbs are planted the seed rate is 1000 to 1200 per Ha. Bulbs of about 1/2 inch diameter are dibbled on one side of ridge or in plain beds.

Spacing: 20X10 cm² for seedlings, 30X10cm² for bulbs

Manuring:

25 tonnes of FYM, 60 to 100 Kg N, 40 to 60 Kg P₂O₅ and 60 to 80 Kg of K₂O. FYM should be incorporated at the time of field preparation, P and K along with half of N should be applied in bands, 5 cm below the seedlings to the side before transplanting. The rest of N may be top dressed 2 months after planting.

Irrigation:

It requires 14-18 irrigations from planting to harvest depending on the season. At bulb formation stage i.e 70 to 75 days after planting irrigation is essential. Drought at this stage results in cracking of the bulb and low yield.

Harvesting and curing:

Green bunch onions are harvested when they are lead pencil thickness with a small bulb, as a direct vegetable. As a bulb crop it comes to harvest after 4 months of planting. A well matured bulb should be harvested by putting out or digging out the bulbs. Maturity is indicated by the tops dropping just above the bulb while the leaves are still green. The outer scales on the bulbs become loose and cloured. The bulb after harvesting are left in the field under shade for 4 days for curing which helps the bulbs in storing well.

Methods of storage:

Onions are stored in special house constructed with thatched hours sides covered with bamboo stick for good air circulation the sides can be covered with the gunny cloth instead of bamboo stick in order to have better circulation. Onions are stored in open shed long spreading in a dry floor.

Important floor should be avoided. Onions are stored in 2 to 3 layers in well ventilated rooms and is desirable for proper storage.

Bolting in onions:

Premature production of seed stalk is known as bolting when such seed stalks are produced which produce will grow bulb purpose. It becomes a problem, the production of bolt takes at the cost reserved foods at stalks and then bulbs keep long storage. Some of the salient features of bolting are the bolts are little bit smaller in size normal inflorescence. Bolting occurred to the extent of 50 to 80% in the onions planted in the later part of kharif or early rabi season. Bolting may be due to one or two more factors like varietal difference, extremes of temperature, growth obstacles, poor seed quality, poor soil, amount of sun shine etc. The number of bolts observed in onion after early sown bajra has less of bolting compare to other crops like beans, sunhemp etc. Between MH 30, 40 formulation, MH 40 was found to be more effective in controlling the problem of bolting. A species of 5% MH 40 in water reduced percentage of bolting to about 3% control when applied 2 ½ months after transplanting.

GARLIC

Botanical name:	<i>Allium sativum</i>
Family	: Amaryllidaceae
Origin	: Central Asia and Southern europe
Plant part	: Cloves

Un injured bulbs of garlic contain an amino acid called as **Allicin**. It is a odourless, colourless water soluble amino acid. On crushing the garlic bulbs, cloves an enzyme called as **allinase** break down allicin to produce alicing.

The other principle ingredient responsible for typical odour of garlicis **Diallyl disulphide**. Ingredients like **diallyl tri/ poly sulphides, allyl, propyl disulphides, diallyl dipropyl disulphides** etc are present in the garlic. The typical flavor of garlic is due to the presence of Chemical **Allecin** plus **Diallyl disulphide**.

Importance:

It is used as a spice. It is important because of its medicinal properties carminative or gastric stimulant in unani or ayurvedic medicines. It is useful against flatulence the allicin has hypo cholesteralic action it reduce the cholesterol level in the blood. Garlic used in the case of TB, rheumatism, sterility, cough, red eyes etc in the form of garlic juice. The crop is other foreign exchange earner from one hectare. It has insecticidal action mainly repellent action. One per cent garlic extract protect from mosquito and flies.

The extract of garlic along with chilli and ginger exhibited strong nematicidal action and it has killed the meloidogyne nematode. Garlic extract also act larvicidal properties the larvae of Culex, spodotera. Extract of garlic acts as fungicial action. It is found to inhibit growth of 200 pathogenic fungi and also control foot rot of French bean. Garlic extracts records number of records on several medicinally important fungi.

It got bactericidal properties. It is found against Staphylococcus bacteria gram positive and gram negative bacteria can be effectively damaged by garlic extract. Garlic kills the colonies of food poisoning bacteria, clostridium perfringens.

Garlic differs from onion in that instead of producing one large bulb it produces small bulbs called **cloves**. This group is covered with a thin skin.

Soil:

The crop should be grown in well drained loamy soil the other requirement, same as that of Onion, except that the garlic develop in richer soils, under loamy soil conditions the bulbs are deformed and difficult to harvesting. The badly drained soils the bulbs get discoloured. Therefore good drainage is essential.

Climate:

Same condition as that of onion as required by Garlic crops. The frost hardy but excess hot drought long not tolerate, prefers moderate temperature in summer and winter. The maturity of bulb should coincide in dry period.

Therefore the crop is planting during winter (September to October) and harvested when the hot season sets in. Bulbing requires longer day in high temperature. Bulbs planted in the western side are found to be larger when compared to the eastern side.

Exposure to low temperature was found to increase the size of the bulb and this can be supported by the above findings. The survival and yield is slightly higher at low temperature. Leaf growth stalks when bulb growth commences.

Short day and low temperature action should coincide the vegetative growth long day condition high temperature. Yield depends on the amount of the vegetable growth by the bulbs. Late planting adversely affects the vegetative growth and bulb yield and increasing day length from 8 to 12 hours increased the bulb weight, diameter and number of cloves and decreased the number of secondary cloves. Critical day length for bulbing is found to be 12 hours. Short day condition suppress bulb formation.

Preparation of soil:

The soil should be thoroughly prepared by repeated ploughing. Well decomposed FYM at the rate of 10 to 20 tonnes per hectare is to be applied.

Propagation: by clove or bulbils or aerial bulbils. Tissue culture techniques as used as explants are also in practice. Propagation by aerial bulbils is preferred when closer space is adopted. Aerial bulbils are more productive compare to clones.

Season of planting:

In Maharashtra, Karnataka and AP August to November. In Northern plains of India September to November. In hilly areas, March – April. In West Bengal November.

Spacing:

The ideal spacing is 15 cm x 10 cm. The seed rate 500 to 600 kg of cloves per hectare. The sowing can be done by three methods

1. dibbling 2 furrow planting 3. broad casting.

When close spacing is adopted the size of the bulb decreases but the total yield increases. When broad space adopted size of the bulb increases but total yield decreases.

Garlic cloves are soaked in cycocel (CCC) at the rate of 1000 ppm for around 24 hours before sowing to enhance germination. Storing of bulbs induce low temperature break the dormancy and increase the yield. The temperature used for storing bulbs would be 0 to 20°C with slight variation of maximum limit of 40°C.

Manuring:

Well decomposed FYM at the rate of 20 tonnes per hectare is applied basally. Nitrogen, phosphorus, potassium are applied at the rate of 60 kg each. Nitrogen is given two split doses one is at the time of planting second is 30 days after planting. Application of borax at the rate of 10 kg/ha is found to increase bulb size and yield.

Irrigation:

Before planting the field should be given light irrigation. Irrigation is repeated on every 3rd day during the initial stages the crop is irrigated more frequently per once in a week but afterwards the crop irrigated once in 15 days. The frequency decreases as the crop reaches maturity the harvesting time the irrigation is stopped.

Interculture:

Manual weeding is expensive and it may cause the damage to the crop. Therefore chemical herbicides like pre emergence weedicide pendimethalin @ 1.5 kg a.i/ ha. Control grasses and broad leaved weeds.

Foliar application of GA3 at the rate 400 ppm are treating the bulbs GA3 50 ppm stimulated the formation bulblets. At inflorescence stage this treatment increase the number of cloves per bulbs and delayed leaf formation in storage. Benry amine at the rate 50 ppm induced lateral bud formation but has less effect when compare to GA3 is reported.

Ethrel at the rate 900 to 1400 ppm was found to inhibit the plant height and storage leaf formation but decrease in the leaf width. NAA 50 ppm inhibited the formation and storage of bulbs. CCC at the rate of 1000 ppm resulted in highest yields when the bulbs are soaked before planting. The bulbs are soaked before planting. When the hormone like NAA, CCC ethrel @ 50 ppm are sprayed at 60 to 90 DAP yield and yield contributing characters were found to increase.

Harvesting:

When the top turn yellowish and they show symptoms of drying and bending we can recognized that the crop is ready for harvesting the crop takes around 6 months after planting for maturation. Bulbs are carefully lifted and clean the leaves tied at the top shade dried in one week, the bulbs are cured for proper drying for about 3 – 4 day in shade for maximum one week period before storage.

Yield: yield varies depending on many factors 40 to 100 quintals per hectare.\

Storage:

Garlic is fairly cured in ordinary ventilated rooms. Tied leaves are hanged in well ventilated rooms. Packing is done singly in polythene bags or small sized bags packed together. Storage temperature is kept at 0 to 1.2 – 20 C. irradiation with Cobalt 60 or gamma rays is recommended. Storage at 1.50 °C and 75% RH decreases the storage loss with reference to ambient temperature treatment with ultra violet light 30 minits and storage 1.50 C also increases storage losses.

Storage losses prolong and loss in weight an be decreasing during storage by spraying Maleic Hydrazide at the rate of 3000 ppm 3 weeks before harvest. **MH @ 2500 to 5000 ppm** when sprayed 2 weeks before harvest inhibited **sprouted** in storage up to 300 days without any adverse effect on yield and also decrease the net loss during storage.

Lect.10.Root crops such as Carrot, Radish, Beet-root

Botanical name:	<i>Daucus carota</i> – Carrot
Family	: Umbelliferae
Origin	: Central Asia
Chromosome number	2n = 18

Introduction:

Carrot is grown all over India and is used for human consumption as well as per forage and particularly for feeding. It is taken raw as well as cooked in curries and is made into pickle and sweet meats.

Black carrots are used for the preparation of a sort of beverage called **kanji**, which is a support to be a good appetite. The orange coloured varieties are rich in **carotin** a precursor of vitamin A and contains appreciable quantity of thiamine and riboflavin.

Edible portions: the edible portions of the root varies in colour and size the outer portion of the tap root is more fleshy contains more sugar than the inner core which is more pithy.

Carrot is grown in winter crop in tropical conditions. In temperate countries it is grown as spring summer crop. It is a root vegetable and has world wide distribution. From economic point of view it ranks after tomato, cucumber, lettuce and sweet corn.

Types:

They are two groups.

1. Asiatic type
2. European type

Asiatic type see suitable tropical and sub tropical. European type are suitable to temperate area. Asiatic type have more red water soluble anthocyanin pigment and less water soluble and lycopin pigment. Vitamin A content less. In European types, vitamin A content is more.

S.No	Asiatic	European
1	They produce roots and seeds, freely in the plains	they produce good roots in plains but fail to produce seeds.
2	The Asiatic type have the root ends conical	root ends are blunt.
3	The roots are red with more anthocyanins	they produce orange coloured roots
4	produce large core heavy tops	rich in carotene and core is comparatively less and soft.
5	More juicy	less juicy
6	Cores are less coloured	cores are self coloured
7	Produce more core	less core
8	Early maturing and have high sugar content	late maturing
9	They are tropical and sub-tropical (annual)	temperate (biennial)
10	No chilling requirement for seed production	chilling requirement is must for seed production

Varieties:

In India long orange coloured smooth rooted types are more popular. For exotic types sown for taking late crop varieties like chanteney, nantes and scarlet horn are the best. Seeds of chanteney nantes half long are supplied for vegetable research station in kulu valley in Punjab. The following varieties released by IARI Delhi

1. **Nantes:** It is a European type, gives skin stumpy orange group, perfect and cylindrical. The core is tender sweet self coloured. Gets ready in 120 days.
2. **Chanteney:** European varieties it has attractive deep reddish orange. It has a tapering but distinct stump end and core sweet and indistinct gets ready in 120 days.
3. **Pusa kesar IARI: local red x nantes half long.** It is a selection from a cross between local red with nantes half long. It has self coloured central core less. Branching of root and short leafy top. A notable feature of this variety, so that the **root stay a month longer in field without any sign of bolting.** Varieties released by Punjab agricultural university.
4. **pusa meghali (temperate type) half long:** nantes pusa yamadagni and coreless are some more improved varieties are

Climate:

Carrot is cool season crop. Some of the tropical types tolerate quite high temperature. A colour development and growth of roots is affected by temperature. At the temperature of 10 to 15 °C they will produce longest roots (but they develop poor colour). At the temperature of 23.95 to 24.44 it will be produce shortest roots (colour will be less bright up). The effect of temperature will vary with the variety and soils. At very low temperature 4 to 7 °C colour development very poor roots become enlarged and they are tapered very little enlarge of very little top growth was observed.

Temperature on germination: Off temperature for germination 7.2 to 23.9 °C. Optimum temperature for better growth 18.3 to 20.90 °C.

Temperature on carotene content: Carotene content is reduced at temperature at below 15.6 °C and above 21.1 °C that is why carrots grown 15 to 20 °C will develop good colour. Poor light is essential for producing carrots. The tropical types produce roots even at high temperature of 25 °C.

Soil:

It requires deep well drained sandy loam. This type of soil is particularly good for early crop. Heavy soils will check the development of roots and cause forked lateral roots. It is grown in rainy season under heavy soil condition or black soil condition even then it will produce abnormal or forked roots. Roots impeded by rich soil strengthened or thickened with increased branching. Carrots does not grow will highly acidic soil. Maximum yield can be obtained at pH of 6.5

Seed rate:

5 to 6 kg per ha it can be up to 6 to 9 kg per ha is the variety Premnath. Sowing time is September (best).

Optimum temperature for germination 7.2 to 23.90 C. it can be sown from June-July, August, September and in hills the month of January – February. Tropical types up to September to October. Temperature types are sown from October onwards. Several successive sowing at interval of 2 to 4 weeks help to ensure a continuous harvest of marketable roots. The seeds are sown on ridges on flat land about 1.5 cm deep.

Method of sowing:

Seeds are directly sown in the field in ridges per furrow or flat bed. In rainy season, ridges and furrows are made. In Rabi season if the soil is loose, sowing is done on flat beds. If the soil is hard, sown on ridge and furrows.

Seeds are thin and light there are mixed with soil or broken rice at 1:1 or 1:0.1 or 10:1 ratio. Seeds are sown to a depth of 1 to 5 cm at a spacing of 30 cm between rows. (within rows 5 cm). the seeds germinate within 5 to 10 days after sowing under good moisture conditions.

Thinning has to be done at distance 5 to 10 cm. thinning is done at 8 to 10 days after germination, the resulted spacing is 30 x 5 to 10 cm.

Manuring:

Carrot is a heavy feeder. Potassium requirement is high. For an yield of 100 tonnes per hectare, crop remove around 14 kg Nitroget, 8 kg P and 45 kg Potassium. For an yield of 200 q per hectare a crop removes 40 kg N, 225 kg P and 125 kg Potassium per hectare.

Depending on soil fertility Farm Yard Manure at the rate of 30 tonnes per hectare is applied at the final ploughing and dosage of 40 to 60 kg nitrogen, 25 to 50 kg of phosphorous and 90 to 110 kg of potassium per hectare is recommended has basal dosage. Excess nitrogen reduces root quality damage the content of sugar, dry matter, carotene and vitamin C. fresh cowdung should not be applied it may cause forking of roots.

Irrigation:

First irrigation is given soon after seed sowing this has to be followed. Four to six days after sowing. Irrigation is given at 6 to 10 days interval.

Earthing up:

Should be done at 60 to 70 days after sowing to help in development of roots the soil is earthed up covered the top of developing roots to prevent loss of colour of tops, the tops becomes green and toxic on exposure to sun light.

Mulching:

Field is mulched with soya bean stubbles or rye mulch comparatively to conventionally tilled plots.]

Harvesting and yield:

Early carrots are harvested when they are partly developed. For the purpose of distinct markets otherwise they are retained in the soil till they reach full maturity stage they should not be retained afterward. They have become puffy core becomes hard and is unfit for consumption.

There are two methods of harvesting.

1. roots are dug out when they are sufficiently moist which khurpa or spade in black. Light irrigation has to be given before harvest, so that easy pulling of roots can be facilitated without damage by holding the leaves.
2. in case of Asiatic variety the roots are harvested when they attain marketable stage 2.5 to 5 cm in diameter, at the upper end after harvesting the roots they are trimmed and washed before sending them to the market. They are packed with gunny bags.

Yield: varies to the variety.

Tropical types give around 20 to 30 tonnes per ha

Temperate type may give 10 to 15 tonnes per ha

RADISH

Botanical name: <i>Raphanus sativus</i> – Radish	
Family	: Cruciferae
Origin	: Europe or Asia
Chromosome number $2n = 18$	

Radish is a popular root crop throughout India. Radishes are useful as inter crop or companion planting between rows or plants or slower growth. Radish is both annual and biennial. The edible portion is **fleshy root** it develops both primary root and hypocotyls.

Varieties:

Radish roots vary greatly in size and colour as well as the length of time for which they remain edible. The radish varieties are divided broadly.

1. European or temperate types
2. Asiatic and sub-tropical

Temperate types are smaller in size mild in taste and are mostly raised as salad crops.

A pure white thin and tender variety becomes ready in about 30 days after sowing.

Pusa himani: a long white variety released by IARI it is suitable for sowing for mid December to late February in plains and during summer in the hills. Rapid red white tipped or scarlet globe or French break fast: globular varieties they become ready in 26 days.

Tropical types: Japanese white: roots grow better when the temperature is low, they are pure white 30 – 45 cm long mildly pungent with blunt end.

Pusa desi: Roots are of same size as though of Japanese type they have green stem end they are more pungent they have tapering root are suitable sowing early August.

Pusa chetki: roots are pure white, wild pungent suitable for sowing from March – August. Roots become ready for harvest within 40 – 45 days.

Pusa reshmi: roots are 30 – 35 cm long tapering white with green stem end it is suitable for sowing in the month of September.

Hill queen: it is a popular variety in North India for sowing in September onwards. The other good indian varieties are Jaunpuri. (Gaint variety more than 1 m long) Bombay red, Chinese rose contain arka nisanth, Punjab safed, Punjab agethi.

Climate:

Radish is tolerant to heat. The develops its flavour, texture and size, the cool season the temperature between 10 to 15⁰ C are required. Varieties differ in their temperature requirements. The crop is grown in hot weather small rooted varieties produce tough and extremely pungent roots.

They should be harvested while they are still young and small rather than allow to reach edible size the long and large varieties can withstand heat and rain much better. Plants bulb normally when the day is 10 – 8 hours long. **Bolting** is quick when the day length increases. Long days as well as high temperature produce premature seedling or formation of stalks without adequate roots.

Soil:

Radish is grown in all types of soil but light friable soils is considered best. Sandy loamy soil with high amount of humus is ideal. It can also be grown fairly on acid soils.

Heavy soils produce rough ill shaped roots with number of small fibrous laterals which decrease market value. The summer crop is preferably grown on cool moist soils such as silty loams.

Time of sowing and seed rate:

Radish is sown on ridges. Spacing varies with the variety. Temperate type becomes ready 25 to 50 days. Therefore they are sown very closely whereas tropical types take a long time and being larger in size they are given a wider spacing. Radish is sown on ridges on 45 cm apart and 22 cm high. A small furrow about 1.25 cm deep is made on the top of the ridge with a sharpened end of stick and seed mixed with fine sand or coarse soil is sown in furrows by hand. The seed is then covered and soil is formed around it.

Radish is often sown other field and vegetable crops such as wheat, grapes, onions, cabbage, cauliflower, methi etc.

Larger variety need earthing up at least once. The first irrigation given immediately after sowing and subsequent irrigations can be applied once a week. One or two weedings in the form of earthing up or weeding manually is sufficient for the whole period of crop.

Manuring:

Radish is average of short duration judicious use of manures and fertilizers highly needed for its production. 120, 60, 120 of N:P:K along 30 kg of MgO will be generally sufficient and basal dressing of 25 to 45 tonnes well decomposed FYM is to be added when the time of soil preparation.

Harvesting:

Radish should be harvested when the roots are still tender. A few days delay of harvesting particularly temperate type may make the roots pithy and quite unsuitable for market. The roots are pulled out along with the tops and washed and then packed for the market. European types should be uprooted 20 day after sowing before they become spongy and hollow inside other types are harvested the roots are till tender and attainment of full size.

All the roots are not uprooted from the whole field at one time, but they are harvested as when they become ripe enough for the market or home consumption. They are pulled out with tops by hand and wash it remove the soil to give good appearance they are sent to market loose in basket or tied in the bunches of 3 – 6 according to variety.

Yield: The European variety yield 8000 to 12000 kg per ha

BEETROOT

Botanical name: *Beta vulgaris* – **Beetroot**

Family : **Chenopodiaceae**

Origin : **Mediterranean area**

Chromosome number $2n = 18$

Varieties: Ooty 1, Crimson Globe, Detroit Dark Red and Red Ball.

Soil: It comes up well in all types of friable soils. The suitable soil pH is 6.0-7.0. It is considered to be a cool weather crop.

Season and sowing: July – August.

Seed rate: Sow 6 kg/ha directly in ridges at a spacing of 10 cm.

Preparation of field: Plough to a fine tilth and form ridges and furrows 30 cm apart.

Irrigation: Irrigate the field copiously immediately after sowing and afterwards as and when necessary.

Drip irrigation

Install the drip system with main and sub main pipes and place the inline lateral tubes at an interval of 1.5 m. Place the drippers in lateral tubes at an interval of 60 cm and 50cm spacing with 4 LPH and 3.5 LPH capacities respectively. Form the raised beds at 120 cm width at an interval of 30cm and place the laterals at the centre of each bed.

Application of fertilizers: Apply FYM at 20 t/ha and 60:160:100 kg NPK/ha as basal and 60 kg N/ha after 30 days.

Fertigation:

Fertigation requirement: 120:160:100kg / ha

Spacing: 30 x30 x10 cm as four rows in each paired row / raised bed system.

LECT.11.Tuber crops such as Potato and Sweet potato

Botanical name:	<i>Ipomaea batatas – sweet potato</i>
Family	: Convolvulaceae
Origin	: South America
Chromosome number:	$2n = 6x = 90$

Sweet potato is very important crop in tropical and sub tropical region of the world.

Importance:

It is a staple food in several tropical countries. It is one of the **world's highest yielding crop with reference to total food production and requires low inputs**. It can be grown throughout year in tropical and subtropical areas even marginal farmer gets some income. Asia accounts for **92% of world's sweet potato** area and production it occupies the seventh rank world statistic, after wheat, rice, potato, barley and cassava. It has drought tolerant attributes and can be grown on lands of low fertility and it has the **ability of the highest solar energy fixing efficiency**.

The creepers are used as cattle feed. The crop also are used making starch, 4% sugar and 20% alcohol producing materials. It is a convolvulaceae for member. It is a morning glory family **n = 15**. generally is seen as **hexaploid** with chromosome irregularities are common leading to infertility.

Varieties:

State variety

Bihar and UP	Pusa safed, pusa lal, rajendra sakarkand.
Maharashtra H –	268
Kerala kemhangad local,	srinandini, sreevardhini
Andhra Pradesh	H – 41, 42, H – 268, H 30
ANGRAU	Kiran and Samrat
Tamilnadu	Co – 1, 2 and 3

Kiran: it is a mid duration variety, photo insensitive variety escapes weevil infection because of its earliness. Tuber is fusiform with red skin the tissue rich in carotene with height orange colouration. In Kharif, 14 tonnes per hectare in Rabi. 17 tonnes per ha in summer 4 tonner per hectare. The crop comes to harvesting in 110 to 120 days.

Samrat: short duration variety. Photo insensitive good for summer and kharif tuber is fusiform with white and cream chromosome flush with pink tinge. Yield: 12 to 15 tonnes per hectare. 120 days duration.

Pusa safed: produces long tubers which are whitish. It is less affected weevils. Average 30 tonnes per hectare; duration is 105 -120 days.

Kalmegh: tubers are round, light brown in colour. Yield is 20 to 26 tonnes per hectare. Duration 90 to 100 days. It is a very early variety.

Pusa Ial: tubers are long with pinkish red flesh. Average 20 – 22 t per ha with 120 days duration.

Among the red and white varieties the white variety is supposed to be less sweeter and more fibrous. The red variety is shorter in duration and has better quality tuber. The tuber weight varies widely depending on the variety from 100 g to 1 kg. IARI released variety **Pusa safed**.

Soil:

Crop prefers sandy loam type of soil. It also grows on clay loam. Best soils are lateritic soil with good depth and better fertility. Cultivation of sweet potato is to be avoided. Heavy clay soils because then hinder the tuber growth and aeration will be a problem. It prefers a pH

Range 5.2 to 6.5. Highly fertile soils should be avoided. Since they favour vegetative growth of vines vine growth will be luxurious at the cost of tuber growth. Harvesting becomes a problem in heavy soil. Selected soil should have good drainage capacity also.

Climate:

Prefers moderate warm climate for at least a period of 4 months. Optimum temperature range 21 to 26⁰ C. Optimum temperature range is 21 to 26⁰ C. Temperature less than 10⁰ C will damage the tubers. Crop requires bright sunshine moderate rainfall ranging and long photoperiods promotes vine growth and reduce tuber formation.

Cultivation of sweet potato can be observed 42⁰ C north and crop can be cultivated above short 20 MSL. Shorter days can favour tuber development. It is one of the most drought resistant vegetable thus not stand frost.

Land preparation:

The roots of sweet potato tend to be long and tend on deep soil hence deep ploughing avoided around 15 to 20 cm depth of land preparation is sufficient. Fresh manure is not applied this crop is produced long and rough roots soil is brought in to fine tilth by discing and harrowing.

Propagation:

propagated by sprouts from **tuber and vine cuttings**. Vine cuttings are preferred when two crops are grown in succession it is faster and cheaper vine cuttings are collected from previous crop and are used for planting the succeeding crop. Cuttings are sown directly in the states like Bihar, UP and Orissa practice of taking and creeps in succession respective year is common under such situation vine cuttings are cheaper and material fence collected uniform shape tuber can be obtained. Terminal vine cuttings are better propagating material but yield is comparatively less with vine cuttings.

Sprouts from tuber:

Sweet potato can yield better when sprout tubers are used for propagation. Medium sized, healthy sprouts free from pests and diseases or branches are selected and they are planted in a nursery.

Primary nursery:

site is well prepared adequately manured provided with good drainage and it is kept raised. The tubers are planted at around 16 to 25 cm row to row 45 cm distance up to around 10 cm depth and is adequately irrigated. After maintaining the sprouts for about 45 days in primary nursery will be shifted to secondary nursery.

Secondary nursery:

Sprouts are shifted to secondary nursery. Sprouts are shifted from primary nursery they will be maintained in secondary nursery up to the time when they reach sufficient length for transplantation secondary nursery same spacing.

Planting:

Healthy sprouts or cuttings are selected for planting time varies depending on the availability of soil moisture and area spring season average planted February to March autumn September to October in case of Bihar and U.P. In West Bengal September and October the planting is done. In Tamilnadu, September is normal season, kerala – may. **Sweet potato is a largely a rabi season crop.**

Method of plantation:

Tamilnadu ridges and furrows prepared. In kerala, raised mounds are prepared over which planting can be taken up. Cuttings or sprouts should be an 5 to 16 cm per with 5 to 7 nodes.

Two method of planting **horizontal and vertical** in case of horizontal planting are node on either exposed and middle portion is buried under the soil. In case of vertical planting 1 to 2 buds are kept out of the soil. In Punjab, planting is preferred in Maharashtra, Bihar, Orissa and U.P vertical planting is preferred. Keeping the vines in shade from about 2 to 3 days before planting will facilitate faster vine growth besides increased in yield. Spacing adopted is 6 x 20 cm and depth of planting can be up to 20 to 30 cm in up to 40 cm.

Manuring:

Well decomposed Farm Yard Manure applied dosage of 20 to 25 tonnes per hectare. Nitrogen @ 20 kg per hectare is applied basally nitrogen same quantity is given after 30 days and 60 days after planting. Phosphorus at the rate of 60 to 80 kg per hectare is complete basal dose, potash at the rate of 80 to 120 kg per hectare is applied in two split dose, basal application and second 60 days after planting.

Irrigation:

One light irrigation is essential before planting. Three days after planting one more irrigation is a must it is called as a life irrigation because it is essential for survival after wards the crop is irrigated once in a week. Critical stage varies from north stress 40 days after planting.

Inter culture:

In early stages perfect weed free situation is provided. Earthing up of soil will control weeds and also improve physical conditions around the plant. The crop requires 3 to 4 weeding after planting. Generally normal weeding is practiced. Chemical weedicides like 2,4 – D. Alachlor, Bromuron can be recommended for application.

Harvesting:

Maturity of a crop is indicated by a change in colour of leaves. Leaves turn yellow gradually dry and or shed. Certain varieties comes to harvesting within 90 to 100 days and certain varieties 120 to 180 days. Tuber also become brittle when fully developed. The surface of a mature tuber if cut and exposed to the air dries up soon while the immature one remain moist and turn dark in colour. Judging the correct stage of maturity crop has to be harvested. If harvested early it cause reduction in yield. If harvesting is delayed up to 8 to 9 weeks it leads to serious loss in yield.'

Method of harvesting:

irrigate the field 3 to 5 days before harvest work with space to uproot the crop and while doing so, avoiding mechanical damage to the tubers. The tubers after harvesting are cleaned, graded and either send to market or stored.

Storage:

Tubers should be cured before storing. Process helps in the formation of callus on the injured portions particularly where the roots have been severed. The tubers are cured at best 180 C at 30%. Relative humidity maintaining for 10 days. Where the these control conditions not available, the tubers are cured either room temperature or dried in the sun for about a week.

Yield:

Yield vary with variety, management, irrigation etc. under irrigated conditions the average yield 35 to 45 tonnes per hectare under rainfed conditions average yield is 10 tonnes per hectare. In India average yield of sweet potato under field condition seems to be 6 to 7 tonnes per hectare and 20 tonnes vine per hectare can also be harvested to use it as green fodder per manure.

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POTATO: *Solanum tuberosum* L.; Solanaceae

Varieties: Kufri Jyoti, Kufri Muthu, Kufri Swarna, Kufri Thangam , Kufri Malar and Kufri Giriraj.

Soil and climate: The soil should be friable, porous and well drained. The optimum pH range is 4.8 to 5.4. It is a cool weather crop. Potato is mostly grown as a rainfed crop. Cultivated in regions receiving a rainfall of 1200 - 2000 mm per annum.

Season and planting**Hills**

Summer : March – April

Autumn : August – September

Irrigated : January – February

Plains: October – November.

Use disease free, well spouted seeds weighing 40 – 50 g. Use Carbon disulphide 30 g/100 kg of seeds for breaking the dormancy and inducing sprouting of seeds. Plant the tubers 20 cm apart.'

Seed rate: 3000 – 3500 kg/ha.

Preparation of field: Prepare the land to fine tilth. In hills provide an inward slope of 1.40 in the terraces. Provide drainage channel along the inner edge of the terrace. Form ridges and furrows with a spacing of 45 cm between ridges.

Irrigation: Irrigate the crop 10 days after planting. Subsequently irrigation should be given once a week.

Drip irrigation

Install drip system with main and sub-main and place the inline laterals at the interval of 1.5. Place the drippers at the interval of 60 cm for 4 LPH or 50 cm for 3.5 LPH in the lateral system. Form the raised beds at 120 cm width at an interval of 30cm and place the laterals at the centre of each bed.

Application of fertilizers: Apply 15 t of FYM/ha, and 2 kg each of *Azospirillum* and *Phosphobacteria* as basal and 120 kg N, 240 kg P and 120 kg K/ha in two splits; half as basal and the balance for top dressing 30 days after sowing. Apply magnesium sulphate at 60 kg/ha as basal dose.

Fertigation

Fertigation requirement - 120: 240:120 kg of NPK / ha. Apply once in every three days throughout the cropping period.

Spacing: 60 x 30 x 20cm in paired row system

Diseases

Late blight: Carefully select seed tubers. Remove ground creepers which serve as a source of infection. Spray Mancozeb 2 g/l or Chlorothalonil 2 g/l on 45, 60 and 75 days after planting. Grow late blight resistant varieties like Kufri Jyothi, Kufri Malar and Kufri Thangam.

Brown rot: Select disease free seeds. Give proper drainage facilities. Remove and destroy the affected plants.

Early blight: Spray Mancozeb 2 g/l or Chlorothalonil 2 g/l at 45, 60 and 75 days after planting.

Virus diseases:

Use virus free potato seeds. Rogue the virus affected plants regularly. Control the aphid vectors by spraying dimethoate 30 EC or methyl demeton 25 EC @ 2 ml/lit

Yield: 15 – 20 t/ha in a duration of 120 days.

Lect.12 Leafy vegetables such as Amaranthus and Palak

Botanical name: *Amaranthus blitum*
Family : **Amaranthaceae**

Amaranthus blitum and *A. tricolor* are most commonly grown in South India. Blitum is dwarfish with small leaves and rejuvenate quickly after each cutting. It is suitable for sowing in summer. Tricolor is taller with thick stems and larger leaves. This type can be grown in summer as well as in the rainy season.

Varieties:

Two varieties green and red. The green variety is more common.

Method of sowing:

The seed is sown broadcast after mixing with fine earth for uniform distribution.

Seed rate: 2-3 Kg /Ha

Manuring:

Small quantities of Ammonium sulphate or urea can be mixed with irrigation water and applied when the seedlings are about 4 inch high to push vegetative growth.

Harvesting:

Usually starts in about 3-4 weeks after sowing. About 5-9 cuttings can be taken at weekly interval. The fully grown side leaves are removed. The top may also be cut leaving the lower leaves to produce new shoots in their axils.

Yield: 7500 Kg /Ha in a span of 2 months

Botanical name: <i>Beeta vulgaris</i> Var. <i>Bengalensis</i> - Palak
Family : Chenopodiaceae
Origin : Indo – china

Desi palak : *Beta vulgaris*

Vilayathi palak: *Spinacea oleracea*

Spinach is grown in north India. Vegetable growers mostly cultivated spinach while spinach is limited to kitchen gardens.

Varieties:

1. varieties with reddish mid rib
2. varieties with greenish mid rib
3. **All green** released by IARI, New Delhi. Leaves are uniformly green given 6 to 7 cuttings at 20 days interval starting from one month after sowing 125Q per hectare. Development of seed stalk take about 75 days.
4. **Pusa palak:** IARI released produce uniform green leaves it is late in bolting and

5. **Pusa jyothi:** evolved treating the seed of all green 2% colchuri it produces large green tender succulent leaves. Leaves are brittle in nature they have flavour and colour it gives 6 to 8 cutting. *Yield:* 50 tonnes per hectare. It is grown throughout the year rich in many nutrients.
6. **Pusa harit:** evolved by hybrid sugarbeet x local palak released by IARI. It was developed to mainly to suit hilly regions. Crop require chilling a character inherited from sugar beet. Leaves are thick green and big sized. It is a late bolting variety. Tolerate alkaline soil.
7. **Jobner green:***** university of udaipur popular in Rajasthan. It is a mutant variety from the popular of local collection it produces large thick tender succulent leaves. The taste is equal to all green gives yield of 300 q per hectare.

Soil:

Prefer sandy loam good drainage. Neutral pH. Cultivar Jobner green can withstand pH of 10.5. crop is regarded as highly salt tolerant and grow saline soil. Spinach is regarded as still more tolerant to saline condition.

Climate:

Prefer autumn and winter mild climate in plain. Withstand frost and some extent and warm weather. If temperature is very high, it results in bolting.

Sowing time:

Sown 2 to 3 seasons and condition are favourable. Sown through out the year. In plains spring season and rainy season are the important times of sowing. In hilly areas it can be sown even the a month of April.

Seed rate:

25 to 30 kg per ha. Beds are made irrigation channel in between. Seeds are soaked in water to hasten up germination. In general the seeds are sown by broad casting. But time of sowing is preferable. Spacing of 20 cm is kept in successive time. Germination commences within 10 DAS will complete in soon. One light irrigation after sowing is essential.

Manuring:

Nitrogen fertilizer is more important. Apply 35 to 40 kg per ha of well decomposed FYM 25 to 50 kg of N, P, M are given as basal and 25 kg Nitrogen after every cutting.

Irrigation:

More number irrigations are essential in light or sand soil. Summer crops requires frequent irrigations at 3 to 4 days interval. In winter, autumn irrigations are given at 10 to 15 days interval.

Harvesting:

Starts at 25 to 30 DAS. It may continue to 15 to 20 days interval. Cut only green succulent tender leaves, winter crops gives more cuttings than summer crop varieties. Jobner green gives more cuttings compared to all.

Yield varies with cultivar 120 to 300 quintals per ha.

Lect.13 Perennial vegetables such as drumstick and pointed gourd

Botanical name:	<i>Moringa oleifera</i> - Drumstick
Family	: Moringaceae
Origin	: North west India
Chromosome number	2n = 28

It is also known as Moringa. It is also known as **ben oil tree**. Economic plant parts: pods, leaves and flowers.

Importance:

Drumstick is considered one of the popular tree vegetable especially in south India as a home garden tree besides also being commercially cultivated on large scale for leaves flowers etc. The tender fruits and buds form the main vegetable part of the tree and are nutritious.

The crop is highly valued for its distinct appealing flavour of fruits. It is a rich source of proteins, minerals and vitamins. Seeds are also very important as they give oil which is known as ben oil or beh oil. This oil is used for illumination and soap making industry. The oil is highly prized; it is used for lubricating wrist watches, computer peripherals and delicate goods. Seeds contain 38 to 40% non-drying oil which is clear odourless and never become rancid. Oil is edible and also used in the making of perfumes and hair dressing. Left over cake is non-edible and used for purifying effluent water from rivers, ponds etc along with Aluminium sulphate.

The wood of the trees gives blue dye and a coarse fibre. A paste of leaves is used as an external application for wounds. Flowers are used as tonic and diuretic. The roots of the plants are used for seasonal pickle and for flavouring ghee. In some parts of Dutch East India the seeds are fried and eaten. The fruits are rich in vitamin B and vitamin C and they have medicinal properties so as to use them in the treatment of rheumatism and cardiac and circular stimulant.

Varieties:

There are **perennial and annual** types in drumstick. Perennial types are mostly suitable for kitchen garden. Ex: **Jafna**.

Kodaikas, Moringii, Kuttu maringi, Palmedu, Moringi, Palamedu moringi.

Annual: they are commonly planted on commercial scale their fruit and seed. KN 1, PKM 1 (conunce), PKM 2 its bigger size (pods).

PKM 1: it is annual type belong to Moringa. Pterigosperma released from TNAU by pure line selection the fruits are long fleshy and tasty. Tree 4 to 6 metres height, comes to flowering 100 to 105 DAP. Each tree 250 pods are pod length well 60 to 75 cm. each fruit weigh 150 grams and contain 70% edible pulp. The tree is headed back after harvest and it is cultivated 3 year after which fresh planting is done. Pods obtain edible maturity 65 DAS. This variety can be grown as inter crop in orchard during their pre bearing period. Estimated yield per hectare 520 Quintals.

PKM 2: it was evolved as hybrid between **MP 31 x MP 28**. the plants are thick growing around 4 to 8 m height each tree has around 12 branches bears flowers in clusters 3 to 4 pods can be obtained in a cluster. Start flower 100 to 120 days after planting.

First harvesting is taken at 170 to 180 DAP. Each pod weight 280 grams with 125 cm length and it girth of 8 cm. seed content decrease with increase in the flesh content. A yield will be around 980 quintals per hectare.

Perennial type:

Jaffna: the fruit length is 60 to 90 cm.

Chavakacheir also a sub type in Jaffna length 90 cm

Chemmurunga: red tipped fruits.

Soil:

Drumstick grown on any soil except stiff clay soil. Sandy loam with good amount of time are mostly preferable.

Climate:

It is a tropical plant will come up will under dry and arid tract said to be drought resistant. The optimum temperature is 25 to 35 degrees Centigrade. It is highly susceptible to frost when the temperature goes beyond 40 degrees centigrade results heavy flower shedding.

Propagation:

Propagated by seeds or limb cuttings. Perennial types on commonly propagated by limb cuttings. Annual types are propagated by **seed**. The yield in perennial is used during first two years. Perennial types are to be planted at wider spacing, annual closer spacing. The **limb cuttings** 1 to 1.4 m length and which having 14 to 16 cm girth are collected from index plants and these limb cutting planted to 60 centimeter cube. **Planting time:** June to October.

In case of seed propagation or 625 g of seed is required for one hectare. Limb cuttings are planted at distance 2.0 x 2.5 m

Pit size and planting:

Around 45 to 60 cubic centimeters pit are dug up and pits are refilled the top soil is mixed with 5 to 10 kg FYM, 25 g of SSP and 50 g of foli-dol dust. The pits are irrigated before planting per sowing either seed and seed per pit are sown and regular watered 15 to 20 days.

Seeds germinate. Thinning is practiced seedling 15 to 20 cm height only vigorous plants are retained other or removed.

Nursery:

In case of seed propagation instead of direct sowing a nursery can also be sown with 15 x 75 cm polythene bags. Bags are filled with mixture of sand, soil, biofertilisers and seeds are sown per bag. They are allowed for one month, till the seedling attains 30 cm height. Seeds are transplanted to the main field. Nitrogen is done in the month of June transplanting in the month of July.

Flowering is attained during the month of January to February and pods will come to harvest from March onwards. Around 75 to 100 additional plants are grown in polythene bags per the purpose of gap filling.

Manuring:

100 grams of Urea, 60 grams of each SSP and MOP are applied for each plant and then they are sufficiently irrigated. The plants are top dressed with 100 g of Urea 3 months after the first application besides these inorganic fertilisers well decomposed FYM @ 10 to 15 Kg may be added for each tree. The manures and fertilizers are applied in ring trenches dug at 60 to 90 cm among from the tree during the rainy season.

Aftercare:

Training and pruning practices are important when seedling attain 75 to 100 cm height they are headed back this practice helps in production of side shoots, strong side shoot are allowed in all direction when the side shoots are 45 cm long they are again headed back. This helps in development of better frame work and increase the yield.

Moringa plant can fell down during heavy wind. Wind movement height rise. The soil trunk mound up 30 to 45 cm height from ground level.

Irrigation:

The basins of trees should be cleaned of all weeds mid slight sowing of soil is very useful. Basin method of irrigation is followed in drumstick. Irrigation management is important in this crop. Since the plant need optimum moisture for good growth and development.

First irrigation is given to pits before sowing next irrigation after 3rd day of sowing light irrigation are given to maintain optimum moisture till the seeds are germinate. Irrigation 10 to 15 day interval in the absence of rains is beneficial however they should not be any water stagnation otherwise they will be a heavy flower drop soil should never become too dry or too wet.

Harvesting:

Annual type comes to first harvest in about 6 months after sowing or planting perennial. Perennial types which are seed propagated may come to first harvest during third year of planting. The pods should be harvested at optimum edible maturity around 60 days after pollination the pods will reach this stage. Flowering and fruit bearing takes place simultaneously and continues uninterrupted for the next 2 to 3 years 1 to 2 harvestings

Yield:

Annual type 200 to 250 fruits per tree per year, during first and second year. Perennial type can yield 80 to 90 fruits during the first 2 years and afterwards the yield will increase up to 500 to 600 fruits per tree per year during the fifth year. The important reasons of harvest in drumstick in March to June, September to October. There is no storage method. Therefore fruits consume immediately after harvest it may be used 1 or 2 days.

INTRODUCTION TO SPICES

Spices – uses – properties and classification – importance of spice industry in India

Spices:

Spices are those plants, the products of which are made use of as food adjuncts to add aroma and flavour (**ex.** Pepper, Cardamom, Clove, Nutmeg etc), while **condiments** are those plants, the products of which are used as food adjuncts to add **taste** only. Both spices and condiments contain essential oils, which provide the flavour and taste. They are of little nutritive value. They are used as whole, ground, paste or liquid form, mainly for flavouring and seasoning food. Most spices increase the shelf-life of food.

Uses of spices:

1. The principal use of spices is to season the insipid foods to impart flavour, aroma and taste.
2. They are also used as preservatives and fumigants.
3. Spices also find use in pharmacy and indigenous medicines.
4. Spices are also used in perfumery, soaps, cosmetics, tooth paste, confectionery, incenses,

Properties of spices:

1. Spices are well known as appetizers.
2. They add a tang (taste) and flavour to otherwise insipid foods.
3. Some of them also possess anti-oxidant properties.
4. Some of them have preservative qualities (clove and mustard)
5. Some have strong anti-microbial and antibiotic activities.
6. Several of them possess medicinal properties.

LECT.14.Seed spices: Coriander, cumin, fenugreek & fennel

LECT.15. Black pepper and Cardamom

LECT.,16. Turmeric & Ginger

GINGER

B.N: *Zingiber officinale*
Family: Zingiberaceae
Origin: South East Asia

Ginger is one of the fine most important major spices of India. India is the largest producer of dry ginger in the world, accounting for more than 60% of world production. 1/3rd of the production of Ginger in the country is exported.

Kerala is the largest producer of ginger accounting for more than 40% of the total countries production. In Andhra Pradesh, it is cultivated in Nellore, East and West Godavari, Medak, Visakhapatnam and Srikakulam districts. Dried rhizomes scraped or peeled are greatly esteemed for their aroma, flavour and pungency.

Uses: Ginger is used as a spice.

It is used for the preparation of Ginger oil and oleoresin.

It is used in the preparation of soft drinks, alcoholic beverages.

Green Ginger is used in the culinary preparations.

Preserved Ginger is used for the manufacture of processed food.

It is used as carminative and stimulant.

Botany:

It is a herbaceous perennial plant but largely grown as annual. Height varies from 30- 90 cm. The under ground stem (Rhizome) grows horizontally, thick, flattened, branched, covered with small scale leaves and fibrous roots.

Varieties: Varieties are named after their places. They are mainly clones. The following are some of the clones grown in the respective states.

Assam : Jorhat, Nadia, Maran.

Andhra Pradesh : Narsipatnam, Siddipet and Tuni.

Kerala : Wyanad local, Ernad, Kuruppampadi, Thoduphuza.

Exotic varieties:

China: Introduced from China. This variety has got 3.43% fibre. It has too much moisture. The yield of dry Ginger from green ginger is 13- 15%.

Rio-de-janeiro: This variety was introduced from **Brazil**. It has 5.19 % fibre. It consists more moisture. The yield of dry to green ginger is 16-18%. It became more popular where Ginger is used in green form.

High Altitude Research Station, **Pottangi** has released the following varieties.

1. **Suprabha:** It is a clonal selection from Kanduli local. Rhizomes have plumpy fingers and bright grey skin. It yields about 3.4 tonnes per ha. It has got 4.4% fibre.
2. **Suruchi:** It is a clonal selection from Kanduli local. The Rhizomes are greenish yellow. Its yield is 2.7 tonnes per ha. It has 3.8% fibre.
3. **Surari:** It was developed as a mutant. Fingers are cylindrical. Yields about 4 t per ha.

Durations of the above varieties vary from **220 to 230 days**.

Propagation:

Propagated by **seed rhizomes**. Rhizomes are cut into small pieces (setts) of 2.5 to 5.0 cm. They may weigh about 15 – 20 grams having one or two buds.

Climate:

It requires warm humid climate. It grows upto 1500 m above MSL. It requires moderate rainfall from sowing to sprouting. It requires fairly heavy and well distributed rains during crop growth period. It requires dry weather from 28 – 35°C for about a month before harvesting. Prevalence of **high humidity** and **shade** are necessary during crop growth period.

Soil:

It requires deep, well drained, humus rich soil. It is sensitive to water logging. It can be grown well on sandy loams, clay loams and laterite soils.

Preparation of land: The land is ploughed repeatedly for 5 to 8 times thoroughly

1. to achieve fine tilth.

Systems of cultivation / planting technique: Ginger is raised by 2 methods viz. bed and ridge system depending upon rainfall and soil type

Heavy rainfall areas : Beds or ridges.

Low rainfall areas : Flat beds or ridges.

Irrigated crop : Ridges & Furrows 40 – 45 cm apart to a height of 30 cm

Rainfed crop : Raised beds of convenient length and 100 cm wide are prepared with an inter space of 30-40 cm in between the beds for irrigation and drainage

Planting:

- i) Season: In Andhra Pradesh, first fortnight of May. Early planting is beneficial.
- ii) Seed rate : 1500 – 1800 kg per ha.
- iii) Spacing : 45 x 15 cm
- iv) Sowing : Treat with Dithane M 45 0.3 percent for 30 min.

After treating, seed rhizomes are kept in moist gunny cloth for 2 – 3 days to promote early sprouting.

The crop is mulched after sowing. Sprouting starts within a week and continues for another 3-4 weeks.

v) Mulching:

In ginger cultivation, mulching the field is an important operation. Mulching has several advantages

1. Source of organic manure
2. Prevents washing of soil
3. Conserves soil moisture
4. Smoothers weed growth.
5. Improves the physical properties of soil
6. Protects the setts and sprouts from hot sun
7. Provides favourable conditions for the sprouting of the setts.

First mulching is to be done at planting with 10-12 tonnes of green leave per ha. Second mulching is done at 40-60 days after first mulching with 5 tonnes of green leaves. Farm yard manure is spread over the green leaves. In irrigated crop – raise cow pea and black gram as live mulches. Remove the mulches 60 days after sowing during rains. Castor seeds are sown on bunds at 9 m intervals, to serve as wind breaks.

Irrigation:

Irrigate at 4- 10 day interval. During mid September to 3rd week of December irrigate at fortnightly interval.

Manuring:

At the time or before planting 25 – 30 tonnes of cattle manure 50 kg P₂O₅; 25 kg K₂O are applied as basal dressing. At 60 and 90 days after planting Nitrogen is applied in two splits @ 37.5 and 37.5 kg per ha along with 25 kg K₂O.

In Andhra Pradesh:

Basal dose: 15-20 tonnes of FYM per ha : 60 kg P₂O₅; 30 kg K₂O per ha.

Top dressing: 120 kg Nitrogen (40 + 40 + 40); 60 kg K₂O (20 + 20 + 20) per ha is applied in three equal splits at 40, 80, 120th day after planting. The plants are earthed up after each application of fertilisers.

Weed control: Reduce competition by weeds. 3-4 weedings are necessary between June to September. After each weeding, mulch is to be replaced. Clumps are removed and earthed up in September.

Harvesting: For Green Ginger: **6 months** after planting. For Dry Ginger: **8 months** after planting, when leaves turn yellow and start drying between 245 to 260 days.

Preserved Ginger: **7 months** after planting. Before they fully mature, after which they become more fibrous. Light irrigation is given before harvesting. Dig the plants carefully. Soak them overnight. Wash thoroughly. Sundry for a day or two before marketing as green ginger.

Yield: Average yield under rain fed condition is 10-15 tonnes per ha.

Average yield under irrigated condition is 15 – 25 tonnes per ha.

Processing:

I. Dry Ginger: Preparation of commercial dry ginger involves a series of steps. Fully developed rhizomes are harvested after **8 months** of planting for preparation of Dry Ginger.

1. **Soaking in water:** The rhizomes are soaked overnight in cement tubs for easy removal of skin.
2. **Trampling:** The rhizomes are trampled under feet in the tub. Avoid damage to epidermal cells containing flavouring oil.
3. **Peeling:** The skin is peeled off, with sharp bamboo knives. Don't rupture epidermal cells. This step hastens drying process.
4. **Washing and Drying:** The peeled rhizomes are washed and sundried for 3-4 days on cement floors.
5. **Polishing:** After drying the rhizomes are polished by rubbing with a coarse cloth to remove all bits of skin or dirt. These are called **unbleached** ginger. To get **bleached** **Ginger**, peeled rhizomes are soaked in **2% lime water** for 6 hours, fumigated with **sulphur** for 12 hours.

Yield of dry Ginger is **16 to 25%** of the fresh Ginger.

II. Preserved Ginger: Ginger is harvested at **7 months** after planted for preparing the Preserved Ginger. It is preserved in syrup or brine.

Rotations and mixed cropping:

Continuous Ginger cultivation – promote exhaustion of fertility and favour the incidence of diseases.

In Irrigated lands – Ginger is rotated with betelvine, banana, turmeric, Onion, Garlic, Chillies, Vegetables, Sugarcane, Maize etc Ginger can be grown as intercrop in Coconut, Arecanut, coffee, Mandarin, Mango, Guava and Grapes.

TURMERIC

B.N: *Curcuma longa* and *Curcuma aromatica*

Family: Zingiberaceae

Origin: South East Asia

Plant part: Rhizome

Underground rhizome is used as condiment, dye stuff, drug and cosmetic. Traditional item of export. **India** is the largest producer of Turmeric. In India, A.P., leads in Area and Production. In A.P., there are five agro-climatic zones.

Duggirala zone, Cuddapah zone, Nizamabad zone, Godavari zone and Agency Turmeric ranks 4th as foreign exchange earner among the spices after Pepper, Cardamom and Ginger.

Botany:

Curcuma has 42 species. *C. longa* is cultivated in large area in A.P. ***C. aromatica*** is grown in some parts locally known as **kasturi** used for the preparation of **kum kum**. Erect herbaceous perennial 60-100 cm rhizome with fingers. Rhizome is brown on outside and dull orange coloured inside. Leaves are broadly lanceolate with long leaf stalk.

Varieties: Three categories.

Long duration types : 9 months duration: Duggirala, Tekurpeta, Armor and Mydukur, CLL 324, 325, 326, 327

Medium duration types: 8 months duration: Kothapet, Krishna, Kesari, CLL 317.

Short duration types: 6-7 months duration: Amalapuram, Dindigam, PCT – 13 (Suguna), PCT – 14 (Sudarshan) Armoor is the popular type in Nizamabad district.

Kasturi and kesari are good in curcumin content, but poorer in curing percentage.

Propagation:

Through rhizomes, mother rhizomes, cut mothers, primary fingers. Planting primary fingers has become a common practice in A.P., because they keep better in storage, more tolerant to wet soil and involve low seed rate.

Climate:

Tropical herb. Grows well 1200 m above MSL. Requires warm and moist climate. Rainfall 100 to 200 cm. Temperature range preferable is 20 to 30⁰ C

Preservation of seed rhizomes:

1. Mature, healthy rhizomes are heaped over a layer of sand (5- 10 cm thickness) under shade of a tree or shed.
2. Covered with turmeric leaves.
3. Heaps are plastered with earth mixed with cow dung.
4. Treat with Dithane M 45 0.3% or Bavistin 0.1% solution for 30 min and shade dried before heaping.
5. Remove rotten rhizomes at the end of storage period.

Soils:

Can be grown on various soils. Thrives best in well drained, friable, rich sandy or clay loam soils. Crop stands neither water logging nor alkalinity.

Preparation of land:

Give 4-6 deep ploughings to get fine tilth up to 20 -25 cm depth. Field is laid out into beds or ridges and furrows. Bed system gives higher yield by 54 to 80%. Beds of 1 m width and convenient length with a spacing of 40 to 50 cm between beds where natural drainage does not exist, ridges and furrows are prepared at 45 to 60 cm spacing.

Time of sowing:

Short duration varieties: second fortnight of may

Mid duration varieties: first fortnight of June

Long duration varieties: second fortnight of June to second fortnight of July.

Seed rate: mother rhizomes 2000 – 2500 kg per ha

Primary fingers 1500 to 2000 kg per ha

Spacing:

Red loamy soils – 30 x 15 cm

Black heavy soils – 46 x 23 cm

Method of sowing: sown behind the plough in ridges and furrow system In bed system, rhizomes are dibbled at 5-10 cm pits. Treat with dithane m 45 0.3% for 30 minutes before sowing. Germination starts in 10-20 days and will be over by 60 days.

Mulching: To protect sprouts.

To conserve moisture|

To reduce weed growth

To enhance germination by mulching with dry leaves. Mulch with dry leaves thickly on which a layer of cow dung is spread. Second mulching is done after weeding and application of fertilizers, after 50 -60 days of sowing.

Irrigation:

A good soaking irrigation is given immediately after sowing .There after, irrigate at weekly interval.

Manuring:

20-25 to of FYM during land preparation 60kg N, 60 kg P, 60 kg K per ha before planting the rhizomes.

60 kg N -----65 kg K per ha 60 das

60 kg N -----per ha 120 das

50% of top dressed Nitrogen may be applied in the form of oil cakes.

Top dressing should be completed within 120 days after sowing. After fertilization, the field is given a light hoeing and the clumps are earthed up and irrigated.

Weeding:

3 – 4 weedings are required at 60, 90, 120 and 150 days of planting. Weedings and hoeings are done simultaneously. Plants are earthed up and mulch is replaced. Provision of shade. By raising castor on the bunds and within the crop. Daincha is also grown for shade purpose.

Intercrops:

Maize or chillies are grown as inter crops. Turmeric can also be raised as an inter crop in coconut and arecanut plantations.

Rotations:

Turmeric is a heavy feeder. Hence depletes soil nutrients. Continuous cropping results in build up of diseases. It is rotated with rainfed paddy, sugarcane, banana, betelvine, vegetables.

Harvesting:

Depending upon the varieties, the crop comes to harvest in 7-9 months.

1. Main season of harvesting falls in February – April.
2. Maturity indication is complete yellowing and drying up of plants
3. Above ground parts are cut close to the ground level.
4. Field is irrigated 1-2 days in advance of harvesting the crop.
5. Crop is harvested by Ploughing or digging.
6. Rhizomes are gathered by hand picking and cleaned.
7. Rhizomes are washed.
8. Mother rhizomes are separated from the fingers before they are cured.

Yield: Indian average yield is 20,000 to 22,000 kg per ha

Processing:

Fresh rhizomes are not useful for marketing. Curing makes fresh rhizomes marketable.

Curing involves **boiling, drying and polishing**.

A. Boiling: is done either by traditional or improved method.

I. Traditional method:

Water is poured to cover rhizomes in the vessels of copper or galvanized iron or earthen material. Mother rhizomes and fingers should be boiled separately, since fingers take long time for boiling. Stop boiling when froth, fumes with typical odour comes. Rhizomes yield to finger pressure. Over cooking should be avoided as it spoils the colour, while under cooking renders the dried product brittle.

II. Improved method:

50 kg of cleaned rhizomes are taken in a perforated trough made of GI sheet. It is immersed in a pan. Alkaline solution **0.1% sodium carbonate/ sodium bicarbonate** is poured in the trough. Boil till fingers become soft.

Alkaline solution helps in imparting orange yellow colour to the core.

B. Drying:

The boiled rhizomes are sun dried in 5.7 cm thick layers for 10 – 15 cm layers. Rack frequently for uniform drying. Dry until they become hard, brittle, break with a metallic sound. After drying they should possess only 8 – 10 % moisture.

C. Polishing:

The dried rhizomes are smoothened by manual or mechanical rubbing. Manually they rubbed on hard surface or trampled under feet. Mechanically they are polished by mechanically operated polishing drums.

D. Colouring:

They are coloured to improve the appearance. Rhizomes are artificially coloured in **two ways**. Dry and wet colouring. Half polished fingers are coloured.

In dry process – turmeric powder is added in the last 10 min to polishing drum.

In wet process – turmeric powder is suspended in water and mixed by sprinkling.

For brighter colour – boiled, dried, half polished fingers are taken in baskets and shaken continuously with an emulsion of

2 kg turmeric powder,
0.04 kg alum,
0.14 kg castor seed oil,
30 g of sodium bisulphate and
30 ml HCl.

Coloured rhizomes are again sun dried before sending to market

BLACK PEPPER

Botanical name: *Piper nigrum*

Family: Piperaceae

Origin: Western ghats of South India (Malabar)

Useful plant part : Fruits / Berry

Introduction:

Pepper is the most important of all spices and popularly known as the '**king of spices**'. Black pepper is a dried mature fruit of perennial ever green climbing woody vine. It is one of the most important earliest known spices produced and exported from India. It is the most valuable and important foreign exchange earner among the important spices earning nearly 50% of the total export earnings from all the spices, because of its importance in the spices and unique position in trade and large share in export earnings, it is popularly referred as **king of spices** and **black gold** in trade.

Area and production:

Outside India it is grown in Srilanka, Malaysia, Indonesia, Brazil Mexico, China, Thailand and Madagascar. India accounts for **54%** of the total area under pepper in the world but its share of production is only 26.6% whereas the other countries like Brazil, Indonesia, Malaysia accounts for lesser percentage of area but with more share in the total production due to their higher productivity.

In India pepper cultivation is confined to southern states only. It is grown mainly in Kerala (96% area), Karnataka, Tamilnadu and pondicherry. Since ancient time pepper is exported from India. On an average of **85%** of the Pepper produced in the country is exported to USA, Canada and Italy. Till 19th century India enjoyed monopolistic position in the world market. However now India lost its top position due to low productivity, poor yield and raise of Indonesia and Malaysia countries.

Botany:

Black Pepper is an evergreen. Perennial woody climber reaching to a height of 10 m or more. It has extensive but shallow root system. The vines branch horizontally from the nodes but do not grow longer. Based on the growth habits morphological characters and biological functions, **five distinct types** of stem portions can be defined in the shoot system of a pepper vine.

1. Mainstem:

It originates from a seed or from a stem cutting. It climbs on a support with the aid of aerial or adventitious roots.

2. Runner shoots:

They are produced from the basal portion of the main stem, growing at right angle to the main stem, usually restricted up to 50 cm from the ground.

3. Fruiting branches (plagiotropics):

They are produced from the nodes of the main stem and they grow laterally more or less at right angles to the main stem, bearing the **spikes**.

4.Topshoots (orthotropics):

vegetative shoots which arise on the top portion of the main shoots. It gives a bushy appearance with stouter, thicker internodes and with large number of adventitious roots at the nodes. They are used for the **propagation**.

5. Hanging shoots (geo tropics):

In a fully grown vine, some of the plagio tropics at the top portions are seen to give rise to special type of shoots which hang down and grow geotropically.

Leaves are broadly lanceolate, alternate, simple, dark green and shiny above, pale green and gland dotted under neath. Size and shape vary according to the variety.

The inflorescence is a **catkin**. Born on opposite side to the leaves on Plagiotropic branches, 5-15 cm long, bearing 50-150 minute flowers. The flowers are monoecious or Dioecious or hermaphrodite (bisexual) forms occurs in different varieties.

High yielding forms should have more percentage of bisexual flowers and in cultivated varieties these plants will be more than 80%. Male flowers are very few 1-19% in different varieties. The fruit is a single seeded berry, sessile, small globose or oval. It has thin pulpy pericarp around the seed. It takes approximately 6 months to mature after flowering. Each spike produce 50-60 fruits. The skin (exocarp) turns from green to red on ripening and black on drying.

Varieties:

More than 75 cultivars of pepper are being cultivated in India. Majority of the cultivated types of Pepper are monoecious.

Karimunda: most popular variety throughout Kerala. A prolific and regular bearer. Dry recovery is 35%. Yields good quality of Pepper. Suitable for **intercropping as well as for high density cropping**.

Kalluvally: This is grown in North Kerala. It is hardy and regular bearer. It withstands water stress and is **moderately tolerant to Phytophthora wilt**. Spikes are medium long and have a characteristic twisting due to very thick setting. Driage is 40%. It is grown either alone or mixed with other cultivars. Kottandan, Narayakkodi, Aimpiriyan, Neelamundi, Kuthiravally, Balankottah, Malligesara and Uddagare.

Recently a number of improved cultivars have been evolved and released. They are Panniyur 1, 2 (Krishna), 3 (shima), 4 and 5 : released from Kerala Agricultural University, Pepper Research Station, Panniyur.

Improved varieties of black pepper :

Name of hybrid / variety	Parentage	Yield / vine(Kg)
Panniyur-1	F1 hybrid Uthirankotah x Cheriya kaniakadan	2.5
Panniyur-2 Krishna	Open pollinated seedlings of Balankottah	4.5
Panniyur-3 (Shima)	F1 hybrid Uthirankotah x Cheriya kaniakadan	4.4
Panniyur-4	Selection from Kuthravally II	2.3
Panniyur-5	O.P progeny of Perum kodi	2.75
Sreekara	Selection from Kanmundu (KS 14)	4.8
Subhakara	Selection from Kariamunda (KS27)	4.2
Panchami	Selecton from Aimpriyan Coll . 856	5.2
Pournami	Selection from Ottaplackal Type coll.No.812	4.7

Sreekara, Subhakara, Panchami, Pournami – released from NRCS, Calicut.

Malabar types: grown in Malabar region in Kerala. Examples are Kalluvally, Balankottah. They are suitable for **shade** Cheriya kody, uthirankottah have high percentage of pistillate flowers.

Malvad types: grown in malvad of Karnataka. Workaiamarata variety is preferred for white pepper other examples include doddagya, karimarata, arasilamarata, malligesara, tisara.

Travancore types: these are hardy, the cuttings root easily and climb the support without external help. **Ex.:** Karimunda, veluthanamban (tolerant to wilt), kuthiravally, chola.

Uses:

Black pepper is used for a variety of purposes. One of the principal values of Pepper is its ability to enhance the seasoning of dishes. It is valued for its characteristic aroma, hot pungent and biting taste and is mainly used for flavouring and seasoning.

Climate:

Pepper is a tropical plant it requires warm humid climate for commercial production. It grows successfully between 20° N and South latitude and from sea level up to 1500 m above MSL. The crop tolerates temperature between **10° C to 40° C**. But it requires an optimum temperature of 25 to 35 degrees Celsius. A well distributed annual rainfall of 125 to 200 cm is considered ideal for pepper. Prolonged droughts stop the vegetative growth of the vines.

Soil:

Pepper can be grown in a wide range of soils such as clay loam, red loam, sandy loam and lateritic soils with a pH of 4.5 to 6.0. It thrives better in soils rich in organic matters.

Propagation:

Pepper is invariably propagated vegetatively (**stem cuttings**). This is because of variation shown by the seedling progenies and also that the seedlings become diocious and come to bearing very late (7-8 years). Pepper is propagated by **cuttings raised mainly for the runner shoots**. Cuttings from lateral branches are seldom used, because reduced number of fruiting shoots, short living and bushy in habit.

Cultural operations:

The pepper vines are tied firmly as and when they grow. The tying is done around the node, so that the nodal region is firmly attached and pressed against the standards so as to allow the roots to cling to the standards.

This is an important operation which has to be attended to carefully as otherwise the vine will have no support. The another practice is that when vine reaches at a height of 75 to 100 cm without branches all the leaves are removed except the terminal 3-4 leaves. The defoliated portions are covered with friable fine soil. This aid in establishing a well developed root system and lateral branches may arise out of them.

Training and Pruning:

The vines are trimmed at the top and prevented from growing too tall for convenience of picking. In Tamil Nadu, pepper vines trained on Silver oak standards are pruned at **6 m** height from ground level for easy picking. The vines are trained up the support to give one main shoot and two lateral orthotropic shoots. These are pruned regularly to encourage the development of lateral fruiting branches but these lateral branches should not be tied to support, as this would discourage the bushy side growth.

The vines are first pruned back to 15-20cm from the ground level, when they developed to 8-9 nodes length. Second pruning is done when a further 9-10 nodes have been produced, to a height of 3-4 nodes of the previous cut. In this way vines are pruned 7-9 times until they reach the top of the support about 3 m high. When this is achieved, the terminal growth of vine is arrested by frequent pinching. The lower portion of vine is kept clean and unbranched at least a meter from ground level.

Manuring:

Judicious and regular manuring is necessary to get good yields. About 10 kg of well rotten cattle manure or compost is given in April – may. Fertilizers to supply 100 g Nitrogen, 40 grams phosphate and 140 grams of Potash per standard for vines at 3 years and above age may be applied annually in 2 split doses in April-May and August – September. During the first year of planting 1/3rd of above dose and in second year 2/3 rd of the above dose may be given.

Manures are applied around the vines at a distance of 30 cm and forked in to the soil. Lime may be applied at the rate of 500 grams per standard. During April in alternate years.

Harvesting:

Pepper vines start yielding usually from the **3rd or 4th** year. The vines flower in May-June. It takes 6-8 months from flowering to ripening stage. Harvesting is done from November to February in plains and January to March in hills. When one or two berries on spike turn red in early the whole spike is plucked. Yields vary with the variety and season. A full bearing vine yields one kg of dry pepper. However, individual vines recorded yields up to 3-5 kg of dried pepper.

Harvesting of pepper is carried out according to the **purpose** for which it is harvested.

For preparation of **white pepper** the berries are harvested at a slightly advanced stage of ripeness i.e when the berries turn red (bright orange). To get **black pepper** the berries are gathered at younger stages.

Yield:

Pepper vine attain full bearing stage in the 7th or 8th year after planting. And yield starts decline after 20-25 years and replanting has to be done thereafter. 7th or 8th year old pepper vine gives 800 to 1000 kg of Black Pepper per ha.

CARDAMOM

Small or lesser Cardamom: *Elettaria cardamomum* (Malabar cardamom)

Large Cardamom: *Amomum subulatum* (native to Eastern Himalayas)

Bengal cardamom: *Amomum aromaticum*

Family: Zingiberaceae

Origin: Western ghats of South India (Kerala)

Useful plant part: Fruit (capsule)

Cultivated cardamom has chromosome number $2n = 48$.

Out of the above species, most popular species occupying a premier position is small cardamom. Large cardamom is mainly cultivated in Dargeeling, Assam, Himalayas, Nepal, Bhutan, Thailand, Indonesia. Bengal cardamom is grown in Northern Bengal.

Introduction:

Cardamom is popularly known as the **Queen of Spices** and also **Green Gold**. It is one of the ancient species of India and is also one of the most valued spices of the world. It is **next only to black pepper** as the largest foreign exchange earner among various Indian spices.

Cultivation of Cardamom is mostly concentrated in the evergreen forests of western ghats in South India. Besides India, Cardamom is cultivated in Guatemala, Tanzania, Sri Lanka, Vietnam, Cambodia and New Guinea.

Among three cardamoms small one is the most popular species. India has the largest area (90% of the world area) and is also largest producer (70%). But of late India is facing still competition from **Guatemala** in the world market for the top position.

In India the cultivation of small cardamom is mainly confined to the southern states viz., Kerala (60%), Karnataka (30%), Tamil Nadu (10%). Among the different spices, exported from India cardamom ranks **second** after black pepper. Nearly 40% of the production is exported to more than 60 countries.

Importance/ uses:

Cardamom is used for flavouring and seasoning various food stuffs, confectionery, beverages and liquors. In Arab countries, beverage cardamom flavoured coffee is called '**Gawa**'. It is generally offered to guests at social and religious functions. In Sri Lanka cardamom is used in manufacturing liquors.

The essential oil of cardamom is used for medicinal purposes both in allopathy and in Ayurveda. It is used as powerful aromatic stimulant, carminative, stomachic and diuretic. Cardamom seeds are chewed to prevent the bad breath, indigestion, Nausea and vomiting.

Eating one cardamom daily with a table spoon of honey improve eye sight and strengthens the nervous system and keeps one healthy. It is believed by some people that excessive use of cardamom causes impotency.

Botany:

Cardamom is a herbaceous perennial plant. A fully grown plant is about 2- 4 m height. The real stem of the plant is the under ground rhizome. The aerial pseudostem is made up of leaf sheaths. Leaves are lanceolate with dark green colour. It has shallow root system inflorescence is a long panicle with racemose clusters arising from the under ground stem but comes up above the soil.

Flowers are bisexual, pale white fragrant flowers. **Fruit is trilocular capsule.** Flower initiation takes place in March – April and from initiation to full bloom it takes nearly 30 days; from bloom to maturity it takes 5 – 6 months. **Honey bee** is the principle pollinating agent.

Varieties:

Based on the size of the fruit, two varieties are broadly recognized in Cardamom

Elettaria cardamomum var major – comprise of wild indigenous type

Elettaria cardamomum var minor – comprise of all cultivated types The cultivated types are identified mainly based on the nature of panicle shape and size of the fruits as follows.

S.No	Particulars	Mysore type	Malabar type	Vazhukka type
1.			Most common and widely cultivated	Natural hybrid between mysore and malabar type
2.	Plant stature	robust	medium	robust
3.	Panicle	erect	prostrate	semi erect
4.	Capsule	bold, elongated	round to oblong	round to oblong
5	Adaptability	high altitudes (900 – 1200 m)	low altitudes (600 – 900 m)	wide range
6	Productivity	More	Less	Less
7	Resistance	Withstand to winds	More tolerant to thrips and less susceptible to drought	
8	Fruits shape	Pods are bold and elongated	Roundish or egg shaped	Roundish to long

Soils:

Grows best on well drained humus rich forest soils. Water logging and excessive soil moisture conditions are detrimental. Moisture level should be 40 to 50% of the field capacity of the soil. An ideal site is a sloppy land with good drainage. In India cardamom is grown on red, deep and good textured laterite forest soils having plenty of humus and leaf mould. Ideal pH is 5. to 6.5

Improved varieties:

Coorg cardamom malabar selection – 1 (CCS – 1)(malabar type): it yields 408 kg per ha dry capsules under rainfed conditions. It was released by National Cardamom Research Station (NCRS), Appangala.

Mudigere – 1(malabar type): it yields 250 to 300 kg per ha dry capsules. It was released by regional agricultural research station, Mudigere (Karnataka).

ICRI – 1(malabar type): it yields 265 to 650 kg of dry capsule. Released by Indian Cardamom Research Institute, Myladumpara.

ICRI – 2 (mysore type): it yields 375 to 760 kg of dry capsule. Released by Indian cardamom Research Institute, Myladumpara.

PV – 1(malabar type): it yields 500 kg of dry capsule. Released by Cardamom Research Station, Pampadumpara.

SKP – 14 (malabar type): it yields 430 to 590 kg dry capsule per ha. Released by ICRI Regional station, Saklespur, Karnataka.

Climate :

Small cardamom is a humid tropical plant. It is grown under natural conditions of ever green forests at an elevation from 600 to 1500 m above MSL. Optimum elevation is 900 to 1200 m. The plant prefers temperature of 10 to 35°C and a well distributed rainfall of 1500 mm per annum.

Summer showers are essential during summer i.e. February – April for panicle initiation. Otherwise it affects yield. It does not stand drought and is highly sensitive to winds. Under exposed conditions, the plant does not attain its full vegetative growth because of sun scorching.

It grows luxuriantly under shade. Shade trees besides providing shade create a congenial micro climate in the plantation. It keeps the surroundings humid and cool. Moderate shade high humidity cool surroundings well distributed rainfall and wind less areas are very essential for the satisfactory performance of Cardamom.

Propagation:

Cardamom can be propagated by seeds, rhizomes and suckers. Out of which seed propagation is most preferred because of certain advantages over vegetative propagation.

By seed:

Propagation by seeds prevents spread of **khatte disease**. This is the most common and widely prevalent method. A large number of seedlings can be raised within a short time. The main disadvantage is that the progeny is highly variable with no uniformity in the yield. The seeds also do not remain viable for longer time.

By rhizomes: Planting material of rhizomes is collected by up rooting 2 to 2 ½ year old clumps

Nursery site and planting:

Seedlings are normally raised in primary and secondary nurseries. The nursery site should be selected on gentle sloppy lands, having an easy access to a water source. Raised beds are prepared after digging the land to a depth of 30- 45 cm. The beds of 1 m width and of convenient length raised to a height of about 30 cm are prepared. A fine layer of humus rich forest soil is spread over the beds. Seeds are to be collected from well ripe capsules.

Immediately after harvesting, the husk is removed and seeds are washed repeatedly in water for removing the mucilaginous coating. After draining the water the beds are to be mixed with wood ash and dried in shade for a day. In order to ensure uniform and early germination, seeds should be sown immediately after extraction. If the sowing is delayed, pre sowing treatment of seeds with 25% Nitric acid for 10 min is advisable to get a quick and higher germination. One kg of capsules may produce 5000 seedlings.

Sowing may be taken up during November – January and is done in rows. Deep sowing should be avoided for better and quick germination. Seeds are mulched to a thickness of 2 cm with paddy straw or any locally available material and are watered regularly. The germination commences in about 30 days and may continue to a month or two. After germination the mulch is to be removed.

Seed rate: 10 g per m² of nursery bed area.

Manuring:

Cardamom is a surface feeder and its growing areas are usually subjected to heavy rain fall conditions, the top soil is subjected to frequent leaching, resulting in the loss and plant nutrients, even though there is annual replenishment of nutrients through the incorporation of fallen leaves of shade trees and cardamom plants. Therefore manuring is very essential.

Under irrigated conditions – 75 kg N; 75 kg P and 150 kg K per ha.

Under rainfed conditions – 30 kg N + 60 kg P and 30 kg K per ha.

Organic manures may be applied at the rate of 5 kg per plant. Two split doses one during May – June for production of suckers, Second during September to October for initiation of panicle. Half dose can be applied during first year. Full dose can be applied from second year onwards. Being a surface feeder deep placement of fertilizer is not advocated. These fertilizers apply 30 cm away from the plant.

Irrigation:

Cardamom is generally raised as rainfed crop. However, it responds well to irrigation. It is necessary to irrigate the crop during dry periods to get increased yields. Since, cardamom is raised under evergreen forests and on undulated terrain, conventional irrigation methods are not useful.

Intercultural operations:

Weeding: 2-3 weedings per year may be necessary during May- June, August – September and December – January. Paraquat @ 625 ml in 500 litres of water may be sprayed.

Mulching: it is an important cultural operation in Cardamom. Fallen leaves at the shade trees are up rooted weeds are utilized for mulching. Mulches should be applied during November – December to reduce ill effects of drought conditions during ensuing summer.

Trashing or clearing:

it consists of removing old and drying shoots of the plant once in an year with the onset of monsoon under rain fed conditions and 2-3 times in high density plantations. The plantation is provided with irrigation facilities. Weeding and clearing may be done simultaneously during May- June and August – September.

Packing and digging:

At the end of the monsoon rains a light raking or digging of soil should be given around the plant up to a radius of 60-75 cm to conserve the moisture to the ensuing dry period particularly in low rainfall areas.

Earthing up: After the completion of monsoon, a thin layer of fertile soil rich in organic matter may be earthed up at the base of plant up to Collar region to encourage new growth.

Cropping:

Cardamom plants start bearing in about 3 years after planting. Flowering starts in April – May and continues up to August – September. Peak flowering will be in the month of May-June. From flowering to maturity the fruit takes 5-6 months.

Harvesting:

Only ripe capsules are harvested at 25-30 days interval, the harvesting is completed in 5- 6 pickings. In most of the areas the peak period of harvest is during October – November.

Yield:

Although the Cardamom plant start bearing from 2nd or 3 rd year of planting, an economic crop can be obtained only from 4th or 5th year. Yield vary with variety and age. Optimum average yield is 50-70 kg of dry capsule per ha.

Yields decline from 10th year to 12th year.

1st year of bearing – 25-50 kg per ha (dry capsules)

2nd year of flowering 50-70 kg per ha

3rd year of flowering 70-100 kg per ha

CORIANDER

Botanical name:	<i>Coriandrum sativum</i>
Family	: Apiaceae.
Origin	: Mediterranean region.
Plant part	: Leaves and Fruits (seeds).

Coriander is cultivated both for seed and leaves. It is a **minor and seed** spice. It is grown for seed in Andhra Pradesh, Tamil Nadu, Karnataka, Madhya Pradesh, Orissa, Rajasthan for leaf throughout the country. In Andhra Pradesh, Coriander is grown in Kurnool, Anantapur, Cuddapah, Guntur and Prakasam, Medak and Nizamabad.

Varieties: The improved varieties released from RARS, Lam.

Lam CS 2 – Sindhu: It is a medium tall variety – more branching – seeds are straw coloured. It withstands pests and diseases – It yields about 1300 to 1400 kg per ha. The duration of the crops is about 105-110 days.

Lam CS 4 – Sadhana: It is fairly tall – good branching – seeds are straw coloured. It withstands pests and diseases – it is a dual purpose – it yields about 1400 to 1500 kg per ha – the duration is about 115 – 120 days.

Lam CS 6 – Swathi: It is a medium tall variety. Much branching. Seeds are straw coloured. It is an early and short duration variety. It yields about 800 to 1200 kg per ha. It has a duration of about 80 -85 days. It escapes Powdery Mildew disease.

Lam CS 7: It was developed as a selection from North India – It is a dwarf plant with more leaves.

Climate:

It can be grown under tropical and subtropical conditions. It require cool climate in early stages and warm dry weather at maturity. It needs dry and cool weather free from rains at flowering and fruit setting stage. Heavy rains effect the crop.

Soil:

Well drained soils, moisture retentive, humus rich soils are essential. It can be grown in sandy loams to heavy black cotton soils. Heavy Black cotton soils of Deccan and South India are particularly suited.

Land preparation:

For rain fed crop plough 3-4 times. For irrigated crop plough twice or thrice. After ploughing is complete, the land is laid out into beds and channels.

Season:

For leaf purpose the crop is grown throughout the year (except March- May).For grain purpose the crop is grown during rabi season in A.P. and north and central India Kharif in Tamil Nadu and Madhya Pradesh. Some times it is grown as a mixed crop in cotton and pulses.

- i) **Sowing time:** for rabi crop mid October to middle of the November. For Kharif crop June-July to August - September.
- ii) **Seed rate:** it needs 12-15 kg per ha in A.P. Crush or trample the seeds gently to separate pericarps. Soak the seeds in water for 12- 21 hours. Treat with Thiram @ 2 g per kg seed. Shade dry before sowing.
- iii) **Method of sowing:** Broadcast sown for rainfed crop. A spacing of 3-40 cm x 15 cm is adopted for an irrigated crop. Germination starts within 10-15 days and will be over by 30 days.

Irrigation: A post sowing irrigation is to be provided. Afterwards, the crop has to be irrigated at Two leaf stage, at branching stage, at flowering stage and at seed filling stage.

Manuring:

On fertile Black Cotton soils, it is not generally manured. During rabi season in A.P. 10-15 tonnes FYM; 20-30 kg N; 40 kg P₂O₅; 20 kg K₂O per ha is given as basal dose. 20-30 kg Nitrogen is given at flowering under irrigated conditions.

Interculture:

Maintain weed free conditions during first 30 days. First weeding is done within 30 days after sowing. Second weeding is done before the rows close up. Inter cultivate twice with a cultivator at monthly interval.

Harvesting:

The crop will be ready for harvest in 80-120 days. Kharif crops come to harvest earlier than Rabi.

Maturity indices:

Green colour turns to straw coloured. If harvesting is delayed – seeds shatter, fruits split. Plants are cut or pulled. Tied in bundles. Piled in shade for drying to avoid grain shattering and loss of essential oil. After 2-3 days of shade drying, the grain is threshed, winnowed and sundried. The moisture content is reduced from **2 to 6 per cent**. The cleaned, dried produce is stored in gunny bags lined with white polythene.

Seed rate:

10 - 12 kg/ha (Irrigated crop)

20 – 25 kg/ha (Rainfed crop)

Seed Treatment: Soak the seeds in water for 12 hours. Treat the seeds with Azospirillum @ 1.5 kg /ha for better crop establishment + *Trichoderma viride* @ 50 kg/ha to control wilt disease.

Pre sowing seed hardening treatment with Potassium Dihydrogen Phosphate @ 10 g/lit of water for 16 hours is to be done for rainfed crop.

FENUGREEK

Botanical name:	<i>Trigonella foenumgreekum</i>
Family	: Fabaceae
Origin	: South East Europe or West Asia
Plant part	: Leaves and Seeds.

Fresh tender leaves, stem are used as vegetable. Seeds are used as spice. It has high medicinal value as it prevents constipation. It removes indigestion, stimulate spleen and liver. Grown in Gujarat, Rajasthan. It occupies 3rd place in area and 4th in production among all minor spices grown in India.

Common methi – *Trigonella foenumgreekum* – is quick growing and produces erect shoots to a height of 40-70cm.

***Kasuri methi** – *Trigonella coriculata* also called as Champa menthi. Champa menthi is a slow growing type and remains in rosette condition during most of its vegetative growth period. Out of these two the former is more commonly cultivated.

Common menthi		Champa menthi	
1	Quick growing habit	1	Slow growing habit
2	Characterized with erect shoots	2	Shoots remain rosette appearance specially when plants are in vegetative condition
3	Reaches to a height of 40-70 cm	3	Reaches < 40 cm
4	Commonly cultivated	4	Rarely ultivated
5	Light to dark green leaves	5	Possess dark green leaves
6	Small bisexual flowers white in colour	6	Light pink flowers
7	Slender pods, long and curl	7	Seeds small, flat and square shaped
8	Characterized by prominent beak pods	8	Deep furrowed pods

Varieties:

Pusa Early Bunching: It is a Bushy variety. 115 – 125 days, seeds are bold.

Lam selection 1: It is a bushy variety, 70 days duration on Black soils under rainfed condition. It yields about 700 -750 kg per ha.

Co 1: It is a selection from TG 2336. It is a dual purpose variety. It is suitable for intercropping. It yields about 650 – 700 kg grain; 4.5 tonnes of green leaves per ha.

Rajendra Kranti : It was evolved by mass selection. Developed by RAU, Dholi. It is suitable for both pure and inter crop. It yields about 1200 – 1400 kg in 120 days.

Rmt 1 : Evolved by RAU, Jobner. It is a pure line selection. It is moderately branched. It is tolerant to root rot and powdery mildew. It yields 1500 kg per ha in 140 days.

Climate:

It is cultivated both in tropics and temperate regions. It is a cool season crop, tolerates warm weather. It **tolerate frost and freezing weather**. It requires low to moderate rainfall. Heavy rains are unsuited.

Soil:

Clayey loams are better. It requires 6-7 pH. It is **tolerant to salinity** problem.

Land preparation: Land has to be ploughed 3-4 times. Prepare uniform beds.

Sowing:

Time: It is sown in the second fortnight of October.

Method: Broad cast over beds and rake in rainfed conditions. In irrigated conditions beds of 200 cm wide 500 – 700 cm length are prepared. Seeds dibbled in lines at 30 x 60 cm spacing.

Seed rate: 30-35 kg for rain fed crop. 12- 15 kg for irrigated crop. Germinate in 3- 4 days and complete in 7 – 10 days.

Irrigation:

The crop is grown as rain fed in heavy moisture retentive soils. Crop requires 3-6 irrigations. First irrigation is given after sowing seeds. Second irrigation is given after first cutting of leaves. Afterwards irrigated at 7 – 10 interval.

Manuring:

The crop is commonly grown on **residual fertility**. FYM @ 10-15 tones per ha may be incorporated in the last ploughing. Top dress with Nitrogen one month after sowing and after each cutting @ 10-15 kg per ha. For grain crop 20 kg Nitrogen, 40 kg P₂O₅ and 20 kg K₂O are applied as basal dose. Depending upon growth 10-15 kg Nitrogen per ha may be applied after one month.

Intercultivation:

Hoeing and weeding during the early stages of plant growth.

Harvesting:

First cut is to be given at 20-25 days after sowing. Later the crop is cut at 12-15 day interval. Afterwards 4-5 cuttings are taken. Then the plant is uprooted at flowering and disposed. For a grain crop, 1-2 cuttings are allowed. The plant flower in 30-40 days after sowing. Pods mature within 60-90 days after sowing.

Maturity indices:

Plant start drying, pods are partially dried. Uproot the plants. Dry in the sun on threshing floor for 2-3 days. Pods are threshed. Seeds are separated by winnowing, cleaned and sun dried. Seeds are stored in gunny bags lined with white alkathene paper.

Yield:

Leaf yield is 700-800 kg per ha. Grain yield is 750 to 800 kg per ha under rain fed conditions. 1200 to 1300 kg per ha under irrigated conditions.

CUMIN

Botanical name:	<i>Cuminum cyminum</i>
Family	: Apiaceae.
Origin	: Mediterranean region.
Plant part	: Fruits (dried seed)

Introduction, uses:

Aromatic fragrance is due to an alcohol 'cuminol'. Used as a spice in curry powder. Ingredient in ayurvedic and veterinary medicines against diarrhea and dysentery. (carminative, stomachic and astringent). Volatile oil content ranges from 2.8 to 4.7

Botany:

Annual with a height of 30-45 cm. Produces many branches. Leaves are long divided deep green. Flowers small, white or rose coloured born as umbels. Seed elongated 6 mm long and light yellowish brown.

Varieties:

RS – 1: Evolved by selection – it is an early maturing variety. It has bold, aromatic seed. It gives an yield of 763 kg per ha.

Gujarat cumin 1 (GC -1): developed by selection. Seeds bold. Yield 735 kg per ha. Duration is 105 -110 days duration. Tolerates blight and wilt.

Gujarat cumin 2:

Rajasthan zeera 19: released by RAU, Jobner. Seeds bold. Yield is 470 to 570 kg per ha. Duration 120 to 130 days. Tolerate blight.

Climate:

Tropical plant and it can be cultivated as Rabi crop in area where Low atmospheric humidity during February to march, when crop flowers and seed formation takes place. High humidity at this stage favour disease. Can be grown from sea level to 3000 m above MSL.

Soil:

Well drained, medium to heavy textured soils. On light textured soils wilt is more. Can be grown in **slightly alkaline soils having a pH of 8.9.**

Land preparation: Land is ploughed repeatedly to bring to fine tilth. Land is laid out into beds.

Seed rate: 8 – 15 kg per ha. Variation is due to method of sowing (line sowing or broadcasting). Soaking of seed for 24 – 36 hours is suggested to enhance the germination percentage. In line sowing, lines are spaced at 20 cm apart. Seeds are covered by fine soil. Irrigate lightly. At 5 cm height, thin the population to 15 cm spacing.

Irrigation:

First irrigation: light irrigation immediately after sowing. **Second irrigation:** after 8 – 10 days, after which the germination is visible. **Third irrigation:** after a week, to help to complete the germination. **After wards:** irrigate at 12-20 day interval. Irrigation at maturity of the crop should be avoided.

Manuring:

FYM @ 15 to 20 tonnes per ha is to be applied at field preparation. Nitrogen @ 30 kg per ha in two splits at 30 and 60 days after sowing is done.

Intercultivation:

At 5 cm tall; first hoeing and weeding. Later one or two hoeing or weedings.

Harvesting:

The crop matures in 100 – 120 days. Uproot the plants. Stake them in sun for drying. Threshed by beating. Clean by winnowing. Store in polythene line gunny bags.

Yield:

Varies from 500 – 800 kg per ha grains. Can be up to 1000 kg per ha under good management.

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